

Firm Pay, Amenities, and Inequality*

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Abstract

We estimate firm-specific amenity valuations using discrete choice experiments in a survey of German workers linked to administrative records. Workers rank hypothetical offers from real firms they would consider joining; randomized wages identify ex-ante, money-metric valuations. Valuations vary across firms and demographic groups, yet a single index predicts firm size and employee satisfaction. High-wage firms provide no worse amenities: valuations rise with wage premia across firms and are approximately orthogonal when weighted by employment. Accounting for amenities widens between-firm inequality by roughly 20 percent in variance terms. Gender differences in valuations explain part of women's sorting into lower-wage firms.

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1 Introduction

Firms differ in pay and in non-wage dimensions (Mas, 2025). A large literature has examined firm-level wage heterogeneity and its implications for inequality. Far less work has asked how non-wage job characteristics vary across firms and how workers value them (Card, 2022).

Identifying the value of firm-provided amenities is empirically challenging. Regressing wages on non-wage characteristics does not identify the value of those characteristics, because such regressions conflate preferences with sorting and with unobserved firm characteristics. A long literature on compensating differentials has emphasized that these approaches often yield “wrong-signed” estimates, and that richer data on job attributes alone cannot resolve this problem (Lucas, 1977; Brown, 1980; Lavetti, 2023). Workers’ transitions between firms can yield estimates of the overall value provided by a firm—given assumptions on offer arrival rates and wage dynamics—but, without added assumptions, these methods recover only the component of amenities orthogonal to wages (Sorkin, 2018). More recent work has turned to stated-preference experiments to estimate willingness to pay for specific job characteristics (see, e.g., Mas and Pallais, 2017; Wiswall and Zafar, 2018; Folke and Rickne, 2022; Maestas et al., 2023), but these studies typically examine specific job bundles and do not yield firm-level estimates of amenity value.

This paper uses discrete choice experiments (DCEs) we embedded in a large-scale survey of German workers, fielded by a group within the German Employment Agency, to estimate firm-specific amenity valuations and examine their role in job choice, sorting, and inequality. We ask workers to rank hypothetical job offers from real firms they themselves identify as likely application targets. We randomize raises across firms, allowing us to recover money-metric valuations of firm amenities, which are not confounded by equilibrium wage differences and which do not rely on assumptions about offer arrival rates.¹ We link the survey to administrative Social Security records and to firm-level data from Kununu, a major employer review platform. The administrative data allow us to study how amenity valuations relate to worker mobility, firm characteristics, and between-firm inequality.

Preferences over non-wage attributes shape job choice. Workers often select lower-wage offers in the discrete choice experiment, implying trade-offs between pay and non-wage attributes even among the firms they are marginal to—those they named as firms they would consider applying to if they switched employers. Valuations of these non-wage attributes (“amenity valuations”) vary widely across workers, even when those workers are marginal to the same firms. For example, the Pearson correlation in amenity valuations between men and women is 0.266, and the correlation between college-educated and non-college workers is 0.231. Formal tests reject equality of male

¹Because the wage variation is randomized rather than equilibrium, the design recovers valuations robust to how market wages are set; as Mas (2025) notes, inferring valuations from choices over exogenous wage variation does not require the underlying wages to be competitive.

and female valuations.

Most models of the labor market assume a uniform ranking of firms, as in single-index models (Card, Cardoso and Kline, 2016; Berger, Herkenhoff and Mongey, 2022; Mas, 2025). The between-group disagreement in valuations would seem to challenge this assumption. Yet a common index—the amenity value shared by all workers marginal to a firm—has strong predictive power for aggregate patterns. Conditional on their wage, firms with higher amenity values are larger and more favorably reviewed by employees—consistent with structural models that infer desirability from firm size or retention, and inconsistent with purely idiosyncratic match effects.

Amenity values vary widely across firms. They are approximately orthogonal to employment-weighted firm wage premia and are positively correlated with firm wage premia when firms are weighted equally. Our results imply non-wage factors compound, rather than offset, the inequality in firm wage premia. Between-firm inequality in total compensation is about 20 percent larger in variance terms than in pay alone. Our results are consistent with models in which firms vary systematically in the utility they provide (Mas, 2025).

Our estimates also reconcile conflicting findings in the literature. Studies based on job-to-job movers sometimes conclude that workers give up amenities to move up the wage ladder; worker-level correlations instead suggest that better-paid workers enjoy better conditions. Neither pattern identifies the firm-level relationship. Our estimates show why both arise. The worker-level correlation reflects sorting: higher-skilled workers earn more and work at firms that both pay higher wage premia and provide more non-wage value. The mover-based pattern reflects selection: applying our firm-level valuations to observed transitions reproduces the negative mover-sample slope even though the firm-level correlation between wage premia and non-wage value is not negative.

Gender differences in amenity valuations explain part of why women sort into lower-wage firms. Female workers make steeper wage–amenity trade-offs when they switch between firms. And among full-time workers, women work at firms that offer lower pay premia. Gender differences in firm-specific valuations work in the opposite direction: women sort toward firms they value relatively more. Whether this fully offsets the pay-premium gap in total value depends on a cross-sex normalization, but the sorting pattern itself—women favoring firms with higher amenities—does not. Consistent with prior work, women in our survey are significantly more likely than men to say that specific non-wage amenities (such as reduced hours or the availability of remote work) are important in their job choice.

This paper contributes to two main strands of literature. First, our work builds on the large literature on amenities and compensating differentials (Rosen, 1986; Mas and Pallais, 2017; Sorkin, 2018; Volpe, 2026), which examines how workers trade off wages for non-wage job attributes. Much of this literature infers the value of amenities from observational wage and mobility patterns, though recent work has introduced experimental methods to value specific job attributes

(Mas and Pallais, 2017). Our approach extends this experimental tradition to estimate workers' valuations of firm-specific amenities. By randomizing wages across hypothetical offers in discrete choice experiments, we identify the money-metric value of firm amenities, avoiding both the endogeneity of hedonic wage regressions and the structural assumptions required by mobility-based approaches. Our results show that the bundle of non-wage characteristics associated with a firm is not, taken as a whole, a compensating differential: high-wage firms are not systematically worse on non-wage dimensions.²

Our results are complementary to those in Maestas et al. (2023), who use a similar stated-preference approach to estimate willingness to pay for a broad set of specific job amenities in a representative US sample. Their design prices individual amenities but does not yield firm-level estimates, and their data are not linked to administrative employment or mobility records. We do not estimate valuations of specific job attributes. Instead, we recover the total non-wage value of specific firms and use the linkage with administrative records to examine whether firms that pay higher wage premia also provide better amenities, whether the gender gap in firm-specific pay premia partly reflects a gender gap in firm-specific non-wage values, and whether job-to-job mobility is related to non-wage factors.

Second, our paper contributes to research on firm wage premia and worker sorting (see, e.g., Card et al., 2018; Card, Heining and Kline, 2013; Song et al., 2019; Lamadon, Mogstad and Setzler, 2022). Prior work has demonstrated that firms differ systematically in the wages they pay, even after controlling for worker characteristics, and that these wage premia are a major source of labor market inequality, including the gender wage gap (Card, Cardoso and Kline, 2016; Sorkin, 2017; Palladino et al., 2025). We know far less about the role of firm-specific amenities in worker mobility and sorting. Our findings show that firm amenities shape which offers workers accept and explain part of the sorting of women into lower-wage firms.³ Overall, accounting for the non-wage value of firms widens, rather than offsets, between-firm inequality.

The paper proceeds as follows: Section 2 describes the data. Section 3 presents the conceptual framework and estimation strategy. Section 4 presents our main estimates. Section 5 examines implications for mobility and inequality. Section 6 concludes.

²Structural models can instead recover a wage–amenity covariance under parametric assumptions; Taber and Vejlín (2020), for instance, estimates that compensating differentials account for only a small share of wage variation.

³Lavetti and Schmutte (2023) also explore whether gender differences in the valuation of amenities can explain the sorting of women into low-wage firms. They find that women sort on physical but not financial risk. Our findings are complementary and focus on the overall non-wage value of a firm, rather than the value of specific job attributes.

2 Empirical Setting and Data

Our analysis combines discrete choice experiments embedded in a large-scale survey of workers, linked to administrative Social Security records and to information on specific amenities provided by firms.

2.1 Data

We obtain data from two primary sources.

CHH Worker Survey. First, we use data from a large-scale survey of German workers—the CHH Worker Survey—to identify workers’ valuations of firm-provided amenities. We fielded this survey to full-time German workers between the ages of 25 and 50 (Caldwell, Haegele and Heining, 2026). The main wave ran in 2022, with a follow-up in spring 2024; we pool data from both waves in most of our analysis.⁴ In previous papers we analyzed modules focused on wage bargaining and on workers’ beliefs about pay (Caldwell, Haegele and Heining, 2026, 2025). Appendix E provides information on the implementation of the survey; Appendix F contains the full survey instruments.

To elicit firm-specific valuations, we designed a survey module in which workers reported where they would apply if they decided to switch firms. This approach allows us to focus on firms that workers actively consider as potential employers. After identifying relevant firms, we asked workers to rank hypothetical job offers from these firms under randomly assigned wage changes. We standardized and cleaned the free-text firm names provided by respondents and matched them to establishment identifiers in the German Social Security records housed at the Institute for Employment Research (IAB) using establishment addresses.

In some robustness exercises we also use data from a survey module in which we provided workers with the names of specific (“researcher-provided”) firms. In this module we asked workers whether they would consider applying to each of seven firms, drawn from a fixed list of 30 large employers; we then asked them to rank hypothetical job offers from three of these firms (chosen at random) under randomly assigned wage changes. We use this module to show both the validity of our estimation strategy and the robustness of our results.

A natural concern with survey elicitation is that workers may respond inattentively or strategically. In a previous paper that used the bargaining modules embedded in this survey, we showed

⁴Both waves were fielded well after the acute phase of the COVID-19 pandemic. To the extent that the pandemic permanently changed firms’ amenity provision—for example, by expanding remote work options—this would drive a wedge between our survey-based amenity valuations and pre-pandemic measures used in some of our external validations. The purpose of the follow-up was to collect data for robustness checks included in this paper and in Caldwell, Haegele and Heining (2025).

that workers’ responses to questions on their bargaining behavior correlate with employer-side responses from a separate survey, fielded by a different organization one year prior (Caldwell, Haegele and Heining, 2026). These and other validations reported in the appendices of that paper and in Caldwell, Haegele and Heining (2025) suggest that workers generally provided truthful, accurate responses in the survey. We validate the specific modules and questions designed for this paper in Section 3.4 and Appendix D.1.

IEB Administrative Records. We use the administrative Social Security records (the Integrated Employment Biographies, IEB) compiled by the IAB to compare amenity valuations to observed pay policies and to link valuations to job-to-job mobility and inequality. These data provide complete employment histories, including job start and end dates, wages, and establishment identifiers. They also include information on demographics and education.⁵ We match the CHH Worker Survey to these records in two ways. First, we match each respondent to their own Social Security records, giving us their complete employment history. Second, we use the crosswalk we generated between firm names and establishment identifiers to match each respondent to data on the firms they named. Appendix B.2 provides more information on how we cleaned the provided strings and generated the crosswalk.

Beyond the data on surveyed workers, we use two extracts from these records. First, we use a 2% random sample of workers to describe the broader labor force and to examine the relationship between firm valuations and the gender gap; we use these data for 2010 to 2019. Second, we use data on all workers who switched firms between 2010 and 2019 to study mobility. In all samples, we focus on full-time workers between the ages of 25 and 50, for consistency with our survey sample. Appendix B.1 provides more information on these extracts.

We obtain additional firm-level data from the IAB, including estimates of firm wage premia. The firm wage premia capture systematic differences in pay across firms, controlling for worker characteristics, and allow us to examine how firm-provided amenities relate to wage-setting (Bellmann et al., 2020).⁶ We use the Betriebs-Historik Panel (BHP) to assign establishments to sectors, and to identify the number of total employees at each establishment. Throughout the paper, we use “firm” to refer to the economic unit, which may span multiple establishments; we construct firm-level measures (for example, wage premia) by taking employment-weighted averages across establishments.

⁵Throughout the text we say a worker is college educated if they have a university degree (BA or more).

⁶While we prefer to use the estimates from Bellmann et al. (2020) for comparability with the literature, a few robustness checks use our own premia estimates from the mobility sample; Appendix B.4 describes the estimation and compares the two series.

Other Data. We also link our survey and administrative data to employer reviews from Kununu, a leading employer review platform in Germany. Like Glassdoor, Kununu allows current and former employees to evaluate their workplaces across multiple dimensions, including compensation, work-life balance, career opportunities, and organizational culture. The primary measures we use from Kununu are the overall firm rating, which summarizes the information that reviewers provide about a firm, and the share of reviewers who would recommend the firm as an employer.⁷ Appendix B.6 describes these data. We also use firm financials from Orbis in the firm-size imputation (Appendix B.5).

2.2 Descriptive Statistics and Sample Selection

We estimate firm-specific valuations using data from the more than 2,000 workers who named at least one firm in either the initial or follow-up survey wave.⁸ Table 1 compares our surveyed workers (Columns 1–2) to two benchmark samples: a 2% random sample of full-time workers ages 25–50 (Columns 3–4) and the universe of full-time job movers between 2010 and 2019 (Columns 5–6). Relative to the full workforce, our sample is younger, more educated, and higher-earning—partly because it overrepresents manufacturing workers. We oversampled workers at firms that participated in a previous survey we fielded on bargaining behavior; many of these firms were in the manufacturing sector (Caldwell, Haegle and Heining, 2026). Compared to a random sample of workers, job movers are somewhat younger and lower-earning. We describe the firms for which we have amenity estimates in Section 4.

Because our survey sample differs somewhat from the German workforce, we take two approaches to ensure our results reflect population-relevant patterns. First, in firm-level analyses we weight firms by employment by default, so that estimates reflect the experience of a typical worker rather than a typical firm among those we observe. Employment weighting changes the estimand across the firms in our sample; it does not by itself correct selection into which firms workers name, which the worker-level reweighting below addresses. We depart from this default when employment weighting is inappropriate—for example, when firm size is itself the dependent variable—and in some specifications we present unweighted results before the employment-weighted ones. We show robustness to alternative weighting schemes throughout. Second, in worker-level analyses that combine our survey with the 2% population sample—such as the gender-gap decomposition—

⁷In unreported results we found this to be highly correlated with the first principal component of the amenity-specific measures.

⁸We cannot estimate valuations using data for workers who declined to provide the names of specific outside firms. For workers who did not provide firms in the initial survey and did not provide three firms in the follow-up, we asked about their reasons for non-response. Appendix Table A1 shows that the majority either had no interest in switching firms or could not identify specific companies. The first group of inframarginal workers is unlikely to be relevant for firms' wage-amenity setting decisions, while the second group would not have provided usable responses in the choice experiment. Our results are robust to using only the valuations collected in the first wave of the survey.

we probe the robustness of our results by reweighting so that our sample matches the demographic composition of the broader workforce. We do so using inverse probability weights estimated from a logit of inclusion in the estimation sample on age, sex, education, labor market experience and its square, and sector.

3 Valuing Firms' Amenities

We estimate money-metric valuations of firm-specific amenities using discrete choice experiments.

3.1 Conceptual Framework

Workers derive utility from wages and non-wage amenities. The utility that worker i receives from working at firm j is

$$u_{ij} = \beta \log w_{ij} + a_j + \epsilon_{ij} \quad (1)$$

where β is the marginal utility of log wages, w_{ij} is the wage worker i earns at firm j , a_j is the value of the firm's non-wage attributes (in utility units), and ϵ_{ij} captures idiosyncratic preferences. Following a large literature, we refer to a_j as “amenities,” though a_j captures the full bundle of non-wage factors that attract workers to a firm—working conditions, career prospects, job security, and culture. This is the standard quantity in models of compensating differentials and job ladders: the total non-wage value a firm provides, not any single non-wage attribute.

A worker prefers firm k to firm j if:

$$\beta \log w_{ij} + a_j + \epsilon_{ij} < \beta \log w_{ik} + a_k + \epsilon_{ik}.$$

Assuming the ϵ_{ij} are independent and identically distributed and follow a Type I extreme value (logit) distribution, the probability a worker prefers firm k over firm j , denoted P_{kj} , is:

$$\begin{aligned} P_{kj} &= \Pr[\epsilon_{ij} - \epsilon_{ik} < \beta(\log w_{ik} - \log w_{ij}) + (a_k - a_j)] \\ &= \mathbb{E}_{j,k} \left[\frac{\exp(\beta \log w_{ik} + a_k)}{\exp(\beta \log w_{ik} + a_k) + \exp(\beta \log w_{ij} + a_j)} \right], \end{aligned}$$

where we take the expectation over the distribution of randomized wage offers. The probability a worker prefers firm k to firm j depends on both the realized gap in log wages ($\log w_{ik} - \log w_{ij}$) and the gap in amenities ($a_k - a_j$). The pairwise share P_{kj} averages this conditional probability over the distribution of randomized wage offers.

Group-Specific Valuations and the Common Index. Equation (1) treats a_j as a single firm-level amenity value. In practice, different workers may value the same firm differently, so we also estimate group-specific valuations a_j^g (e.g., for men and women separately) in Section 5.3. These group-specific valuations are the relevant object for questions about disagreement across groups and for exercises, such as the gender-gap analysis, in which workers’ own valuations matter directly. Heterogeneity across groups does not imply that a common firm component is absent. Rather, amenity valuations may contain both a common component and group-specific or match-specific deviations.

Workers’ valuations may contain a common firm component or be driven primarily by match-specific factors. The common index a_j —the single ranking that best fits the pooled pairwise comparisons of the workers who consider firm j —is the relevant object for questions about firm-level inequality, mobility, and the relationship between amenities and firm wage premia. We provide direct evidence for this common component in Section 4.4, where we show that the common index predicts firm size and employee satisfaction. The headline results on the wage–amenity relationship (Section 5.1) and mobility (Section 5.2) therefore use the common index a_j . Section 4.3 characterizes the extent of this heterogeneity, and Section 5.3 examines its implications for the gender wage gap.

3.2 Recovering Amenity Valuations Through Discrete Choice Experiments

We estimate firm-specific amenity valuations using DCEs embedded in our worker survey. DCEs have become standard in the literature on valuing non-wage job characteristics. In our context, randomizing the wage offer associated with a specific firm lets us identify the value workers place on the non-wage attributes of that firm (rather than the value they place on a particular amenity, such as flexible work). Because DCEs are take-it-or-leave-it offers, we can estimate these valuations without assumptions about offer rates, search frictions, or endogenous wage-setting.

The challenge with implementing DCEs at the firm level is that we must provide workers with a set of firms. The ideal set includes firms a worker might reasonably consider moving to (Mas, 2025). To construct realistic choice sets, we therefore first elicited workers’ consideration sets: we asked each worker to name three firms they would consider applying to if they were to leave their current employer. We then presented these three firms—alongside the worker’s current job—in a personalized DCE (Figure 1 reproduces the elicitation). We paired each outside firm with a randomly assigned wage change, ranging from a 5% pay cut to a 20% raise.⁹ We asked workers to

⁹We did not design the survey to have power to test whether pay cuts matter differently from pay increases. The initial wave randomized raises of 5% to 20% for outside firms; the follow-up wave added a 5% pay-cut option and removed the 20% raise. Because we did not provide pay cuts in the initial wave, robustness checks using data from only this wave provide a joint test that the valuations are stable between waves and that the valuations are not affected

rank the four job options from most to least preferred. The instructions read:

“Suppose you can remain at your current company or switch to any of the companies you provided and immediately receive the raise specified below. Please rank the following job offers from 1 (most likely to take) to 4 (least likely to take).”

Three of the options were the firms the worker provided; one was the worker’s current employer. We exclude this option from the estimation of amenity values entirely: all of our estimates use only the rankings among the three outside firms, and we drop every comparison with the current job.

We recover amenity valuations in two steps.

Step 1: Amenity Values in Utility Units. For each pair of firms j and k , we compute the share of workers who—having listed both—rank k above j . This yields an empirical matrix of pairwise choice probabilities. For firms in the strongly connected set, the fixed point of the associated rank-centrality system—derived in Appendix C.1—recovers a term proportional to $\exp(a_j)$, identifying a_j up to a constant. This is the rank-centrality logic of Sorkin (2018) applied to experimental choices rather than realized job-to-job moves. However, there is one key difference: because we randomized the wage offers, the wage terms cancel in expectation to first order. The eigenvector recovers amenity values alone rather than the combined value of wages and amenities. Appendix C.2 provides proofs and shows that, in our data, this first-order approximation introduces little bias.

Appendix Table A2 confirms that the randomized wages assigned to different firms are uncorrelated with firm characteristics, including actual wage premia, Kununu pay scores, and brand recognition. The randomized wage difference between two firms a worker ranked is also uncorrelated with the difference in the worker’s expected pay at those firms ($p = 0.58$). This confirms that prior beliefs about firm pay do not drive the design.¹⁰ As an additional check, we show that our main results are unchanged when we control for each firm’s mean randomized raise.

We can identify valuations only up to a normalization: we set the valuation of one reference firm to zero. We select this firm so that it has an amenity value near the median; we hold it constant across specifications. The log of the normalized eigenvector then gives a_j in utility units, relative to that firm. Appendix C.2 gives the full derivation.

Step 2: Money-Metric Units. We convert amenity values to money-metric units by dividing by β , the marginal utility of log wages. We estimate β from the structural logit coefficient on the

by the range of wages provided.

¹⁰A remaining concern is that workers’ prior beliefs about firm pay could contaminate the amenity estimates. Two features of the design address this. First, each offer is framed as a raise on the worker’s current pay, not as a level of pay at the outside firm; prior beliefs about firm pay therefore do not enter the choice scenario directly. Second, within-firm wage dispersion is large (Caldwell, Haeghele and Heining, 2026). Raises of 5–20% are within the range workers observe at their own firm, making the offers credible regardless of beliefs about the firm’s average pay.

randomized wage difference (Table A3).

Validity of Approach. The eigenvector approach is commonly used to recover firm values from job-to-job transitions, and is well suited to our setting because we value a large number of firms: linking firms through chains of overlapping comparisons lets us value firms that no single worker ranks directly against one another. It is not, however, standard in the discrete choice literature, which typically recovers valuations from logit models (again dividing by the wage coefficient to obtain money-metric values). For transparency, we validate the eigenvector approach against the logit alternative using the researcher-provided firm module. Because that module includes only a small number of firms, we can estimate valuations using either method.¹¹ Valuations from the two approaches agree: their correlation is 0.95, and a regression of the exact estimates on the eigenvector estimates gives a slope of 1.06, which we cannot reject as equal to one ($p = 0.34$). Appendix C.3 describes this validation.

Limitations. Our approach shares two limitations with the estimation of firm wage premia. First, as in Abowd, Kramarz and Margolis (1999) and Sorkin (2018), we estimate values only for firms in the strongly connected set: each retained firm must be linked to every other through chains of observed comparisons running in both directions. Second, as noted above, we recover amenity values only up to a normalization (Appendix C.4).

3.3 Accounting for Measurement Error

Our approach generates amenity valuations that are noisy, especially for firms that are ranked by only a few workers. We address this in several ways: weighting firm-level regressions by employment, which is positively correlated with the number of survey mentions; presenting estimates that explicitly reweight by the number of mentions; and using a split-sample approach that naturally accounts for measurement error. Our results are robust across these approaches. Measurement error in amenity estimates would, if anything, push us toward over-interpreting common preferences as reflecting idiosyncratic worker–firm match effects, since noise attenuates the correlation across split samples.

These approaches allow us to deal with three potential sources of measurement error. First, classical noise: a worker’s choice between two firms is a noisy draw. Second, worker-level noise: some workers may rank firms less carefully than others. Third, sparsity: some firms are ranked by only a few workers, so their valuations are estimated less precisely. The split-sample approach addresses the first two issues: because we split at the respondent level, each worker appears in

¹¹Because workers ranked multiple offers, we fit a rank-ordered logit, which naturally accommodates worker-specific heterogeneity.

only one half of the data and any error specific to a worker is independent across the two halves. Employment weighting limits the influence of sparsely mentioned firms, since employment is correlated with the number of mentions. We also show that our between-group correlation and mover classification results hold when we restrict to firms mentioned a minimum number of times (Appendix Table D3 and Appendix Figure D12).

Several of our exercises use estimated amenity values as data in subsequent regressions. In these cases, standard errors do not account for first-stage estimation uncertainty in the amenity values. We account for this in different ways. The split-sample variance decomposition (Section 5.1) is designed to remove estimation noise: because the respondent-level split makes half-sample errors independent, the covariance between the two half-sample estimates is a consistent estimator of the signal variance. The IV binscatters in Section 5.1 similarly use one split-half as an instrument for the other, so that first-stage noise does not attenuate the slope.

When exercises treat estimated amenities as data, estimation noise will tend to attenuate the sharpness of the patterns we document. In exercises where amenity values enter as regressors—the size and quit-rate regressions (Section 4.4) and the Kununu correlations (Appendix D.3)—first-stage noise attenuates coefficients toward zero; these exercises therefore provide conservative estimates of the true relationships. In exercises where the amenity values enter as the dependent variable—the gender-gap regressions (Section 5.3)—measurement error inflates standard errors but does not attenuate the point estimates.

3.4 Interpretation of Estimates

Our design identifies the ex-ante value of a firm’s non-wage bundle, for workers on the margin of applying. By “ex-ante” we mean the value a worker assigns to joining a firm before she experiences the match; it can differ from an ex-post assessment, because workers accumulate firm-specific capital, form workplace social ties, and may grow attached to their coworkers and managers over time (Caldwell, Haegele and Heining, 2025). By “marginal” we mean a worker on the margin of applying—a potential hire who treats the firm as a realistic option—rather than a random worker in the population. This ex-ante, marginal-applicant value is the object that governs firm wage and amenity setting in standard models: firms compete for workers who are choosing where to work, and differentials are priced at that margin rather than by the preferences of the workforce at large (Rosen, 1986).¹²

¹²These models are typically static and do not differentiate between the marginal hire and the marginal separator. The fact that our valuations predict ex-post outcomes—firm size and employee satisfaction (Section 4.4)—suggests that ex-ante valuations predict ex-post valuations.

Ex-Ante Valuations. Because the experiment varies only the wage, our non-wage valuation of firm j , a_j , captures workers’ valuations of everything else a firm offers—tangible benefits like flexible schedules, career development, and job security, and harder-to-quantify attributes like culture and managerial quality. We view this as the ideal quantity: it captures all that workers value beyond the wage. We cannot, however, use these values to identify which specific attributes workers value at a firm. Separating the components would require us to cross-randomize wages and amenities at the firm level.¹³ We view our results as complementary to studies that have examined how much workers value specific amenities, such as Maestas et al. (2023); our empirical object differs from the object in those studies.

Because we vary only the initial wage offer, in theory a firm’s valuation could also reflect dynamic considerations—expectations about promotions, wage growth, or firm-specific human capital. We asked a random subset of workers to re-rank the same offers under the assumption that career growth—future raises and promotions—is identical across all options. We refer to the resulting valuations as “no-growth” where relevant.

The Marginal Workers. A natural concern is that workers would name firms that were unrealistic options for them. They do not: the share of workers from a given origin firm who name a destination firm rises with the share who have actually transitioned there (Appendix Figure A1), and the named firms resemble, in their sectoral composition, the firms that workers from the same employer have historically moved to (Appendix D.1).¹⁴

Familiarity also matters for two concerns specific to discrete choice experiments. The first is hypothetical bias: choices over hypothetical options may not match real behavior. Hypothetical bias is most severe when respondents evaluate unfamiliar or abstract scenarios; our workers rank real firms they named themselves, at raises specified relative to their own current pay. A second, related concern is that a higher wage offer could signal that the offer is from a better firm. Because workers evaluate firms they already know, the randomized raise changes the wage at a known firm, not what the worker believes the firm is like. For German workers, raises of this size are within the range of what can be achieved through bargaining (Caldwell, Haegele and Heining, 2026).

Comparison with Firm Wage Premia. In some specifications we relate our non-wage valuations to firm wage premia ψ_j —the firm fixed effects from two-way (AKM) wage regressions

¹³For example, suppose a firm’s dog-friendliness happens to correlate perfectly with other workplace attributes that workers do value, such as flexible hours. A regression of our valuations on dog-friendliness would find a strong positive coefficient even if workers place no value on the policy itself—or even view it as a disamenity. More generally, a positive correlation between a characteristic and our valuations tells us only that the characteristic tends to co-occur with valued workplace attributes, not that the characteristic itself is valued.

¹⁴Because we use job transitions from 2010–2019 and we started fielding our survey in 2022, this relationship is not mechanically driven by workers reporting their future moves.

(Bellmann et al., 2020). These estimates are identified from different populations: amenity values from workers who would consider applying to a firm, and wage premia from workers who move between firms. This does not pose a problem for our analysis because a broad set of papers has found that wage premia are fairly homogeneous across worker groups (Bellmann et al., 2020; Card et al., 2018; Kline, 2024). Because premia carry over across groups, the wage premium a marginal applicant would earn at firm j is the same premium identified from movers; both objects in our comparison apply to the same worker. Applying mover-identified premia to the broader workforce is standard in this literature: decompositions of wage inequality and of the gender gap assign each firm’s premium to all of its workers, movers and stayers alike.

Discussion of Alternative Approaches. The primary advantage of our approach is that we can estimate non-wage valuations using exogenous variation in wages (introduced in the survey), without making assumptions about firm behavior. One challenge with using observed job-to-job transitions is that they reflect both preferences and offer rates: that a worker does not move from firm j to firm k need not mean she prefers j , since she may not have received an offer from k . While it is possible to generate proxies for the firm-specific arrival rate (using hires of non-employed workers), these proxies typically assume that offer probabilities do not condition on employment status or current firm identity. If firms target different job seekers, the proxy is invalid (Kroft, Lange and Notowidigdo, 2013; Caldwell, Haegele and Heining, 2025).¹⁵ Revealed-preference estimates also understate willingness to pay when high-amenity firms have queues of applicants.

Discrete choice experiments are the standard method for inferring valuations from stated preferences. An alternative would have been to directly ask workers how much they would forgo to leave their firm, or require to join one. In theory, such an approach would yield each worker’s exact willingness to pay rather than the bound revealed by a choice at a given wage difference. Because stated willingness-to-pay measures are unreliable, sensitive to framing, reference points, and hypothetical bias (Johnston et al., 2017; see also Mas, 2025), and can diverge from experimental estimates by an order of magnitude (Rodemeier, 2026), this is not the standard approach in the literature. Provided that workers evaluate familiar job bundles, discrete choice experiments produce values comparable to those from revealed-preference approaches (see, e.g., Hainmueller, Hangartner and Yamamoto, 2015; Maestas et al., 2023; Mas and Pallais, 2017; Wiswall and Zafar, 2018).¹⁶

¹⁵Kroft, Lange and Notowidigdo (2013) use an audit experiment to document that employed job seekers receive callbacks at a different rate than unemployed job seekers. Caldwell, Haegele and Heining (2025) show that, among employed workers with the same occupation, education, and experience, the productivity of search is increasing in the firm pay premium of a worker’s current firm, even after controlling for the worker’s own wage premium, a common proxy for unobserved skill.

¹⁶In a recent handbook chapter, Mas (2025) concludes that survey-based discrete choice experiments are a viable way to elicit preferences over workplace characteristics. Firms use them to design benefit plans and schedules in

We fielded the survey to a broad sample of workers rather than to recent job-to-job movers. Conditioning on movers can generate a spurious negative correlation between wages and amenities even when the population correlation is positive (Bonhomme and Jolivet, 2009; Lavetti, 2023; Lavetti and Schmutte, 2023), which would lead us to conclude that amenities dampen between-firm inequality. We show evidence of this bias in Section 5.2.

4 Firm Wages, Amenities, and Valuations

Non-wage values are highly dispersed across the firms workers name. Despite heterogeneity in valuations, a single-index model still has predictive power.

4.1 Characteristics of Valued Firms

Our estimation strategy rests on the assumption that workers perceive a trade-off between wages and non-wage characteristics when choosing among firms. Because we randomized raises across outside firms, any systematic preference for one firm over another reveals a difference in amenity value. If, on average, workers prefer a lower-paying firm, this implies that they perceive it as offering superior non-wage attributes.

The surveyed workers named 3,899 unique firms, described in Table 2. Like the workers in our sample, these firms span all parts of the wage distribution and all major sectors. Firms in the strongly connected set have similar wage premia and sectoral composition to the full set (Appendix D.2). This holds regardless of the weighting scheme (Appendix Table D2). Of the 3,899 firms workers named, 949 are in the strongly connected set.¹⁷ These firms account for 70.0% of total employment among the firms workers named. Because workers can only sort across firms they know, the consideration set is the relevant object for studying wage–amenity trade-offs. Firms outside this set are typically mentioned by only one worker, which provides too few overlapping comparisons to place them on a common scale. Firms in the strongly connected set are mentioned a median of 3 times (mean 12.9; interquartile range 1–7), providing sufficient pairwise comparisons for estimation.

Because we ask workers to choose between offers from firms they said they would apply to, the firms they compare are likely more similar on non-wage attributes than randomly chosen firms in their labor market. Yet even among this set, pay is not the only determinant of workers’ choices. On average, 35% of workers selected the lower-wage firm. This matches observational evidence that many job-to-job transitions involve wage cuts, a pattern we confirm in Section 5.2: workers

real-stakes environments (DeBellis, 2018).

¹⁷This limitation is similar to that in the AKM literature, where firm wage premia can only be estimated within a connected set of firms.

value a bundle of job characteristics broader than pay. Variation in amenity valuations is large even among the firms workers plan to apply to.

Workers' choices do respond to relative pay differentials. Table A3 reports the coefficient on the experimental wage difference from an OLS regression. The pooled estimate is 2.86 (SE 0.11): a one log-point increase in the wage offered by firm j relative to firm k raises the probability of choosing firm j over firm k by 2.86 percentage points. Inattentive or random responding would produce no relationship between the randomized raise and choices. We cannot reject equality of this coefficient across men and women ($p = 0.88$) or across workers with and without a college degree ($p = 0.48$), consistent with a common response to wages across demographic groups. The structural logit coefficient on the wage difference—the preference parameter β that enters the utility function—is 12.08 (SE 0.50). As with the linear probability model, we cannot reject equality of this coefficient across men and women ($p = 0.89$) or across workers with and without a college degree ($p = 0.50$). Because the eigenvector recovers amenity values in utility units (Section 3.2), converting to log-wage equivalents requires dividing by $\hat{\beta}$.¹⁸

4.2 Distribution of Valuations

Amenity values vary widely across firms. Figure 2 plots this distribution, weighted by the size of the corresponding firm. Recall that we normalize the valuation of a reference firm—chosen to be near the median—to zero. The horizontal axis is in log-wage-equivalent units, obtained by dividing the raw utility-unit valuations by $\hat{\beta}$. The figure shows that amenity values are highly dispersed: the spread between the highest- and lowest-valued firms is about a third of a worker's pay. Our estimates do not simply reflect noise. When we weight by firm size, the correlation between estimates from two random halves of the data is 0.66.

Static and Dynamic Considerations. In many models of job choice, workers trade off current wages for expected future wage growth. If firms offering high non-wage values also provide steeper wage-growth profiles, part of what we measure may reflect dynamic wage considerations. To examine whether dynamic considerations affect our estimates, we use an additional discrete choice experiment in which workers re-ranked firms under the assumption that career growth—future raises and promotions—is identical across all options (one of which is remaining at the incumbent firm).¹⁹ Figure 3 shows that the two sets of estimates are highly correlated and that the dispersion

¹⁸As a calibration check, the same discrete choice experiments yield separation elasticities of 2–4 when used to estimate the responsiveness of job-to-job transitions to wage differentials (Caldwell, Haegele and Heining, 2025), consistent with the range reported in recent meta-analyses (Sokolova and Sorensen, 2021; Caldwell, Dube and Naidu, 2026).

¹⁹For our analysis, what matters is that workers perceive dynamic considerations as constant across the outside firms. Anchoring on the incumbent firm makes this cognitively easy: workers already have information and views

of amenity values net of wage growth closely matches that of the baseline.

Correlates of Firm-Specific Amenities. Because we did not cross-randomize provision of specific amenities, we cannot “price” individual amenities. As noted in Section 3.4, without exogenous variation in the provision of individual amenities, regressions of valuations on characteristics conflate the direct value of those characteristics with the value of correlated but potentially omitted workplace attributes. Appendix D.3 shows that our valuations are positively correlated with a range of firm characteristics and policies. This analysis is suggestive both because the Kununu ratings are self-reported and because the correlations do not identify causal effects of specific amenities. Nonetheless, the positive correlations suggest that our valuations capture meaningful differences in workplace quality.

4.3 Between-Group Heterogeneity in Valuations

Our valuations come from the preferences of workers who are marginal to a firm. As noted in Section 3.2, however, our model assumes a uniform ranking of firms (apart from idiosyncratic valuations). To probe the realism of this assumption, we re-estimate firm-specific amenity values separately for women, men, college-educated workers, and workers without a college degree. Table 3 reports pairwise correlations between these subgroup-specific amenity valuations.

Between-group heterogeneity in amenity valuations is large. The Pearson correlation between amenity valuations for women and men is 0.266, indicating that the types of firms women and men view as high-amenity differ markedly. This low correlation arises even though the estimates come from workers who are marginal to the same firms. We find similar divergence by education: the correlation between valuations of college-educated and non-college workers is 0.231. Spearman rank correlations are higher (0.454 and 0.359, respectively): groups agree more on the ordering of firms than on the magnitude of money-metric valuations.

This divergence contrasts with the literature on firm wage premia. In a broad set of countries including Germany and the United States, firm wage premia for different demographic groups (including men and women and college and non-college workers) are nearly identical (Card, Cardoso and Kline, 2016; Sorkin, 2017; Palladino et al., 2025). Similarly, workers’ perceived wage premia do not differ substantially by gender or education (Caldwell, Haegele and Heining, 2025). Workers agree much less on the ranking of firms by non-wage attributes.

Robustness. A natural concern is that measurement error could attenuate between-group correlations, making groups appear to disagree more than they actually do. If amenity estimates are noisy, about their experience there.

the noise in male- and female-specific estimates may be independent even when the true values are similar, mechanically reducing the correlation. However, if noise in rarely mentioned firms drives the low correlations, then upweighting well-measured firms should increase them. Appendix D.4 shows that weighting by the number of times a firm appears in the data yields similarly low correlations. Restricting the sample to firms mentioned at least five or ten times—which substantially reduces noise—also has little effect (Appendix Table D3). Paired t -tests provide the same answer: we reject equality of male and female valuations ($p = 0.032$) but cannot reject equality by education ($p = 0.687$).

To address measurement error more directly, and to probe the robustness of this finding, we turn to a separate module of the survey in which we selected a set of large firms and asked all respondents to evaluate them, regardless of whether they had named these firms as application targets. Because many workers evaluated the same researcher-provided firms, the resulting amenity estimates are more precise than those from the worker-provided module. Using all valuations from this module—normalized relative to a common base firm (a large manufacturing firm)—we reject equality of male and female amenity levels ($p < 0.001$, Appendix Table A6, Panel A). Panel B restricts to firms workers indicated they would consider applying to. With the smaller worker sample, the joint χ^2 test no longer rejects equality ($p = 0.758$), but between-group correlations remain low.

4.4 Predictive Power of the Common Index

The between-group heterogeneity in Section 4.3 poses a challenge for models that represent amenities as a single firm-level index. Many structural models of the labor market assume such an index (Lamadon, Mogstad and Setzler, 2022; Berger, Herkenhoff and Mongey, 2022; Volpe, 2026; Mas, 2025), and for many policy questions—such as whether to mandate amenity disclosure or how to evaluate firm-level regulations—a common index is the relevant object. Our inequality question also requires a single index: between-firm inequality is about common firm-level components, not match quality.

This challenge parallels a central question in the wage-setting literature: whether firm wage premia reflect a common component or are dominated by worker–firm match effects (Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013; Bonhomme, Lamadon and Manresa, 2019; Song et al., 2019). In the wage context, various approaches have established that a meaningful common firm component exists (Kline, Saggio and Sølvssten, 2020). The analogous question for amenities—do firms offer a common non-wage value, or are amenity valuations primarily match-specific?—has received far less attention.

Several forces could weaken the link between our valuations and observed labor market out-

comes (and bias us against finding a relationship). First, measurement error in amenity estimates—particularly for firms mentioned by few workers—could attenuate any relationship. Second, our valuations are elicited from workers considering hypothetical job changes; if workers’ ex-ante expectations of a firm’s amenities diverge substantially from their ex-post experience, valuations may not predict realized outcomes such as retention or firm size. Third, if idiosyncratic worker–firm match effects dominate the common component of amenity value, firm-level averages of our estimates could be uninformative about firms’ realized non-wage amenities. Our split-sample correlation for amenity valuations (approximately 0.66) bounds the first and third concerns directly. We find that the common component accounts for roughly two-thirds of the cross-firm variance in amenity estimates.²⁰

Table 4 presents two tests of whether the common index predicts realized firm outcomes. For each test we use three alternative wage controls: the observed wage premium (Panel A), log median pay (Panel B), and log mean pay (Panel C). These controls allow us to test whether, conditional on the firm’s wage policy, a firm that offers higher non-wage value attracts more workers or is more favorably reviewed.

The first row in each panel shows that, conditional on the firm’s wage policy, firms that offer higher non-wage value are larger ($p < 0.01$). This supports a key assumption in the structural labor literature. In many models, amenity values are derived from the assumption that firms that are large relative to their wage must be providing non-wage value that attracts and retains workers (Lamadon, Mogstad and Setzler, 2022; Volpe, 2026).²¹ Our estimates validate this logic independently: firm-specific amenity values recovered from a discrete choice experiment predict administrative employment. The second row in each panel provides more evidence in support of a common index: conditional on the wage, firms with higher non-wage valuations are recommended by a larger share of employees on Kununu. Finally, Appendix D.5 shows that quit rates are lower at higher-amenity firms in every baseline specification, conditional on wages, though the estimates are imprecise.

The split-sample correlation is direct evidence of a common amenity component. If valuations were purely idiosyncratic, the two half-samples would produce uncorrelated firm rankings, and there would be no split-sample first stage. Instead, two independent halves of respondents produce rankings that are highly correlated ($\rho = 0.66$), and the common component predicts firm size and employee satisfaction conditional on pay. The researcher-provided module allows us to show the

²⁰Match effects are idiosyncratic to each worker–firm pair and therefore independent across the two random half-samples; only the common component a_j generates covariance between the split-sample estimates. Match effects and measurement error together account for the remaining third. Because experimental noise from the discrete choice design also contributes to the residual, the match-effect share is strictly less than one-third.

²¹A common critique of such approaches is that firms may also vary in recruiting standards or intensity. A low-wage firm may be large simply because it is willing to hire a large share of workers, not because it provides a good working environment.

same thing using a dense network of firms. When many workers rank offers drawn from the same fixed list of 30 firms, we find that a common firm-level component exists and is quantitatively large.

Together, these pieces of evidence rule out a pure match-effects interpretation of our estimates. If amenity values were dominated by idiosyncratic worker–firm match effects, they would not predict firm size conditional on wages. They would also not predict aggregate recommendations from a third-party platform. This evidence also implies that ex-ante valuations are informative about realized conditions. If workers’ ex-ante expectations of a firm’s amenities diverged substantially from their ex-post experience, we would expect our valuations to predict (unobserved) acceptance rates, but not realized firm size or employee satisfaction. These findings also rule out the idea that our estimates reflect only the preferences of a small subset of workers.

These patterns are also inconsistent with the idea that workers are inattentive in the survey or are uninformed about the firms they provide. Inattentiveness would produce noise in the rankings, which would not be correlated across samples or with realized outcomes. Misperceptions—if shared across workers—could produce a common component, but would not predict realized outcomes. Ruling out these considerations is important because our estimates are based on workers’ stated preferences, and many of these workers are not currently searching for (or gathering information about) new jobs.

The existence of a common amenity component motivates our subsequent analysis of inequality: because amenities are not purely idiosyncratic, it is meaningful to ask whether they offset or compound between-firm inequality.

5 Amenities and Inequality

Amenity values do not decline with firm wage premia. Total compensation is therefore more dispersed across firms than pay alone—by roughly 20 percent in variance terms. Mobility-based methods reach the opposite conclusion because movers are a selected sample. Women sort toward high-amenity firms.

5.1 Firm Wage Premia and Inequality in Utility

Whether amenities offset or widen between-firm inequality is a question about how pay and amenity value covary across the firms where workers actually work. It is distinct from whether workers are willing to pay for non-wage characteristics. Even when workers value amenities highly—trading pay for them at a given firm—amenities can widen between-firm inequality if firms vary in the overall value they provide to workers (Mas, 2025).

We first ask how amenity values covary with firm wage premia. Because the literature on pay premia focuses on the common firm component ψ_j , we measure amenities at the same level, using our baseline estimates. We then calculate the contribution of amenities to the between-firm variance of total compensation, which includes both pay and amenity value. Full compensating differentials imply a slope of -1 in log-wage-equivalent units. A slope of zero or above instead leaves total firm value at least as dispersed as pay. Throughout, we take the allocation of workers to firms as given.

Relationship with Firm Wage Premia. Figure 5 plots firm-specific amenity valuations against firm wage premia, using the split-sample approach described in Section 3.2 to account for measurement error. Panel A shows that, with each firm weighted equally, there is a positive relationship: firms with higher pay premia offer better amenities. Panel B shows that this relationship shrinks to zero when we weight by firm size, which captures the relationship experienced by the average worker. The same non-negative relationship is visible across broad sectors (Figure 6).

That weighting by firm size pushes the correlation toward zero rules out one obvious concern: that measurement error is attenuating a strong positive true correlation. There is measurement error in both our amenity valuations and the firm wage premia, and this error is larger for smaller firms (which are mentioned less frequently). If the positive unweighted correlation were genuine but attenuated by measurement error, weighting by firm size (or by mentions) should strengthen it, since larger and more frequently mentioned firms have less noisy estimates. Instead, weighting pushes the correlation toward zero—the opposite of what measurement error correction would produce.²² We obtain similar correlations across the size distribution: across firm-size terciles the slope is 1.560, 1.700, and 0.896 from the smallest to the largest firms. This pattern addresses a residual concern: that the non-negative slope merely reflects the fact that more productive firms offer both higher pay and higher amenities.²³

Because workers name only firms they view as realistic outside options, our results are identified among firms that compete for the same workers rather than across the full universe of firms. We view this as the relevant object for firm wage and amenity setting. In standard models, firms choose wage–amenity bundles to attract workers on the margin of accepting or rejecting an offer, so the relevant competitors are firms workers plausibly consider, not firms outside their consideration set. More broadly, some form of selection is unavoidable in any study of wage–amenity trade-offs: realized mobility studies condition on accepted offers, and cross-sectional studies condition on observed matches. Our design makes this selection explicit: we study the firms a worker would actually consider, which is the margin firms compete on.

²²There is also measurement error in wage premia, though these are estimated based on the population of German workers. Measurement error in wage premia would inflate the standard errors, but would not bias the slope.

²³The size regressions from the previous section show a related point.

However, we see similar results when we use valuations of firms we assigned to workers. In the researcher-provided module, workers did not choose which firms to evaluate: each worker ranked three firms drawn at random from a fixed list of 30 large employers, so selection into naming cannot operate. The choice tasks recover amenity values from randomized wage offers rather than from the eigenvector (Appendix C.3). There too the amenity–pay correlation is positive when firms are weighted equally and falls toward zero when weighted by imputed firm size (Appendix Figure D8). No design and no weighting produces a significant negative slope, let alone one near -1 (which is what a pure compensating-differentials story would require).

The robustness check using the researcher-provided firms also helps us address potential collider bias. If workers direct their search on the basis of the total value of a firm, they may name a firm due to either high wages or high amenities. Conditioning on being named would, if anything, induce a more negative correlation between wages and amenities. While a non-negative estimate is conservative evidence against the idea that amenities offset wage inequality, the fact that we also see this with the researcher-provided firms is comforting. Further, Appendix Figure D7 shows that firms named most frequently—those most selected on desirability—do not show a stronger relationship in either direction. Together, these results suggest that selection into naming is not driving our results.

The results are robust to a variety of alternative specifications and samples and do not reflect a perceived positive correlation between wage levels and wage growth. We find similar results when we use the alternative amenity estimates that hold career growth expectations constant (Appendix Figure D10), and when we restrict to Wave 1 of the survey, in which randomized raises ranged from 5% to 20% with no pay cuts offered (Appendix Figure D11). The latter rules out the possibility that the introduction of pay cuts in the follow-up wave drives our findings. Because raises were randomized conditional on listing position, adding a control for each firm’s mean randomized raise leaves the slope essentially unchanged (1.498 unweighted, 0.142 employment-weighted). The employment-weighted orthogonality also holds when we decompose by demographic group: $\text{Cov}(a_j^M, \psi_j)$ and $\text{Cov}(a_j^F, \psi_j)$ are each approximately zero, ruling out the possibility that this is an artifact of aggregating groups with offsetting correlations.

Workers themselves do not believe that higher pay is offset by worse non-pay characteristics. In the follow-up survey we asked workers directly whether a firm paying well above market offers better or worse amenities than one paying slightly above market. Only 29.2% of workers said worse; many (42.8%) said the amenities would be the same (Appendix D.7).

Variance Decomposition. These results suggest that amenities do not offset—and may indeed augment—between-firm inequality. We next calculate the contribution of amenities to between-firm inequality directly. We bias-correct our estimates of variance using a split-sample approach:

the covariance between estimates from two independent random halves of survey respondents provides a consistent estimate of the true variance, analogous to leave-out estimators for AKM firm effects (Kline, Saggio and Sølvesten, 2020).²⁴ When we convert variances to money-metric (log-wage-equivalent) units by dividing by $\hat{\beta}^2$, where $\hat{\beta} = 12.08$ is the structural logit estimate from Table A3, the signal variance of amenity valuations is about one-third that of firm wage premia under our baseline employment weighting.

The variance of total compensation, $\text{Var}(\psi_j + a_j/\hat{\beta})$, is 1.22 times the variance of pay premia when we weight firms equally (95% confidence interval [1.03, 1.41], bootstrapped over firms) and 1.20 times when we weight by employment ([0.77, 1.64]). Correcting the wage-premium variance for measurement error raises these ratios to 1.26 and 1.24. Appendix Table A4 reports the components. Amenities widen between-firm inequality by roughly 20 percent in variance terms. The employment-weighted interval is wide and includes one; for the workforce of large firms, the calculation rules out a full offset but does not establish widening.

It is worth noting that the variance of amenity valuations within consideration sets is likely smaller than across all firms in the economy. Workers will likely exclude firms where obvious disamenities—such as long commuting distances—dominate. Compressed variance is the expected result when studying firms that compete for the same workers. While this selection does not lead to a mechanical positive correlation between wages and amenities, it does mean our statements about the variance of non-wage values may be conservative.

Relationship with Other Work. Our finding of a non-negative (and potentially positive) relationship between wages and amenities contrasts with some studies based on movers (Humlum, Rasmussen and Rose, 2025). Two features of our design plausibly contribute to this divergence. Bonhomme and Jolivet (2009) and Lavetti and Schmutte (2023) demonstrate that analyses focusing on job-to-job movers can generate a spurious negative correlation even when the true population correlation is zero or positive.²⁵ In these models, workers who move to lower-wage firms disproportionately gain in amenities, inducing a negative correlation in the movers sample.²⁶ The belief

²⁴The reliability ratio for firm wage premia is 0.85, the split-sample correlation between our own AKM firm-premium estimates from two independent random halves of the data (Appendix B.4). Because our main analysis instead uses the wage premia of Bellmann et al. (2020), which are estimated on a larger sample and are therefore less noisy, 0.85 is a lower bound on their reliability. This makes our comparison conservative: a lower wage-premium reliability shrinks the estimated signal variance of wage premia, so if anything it overstates the amenity-to-wage variance ratio we report below. Our analogous split-sample correlation for amenity valuations is approximately 0.66, implying a signal-to-total variance ratio of roughly two-thirds.

²⁵We confirm this bias in simulations reported in Appendix C.6.

²⁶A related selection issue arises with reservation-wage designs that elicit valuations from movers: a worker who accepts a lower-paying job must value the destination’s amenities enough to justify the move. The resulting sample can overrepresent transitions in which wages and amenities move in opposite directions. Humlum, Rasmussen and Rose (2025) use survey data on a large number of job movers in Denmark to study how movers trade off pay and non-pay characteristics.

data show the same selection: workers who moved to lower-premium firms are the most likely to believe that high pay comes with worse amenities (Panel B of Appendix Figure A15).

First, because our amenity estimates come from a design fielded to a broad sample of workers—not from the transitions of job changers—our estimates are not subject to this selection by construction. Second, our focus on outside firms means that all of our valuations are “ex-ante” valuations. Previous work has shown that ex-ante and ex-post valuations can diverge substantially (Caldwell, Haegele and Heining, 2025; Humlum, Rasmussen and Rose, 2025). That ex-ante valuations predict firm size conditional on the wage indicates that these measures capture firm-level conditions relevant for labor market equilibrium.

Our finding that amenities do not decline with firm wage premia is consistent with previous work showing that higher-wage workers report better working conditions (Maestas et al., 2023). That worker-level finding reflects sorting: higher-skilled workers earn more and work at firms with better amenities. Our estimates isolate the firm-level relationship and show that, across firms, higher wage premia do not come at the cost of worse amenities.

5.2 Amenities and Mobility

Because our estimates of amenity values are not derived from worker flows, we can ask how job-to-job transitions move workers up or down the firm wage and amenity ladders. We link administrative records on job-to-job moves in Germany to our survey-based amenity valuations; the sample covers moves between firms with estimated amenity values, and we drop moves that occur within firms (establishment switches leave firm-level premia and amenity values mechanically unchanged). We classify each move by whether the worker’s wage increased, decreased, or remained similar (within $\pm 2\%$), and separately by whether the worker moved to a firm with higher, lower, or similar amenity value.

Transition Matrices. Table 5 presents transition matrices that categorize job movers based on changes in both wages and estimated firm amenity values. Panels A and B use our baseline amenity valuations, while Panels C and D use valuations estimated under the assumption that career growth is identical across all options. Within each pair, the first panel examines changes in workers’ own wages and the second considers changes in the firm wage premium. Within each panel, the cells in a row are conditional on the wage-change category and sum to 100%; the final column reports the unconditional share of all movers in that category.

A substantial share of workers who take wage cuts move to firms with similar or better amenities, rather than experiencing losses in both dimensions. Focusing on Panel A, which tracks changes in a worker’s own wage, 18.5% of cross-firm movers take a wage cut of more than

2%. Among these downward wage moves, 28.8% involve a simultaneous decline in amenities, 22.4% leave amenities roughly unchanged, and 48.8% move to firms with at least a 2% increase in amenities. Most downward wage moves leave workers with similar or better non-wage attributes. Workers who take cuts in the firm wage premium (Panel B) show the same pattern: amenity gains outnumber amenity losses.

Workers are more likely to move up the own-wage ladder than up the amenity ladder, though the gap is modest. About 60.3% of cross-firm movers move to firms that offer them higher own wages (>2% increase), 41.1% to firms with higher firm wage premia (>2% increase), and 46.1% to firms with substantially higher amenities (>2% increase). Panels C and D repeat the analysis using amenity estimates that hold wage-growth expectations constant. The pattern is similar: 58.7% of movers experience an own-wage increase, and 48.0% move to firms with higher amenities. Wage gains accompany the largest share of job-to-job moves, but nearly half of movers—46.1–48.0%—move to firms with higher amenity values, consistent with workers also weighing non-wage conditions when they switch employers.

Of course, this classification uses firm-level amenity values: each move is judged by the change in the common component a_j between the origin and destination firms. The classification misses the idiosyncratic match component—a worker may move to a firm with a lower common amenity value because her own fit there is good. As a result, this classification will, if anything, understate the share of moves that improve the mover’s non-wage position.

Although amenity values do not decline with wage premia in the cross-section of firms (Figure 5), a negative relationship emerges among movers—a pattern consistent with selection. To quantify the magnitude of this selection-induced slope, we regress changes in sex-specific amenity valuations on changes in the firm wage premium, allowing the slope to vary by gender. OLS estimates yield a negative slope: the larger a mover’s wage-premium gain, the smaller the amenity gain (Appendix Figure D13). Because the amenity change enters as the dependent variable, measurement error in it inflates the standard errors rather than biasing the slope. The negative slope reflects selection: workers who move to firms with higher amenities tend to accept lower wages and lower firm premia.

Robustness. These patterns are robust to several alternative specifications. First, we repeat the transition-matrix analysis using our alternative measure of amenities that equalizes wage-growth expectations across firms; the results remain broadly similar, suggesting that the observed trade-offs are not merely a reflection of forward-looking wage considerations. Second, we restrict the sample to likely voluntary movers—workers who transition directly from one employer to another without an intervening unemployment spell—and find similar patterns (Appendix Table D4). Third, in some specifications we require that both the origin and destination firms were mentioned

at least five (or ten) times, ensuring that the amenity estimates used in the classification are precisely estimated; the results are again robust (Appendix Figure D12). We also see similar results when we repeat the analysis using a 10% threshold to define large changes (Appendix Table D5).

Implications for Estimating the Wage–Amenity Correlation. These transition patterns illuminate why analyses that condition on movers may find a negative wage–amenity correlation even when the population-level correlation is approximately zero. Among downward wage movers in our data, 71% move to firms with similar or higher firm-level amenity valuations—they accept lower wages while moving to firms that offer, on average, better non-wage attributes. Among upward wage movers, amenity gains and losses are more evenly split. In our data, conditioning on the sample of all job movers generates a negative slope between wage changes and amenity changes, even though the underlying firm-level relationship is, if anything, positive—near zero only when firms are employment-weighted. This is the selection mechanism formalized by Bonhomme and Jolivet (2009) and emphasized in Lavetti and Schmutte (2023).

5.3 Amenities and the Gender Wage Gap

Many papers document that, across countries and institutional settings, women sort into firms that pay all workers less (Card, Cardoso and Kline, 2016; Sorkin, 2017; Palladino et al., 2025; Datta Gupta et al., 2025). A leading explanation is that women place greater weight on non-wage amenities when choosing where to work.

Prior work confirms that women value specific amenities—such as flexible scheduling and reduced hours—more than men (Wiswall and Zafar, 2018; Mas and Pallais, 2017). Our own survey data are consistent with this: women are significantly more likely than men to rate options for home office, reduced hours, and predictable schedules as “very” or “extremely” important (Appendix Figure A2). We conclude our analysis by using our firm-level valuations, linked to samples described in Section 2, to measure how accounting for non-wage characteristics changes the gender gap in job quality.

Trade-offs Among Movers. We first examine the experience of job movers and find that men and women make different trades when switching firms. Figure 7 and Panel B of Table A5 classify job movers by the change in their sex-specific amenity valuation and the change in the firm wage premium. Among men, amenity increases dominate amenity decreases regardless of the direction of the wage change: men who move to higher-premium firms tend to gain amenities, as do men who move to lower-premium firms. Among women, the pattern reverses with wage direction—women who move to lower-premium firms disproportionately gain amenities, while women who

move to higher-premium firms disproportionately lose them—so the wage–amenity trade-off is correspondingly steeper for women. In log-wage-equivalent units, the fitted slope is -0.164 for women versus -0.049 for men (Appendix Figure D13), though the smaller number of female movers leaves the difference imprecisely estimated.

Gender Gap in Firm-Level Outcomes. Amenities do not augment (and may offset) the gender gap in pay. We estimate the gender gap in firm-level outcomes by regressing each outcome on a female indicator and progressively richer controls:

$$y_{ij} = \theta_0 + \theta_1 \text{Female}_i + X_i' \gamma + \eta_{ij}, \quad (2)$$

where y_{ij} is the outcome (firm wage premium ψ_j or amenity valuation in log-wage-equivalent units, $a_j/\hat{\beta}$; Columns 4–6 of Panel B use sex-specific valuations $a_j^{g(i)}/\hat{\beta}$) for worker i at firm j , X_i includes Mincer controls and fixed effects depending on the specification, and standard errors are clustered at the firm level.

Table 6 presents the main results. Panel A documents the gap in pay: the raw gender gap in the firm wage premium is -0.014 (Column 1); it shrinks toward zero and loses statistical significance with Mincer controls (Column 2, -0.010) and occupation fixed effects (Column 3, -0.003). Women work at systematically lower-paying firms.

Under a common ranking of firms, amenities neither offset nor widen the wage gap. Columns 1–3 of Panel B use our common amenity index. This assumes that men and women who are marginal to a firm value that firm equally. These columns show that the gender gap is statistically indistinguishable from zero across specifications (-0.003 to 0.001): assuming that men and women have the same preferences over firms, non-wage amenities do not amplify (or dampen) the gender gap in utility. Of course, this conclusion depends on the assumption that men and women value the same firms equally. Figure 4 shows they do not: the Pearson correlation between male- and female-specific valuations is only 0.266 (Table 3).

As we discuss in Appendix C.5, putting a magnitude on how much more women value their firms' amenities requires us to make a cross-sex normalization. Columns 4–6 of Panel B re-estimate the gap using sex-specific valuations, under a variety of normalizations. Under our base-line normalization—setting the employment-weighted mean valuation in professional services to zero within each sex—women work at firms whose amenity value is 0.014 log-wage equivalents higher than men's, with Mincer controls and occupation fixed effects (Column 6, $p < 0.05$). This is comparable in magnitude to the oppositely signed wage gap on the same sample (-0.013 , Panel A, Column 6), so under this normalization amenities roughly offset the wage gap in total firm-provided value.

While the magnitude varies across normalizations, the sign of the amenity advantage is positive under every normalization we consider, ranging from 0.003 under the overall employment-weighted mean to 0.014 under professional services. At the upper end, accounting for sex-specific valuations roughly closes the gender gap in total firm-provided value. Table A6 and Appendix Table A7 show additional specifications—including those that use the no-growth valuations. These confirm that our results hold even when we allow for the possibility that part of what workers value about firms is wage growth (or promotion opportunities). Adding each firm’s average randomized raise as a control leaves the sex-specific coefficient essentially unchanged (0.014).

Discussion. Our results indicate that gender differences in amenity valuations help explain women’s sorting into lower-wage firms. However, this sorting does not imply that the gender wage gap is welfare-neutral. Our estimates reflect preferences over the existing distribution of jobs. Some of the amenities women value at their current employers—such as the absence of sexual harassment, a known determinant of women’s labor supply (Folke and Rickne, 2022), or a less male-dominated work environment—are likely shaped by the equilibrium itself. Changing a firm’s gender composition could alter its culture and non-wage attributes, potentially making higher-wage firms more attractive to women. But moving a single woman to a higher-wage firm, holding the firm’s characteristics fixed, would not necessarily enhance her welfare: it would separate her from the non-wage attributes she currently values without changing the environment at the destination firm. In addition, our analysis is restricted to full-time workers. Because many women in Germany shift to part-time work after having children, this restriction means we focus on a selected set of women.

6 Conclusion

This paper provides new evidence on how workers value firm-specific amenities and how those valuations shape mobility, sorting, and inequality, including the gender wage gap. Using discrete choice experiments embedded in a large-scale survey of German workers, we estimate the money-metric value of firm amenities as perceived by marginal applicants, workers who would consider moving to a given firm. By randomizing wages across hypothetical job offers, we recover firm-specific valuations that are independent of observed wage premia, offer rates, or hiring frictions.

Non-wage attributes shape job choice: workers frequently choose lower-paying offers. Amenity valuations are highly dispersed across firms; they rise with wage premia when firms are weighted equally and are approximately orthogonal when weighted by employment, so accounting for amenities widens rather than offsets between-firm differences in total compensation. In money-metric units, the employment-weighted signal variance of amenities is about one-third that of wage premia. The between-firm variance of total compensation is about 20 percent larger than that of pay.

Conditional on the wage, high-amenity firms are larger and more favorably reviewed by employees, and have lower (though imprecisely estimated) quit rates.

Amenity preferences vary across demographic groups. Men and women do not value the same firms equally: the correlation between their firm-specific valuations is 0.266. Women work at firms that pay less. They also make steeper wage–amenity trade-offs when they switch firms. Our results suggest, however, that women work at firms that offer them higher amenity value.

Our results motivate several directions for future work. First, while we document substantial heterogeneity in amenities, we do not take a stand on which specific attributes drive worker preferences. Experimental variation in individual amenities at the firm level would allow precise pricing. Second, we abstract from the firm side of the market. Understanding how and why firms choose particular wage–amenity bundles, and how these decisions respond to worker preferences, is a natural next step. Third, our analysis is restricted to full-time workers; extending it to the part-time margin, where trade-offs among hourly wages, hours, and workplace amenities are simultaneous, would be particularly valuable for understanding the gender gap in job quality.²⁷ Fourth, our analysis is restricted to the private sector; understanding how amenities shape sorting and inequality in the public sector is a natural extension.

²⁷It would be important to also collect information on which hours packages are actually feasible at a given firm.

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7 Tables and Figures

Table 1: Descriptive Statistics

	Survey Sample		2% Sample		Job-to-Job Movers	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Demographics</u>						
Female	0.32	(0.47)	0.35	(0.48)	0.34	(0.47)
Age	32.96	(5.93)	39.39	(11.80)	35.80	(7.62)
German Citizen	0.91	(0.29)	0.95	(0.22)	0.92	(0.27)
College Degree	0.62	(0.48)	0.15	(0.36)	0.26	(0.44)
Apprenticeship	0.30	(0.46)	0.75	(0.44)	0.65	(0.48)
<u>Earnings</u>						
Daily Pay	173.11	(56.77)	107.58	(62.49)	103.54	(55.19)
Censored Pay	0.23	(0.42)	0.05	(0.23)	0.05	(0.21)
AKM Firm Effect	0.46	(0.15)	0.36	(0.21)	0.28	(0.23)
<u>Sector</u>						
Manufacturing Sector	0.47	(0.50)	0.31	(0.46)	0.19	(0.40)
Retail Sector	0.08	(0.28)	0.14	(0.34)	0.13	(0.33)
Professional Sector	0.13	(0.34)	0.05	(0.22)	0.08	(0.27)
Observations	2383		20339174		25361184	

Note: This table compares the characteristics of workers in our survey (Columns 1–2) to the set of full-time workers ages 25–50 (Columns 3–4) and to full-time workers ages 25–50 who moved between firms (Columns 5–6). We describe how we constructed these samples in Appendix B.1.

Table 2: Characteristics of Provided Firms

	All Firms		Connected Set	
	Unweighted	Weighted by	Unweighted	Weighted by
		Firm Size		Firm Size
	(1)	(2)	(3)	(4)
<u>Linkages</u>				
To Wage Premia	95%	100%	97%	100%
To BHP	97%	100%	98%	100%
Size	29682	149565	34701	149891
(Std. Deviation)	(59653)	(93273)	(63227)	(93118)
Size (Imputed)	22733	140018	28385	140730
(Std. Deviation)	(51637)	(94320)	(56473)	(94036)
<u>Size (Binned)</u>				
1-10 Workers	3%	0%	2%	0%
11-50 Workers	4%	0%	3%	0%
51-200 Workers	5%	0%	3%	0%
201-1000 Workers	33%	1%	26%	1%
1001-10000 Workers	20%	3%	24%	3%
>10000 Workers	32%	96%	42%	96%
Firm Wage Premium	0.50	0.48	0.51	0.48
(Std. Deviation)	0.11	0.07	0.09	0.06
Firm Wage Premium (CHH)	0.16	0.06	0.16	0.05
(Std. Deviation)	0.22	0.16	0.21	0.16
<u>Sector</u>				
Manufacturing	40%	43%	42%	43%
Retail	9%	1%	8%	1%
Professional	24%	52%	25%	52%
Information	10%	2%	10%	2%
Finance	6%	0%	5%	0%
Number of Firms	3899	3899	949	949

Note: This table describes the firms that workers provided (Columns 1–2) and the subset of these firms in the strongly connected set (remaining columns). Each column indicates how the statistics are weighted. We impute firm size when missing; Appendix B.3 describes this process. Appendix Table D2 presents results for alternative weighting schemes.

Table 3: Correlation Between Valuations of Different Demographic Groups

	Female (1)	Male (2)	College Degree (3)	No College Degree (4)
Female	1.000	0.454	0.666	0.584
Male	0.266	1.000	0.736	0.575
College Degree	0.629	0.570	1.000	0.359
No College Degree	0.545	0.591	0.231	1.000

Note: This table examines the pairwise correlation in valuations of amenities for different demographic groups. The numbers above the diagonal (shaded in grey) are rank correlations; the numbers below the diagonal are raw correlations. Firms are weighted by firm size. Robustness to alternative weighting schemes and minimum-mentions restrictions is in Appendix Table D3.

Table 4: External Validation: Amenities, Firm Size, and Employer Reviews

	Baseline Valuations		Valuations Excluding Wage Growth	
	(1)	(2)	(3)	(4)
A. Controlling for Observed Wage Premium				
Log Firm Size	4.107*** (1.010) 552	3.784*** (1.001) 552	5.126*** (1.720) 305	4.894*** (1.793) 305
Percent Recommend (0-100)	90.788** (35.507) 420	65.408*** (22.684) 420	86.887** (38.024) 242	39.642 (25.824) 242
B. Controlling for Log Median Pay				
Log Firm Size	4.734*** (1.103) 577	4.374*** (1.092) 577	6.672*** (1.719) 318	6.436*** (1.732) 318
Percent Recommend (0-100)	81.697** (33.014) 434	56.025*** (20.908) 434	82.473** (38.498) 251	23.804 (25.517) 251
C. Controlling for Log Mean Pay				
Log Firm Size	4.566*** (1.109) 577	4.194*** (1.099) 577	6.447*** (1.727) 318	6.239*** (1.748) 318
Percent Recommend (0-100)	83.211** (32.424) 434	57.353*** (20.655) 434	84.372** (37.942) 251	26.170 (25.135) 251
Fixed Effects	—	Sector	—	Sector

Note: This table examines the relationship between amenity valuations and two outcome variables, indicated in the left column: log firm size from the Betriebs-Historik Panel (BHP), and the percentage of employees who recommend the firm on Kununu. Each cell reports the coefficient on workers' valuation of firm-specific amenities from a regression of the outcome on this valuation, a measure of the firm's wage policy (indicated in the panel), and a constant. Columns 1–2 use our baseline valuations; Columns 3–4 use the valuations computed under the assumption that career growth is identical across all options. Within each pair, the second column (Columns 2 and 4) adds sector fixed effects; the first (Columns 1 and 3) does not. The log firm size regressions are unweighted; the Percent Recommend (0–100) regressions are weighted by firm size. Levels of significance: * 10%, ** 5%, and *** 1%.

Table 5: Classifying Job-to-Job Moves

	Change in Amenities			All Movers
	>2% Cut	<2% Change	>2% Raise	
	(1)	(2)	(3)	
A. Change in Own Wage				
Decrease ($< -2\%$)	28.8%	22.4%	48.8%	18.5%
Similar ($\pm 2\%$)	35.7%	14.0%	50.3%	21.2%
Increase ($> 2\%$)	34.4%	21.9%	43.7%	60.3%
All Movers	33.6%	20.3%	46.1%	100.0%
B. Change in Firm Wage Premium				
Decrease ($< -2\%$)	33.7%	19.1%	47.2%	45.6%
Similar ($\pm 2\%$)	34.4%	15.6%	50.0%	13.3%
Increase ($> 2\%$)	32.3%	25.1%	42.7%	41.1%
All Movers	33.2%	21.1%	45.7%	100.0%
C. Change in Own Wage				
Decrease ($< -2\%$)	25.6%	16.7%	57.7%	18.6%
Similar ($\pm 2\%$)	33.7%	14.7%	51.6%	22.7%
Increase ($> 2\%$)	32.5%	24.0%	43.5%	58.7%
All Movers	31.5%	20.5%	48.0%	100.0%
D. Change in Firm Wage Premium				
Decrease ($< -2\%$)	26.5%	21.9%	51.5%	46.8%
Similar ($\pm 2\%$)	34.9%	12.7%	52.4%	15.0%
Increase ($> 2\%$)	35.0%	21.9%	43.1%	38.2%
All Movers	31.0%	20.5%	48.4%	100.0%

Note: This table classifies job transitions by the change in wage and amenity values. Within each panel, the cells in a row are conditional on the wage-change category and sum to 100% across the row; the final “All Movers” column reports the unconditional share of all movers in each wage-change category, and the bottom “All Movers” row reports the unconditional share by amenity change. The sample includes all full-time job movers between 2010 and 2019 who moved between firms for which we observe amenity values. We drop moves that occur within firms: establishment switches within a firm leave firm-level premia and amenity values mechanically unchanged. Panels A and B use baseline amenity valuations; Panels C and D use valuations computed under the assumption that career growth is identical across all options. Amenity changes are converted to log-wage-equivalent units before applying the thresholds. The greyed-out cells on the main diagonal are concordant transitions, in which wages and amenities move in the same direction. Cells above the diagonal are transitions in which amenities improve relative to pay; cells below the diagonal are transitions in which pay improves relative to amenities.

Table 6: Amenities and the Gender Value Gap

	(1)	(2)	(3)	(4)	(5)	(6)
	All Workers			Sex-Specific		
A. Firm Wage Premium (ψ_j)						
Female	−0.014*	−0.010	−0.003	−0.022**	−0.019**	−0.013**
	(0.008)	(0.007)	(0.003)	(0.009)	(0.008)	(0.005)
B. Amenity Value ($a_j/\hat{\beta}$, log-wage equivalent)						
Female	−0.003	−0.003	0.001	0.010	0.010	0.014**
	(0.004)	(0.004)	(0.002)	(0.007)	(0.006)	(0.006)
Mincer controls		Yes	Yes		Yes	Yes
Occupation FE			Yes			Yes
<i>N</i>	236,674	236,674	236,661	201,001	201,001	200,982

Note: This table examines the gender gap in firm wage premia (Panel A) and amenity values in money-metric (log-wage-equivalent) units (Panel B). The firm wage premia stem from Bellmann et al. (2020). The amenity values are estimated using the full sample of workers (Columns 1–3) or using sex-specific models (Columns 4–6). The “Mincer controls” include age, a quadratic in experience, and a dummy for whether the worker has a college degree. Results are unweighted. Additional specifications are in Appendix Table A6. We cluster standard errors at the firm level. Levels of significance: * 10%, ** 5%, and *** 1%.

8 Figures

Figure 1: Key Survey Elicitations

“Suppose you planned to move to a new company in the next {one/three/six} months. What are companies that you would consider applying to?

Please list three companies that you would consider applying to and that hire employees in positions like yours (e.g. “PlaceHolder Inc”). These can be companies without current job vacancies.”

[Fill in Company 1] → {Company 1} with a X% raise _____

[Fill in Company 2] → {Company 2} with a Y% raise _____

[Fill in Company 3] → {Company 3} with a Z% raise _____

☐ I do not want to answer this question

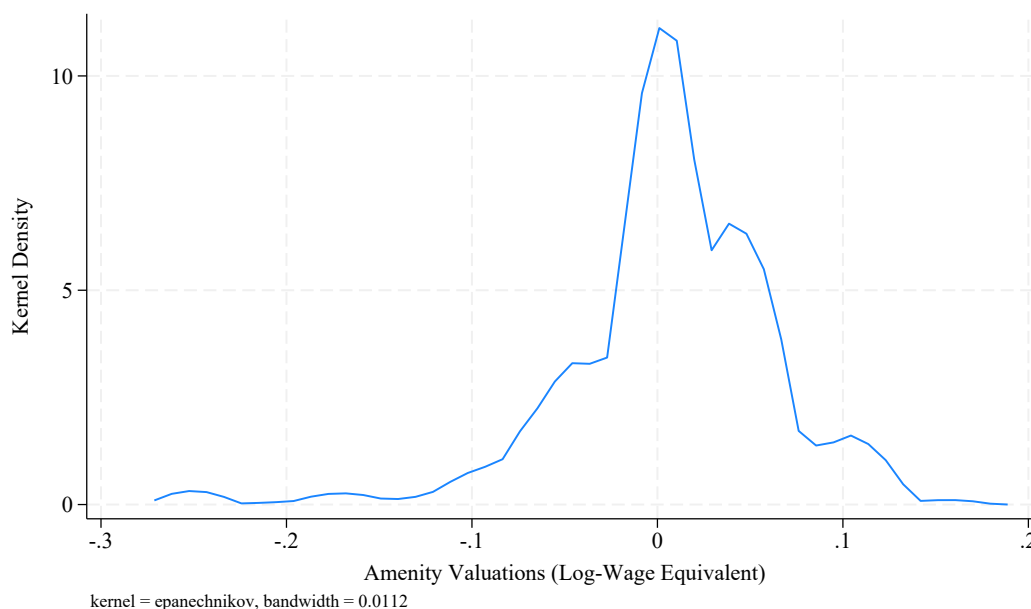
Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the raise specified.

Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Remain at current firm at current pay _____

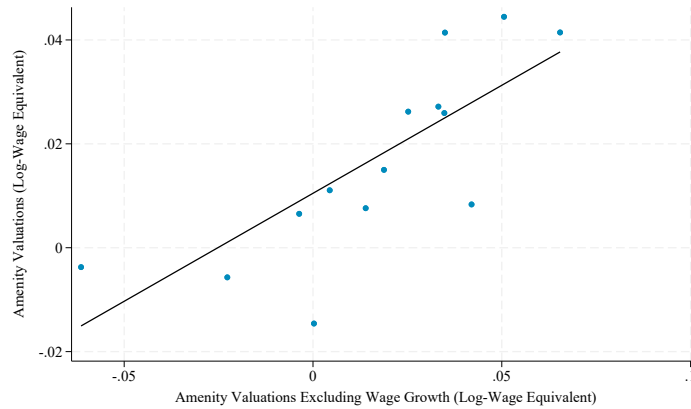
Note: This figure shows the two survey questions we use to estimate valuations. As noted in Section 3, we do not use information on the ranking of the current firm. Appendix E provides more information on the survey.

Figure 2: Distribution of Amenity Values



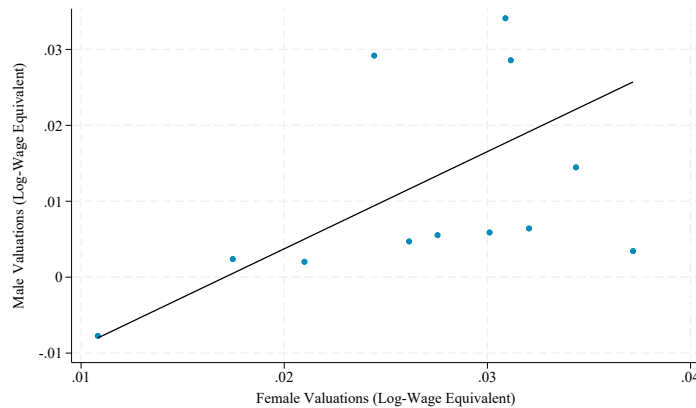
Note: This figure plots the distribution of estimated amenity values, weighted by firm size. Values are expressed in log-wage-equivalent units, obtained by dividing the raw utility-unit estimates by the structural wage coefficient $\hat{\beta}$. Section 3 and Appendix C provide details on estimation. Versions that use alternative weighting schemes (unweighted, number of mentions, and imputed firm size) are in Appendix Figure D4.

Figure 3: Relationship Between Valuations with and without Wage Growth



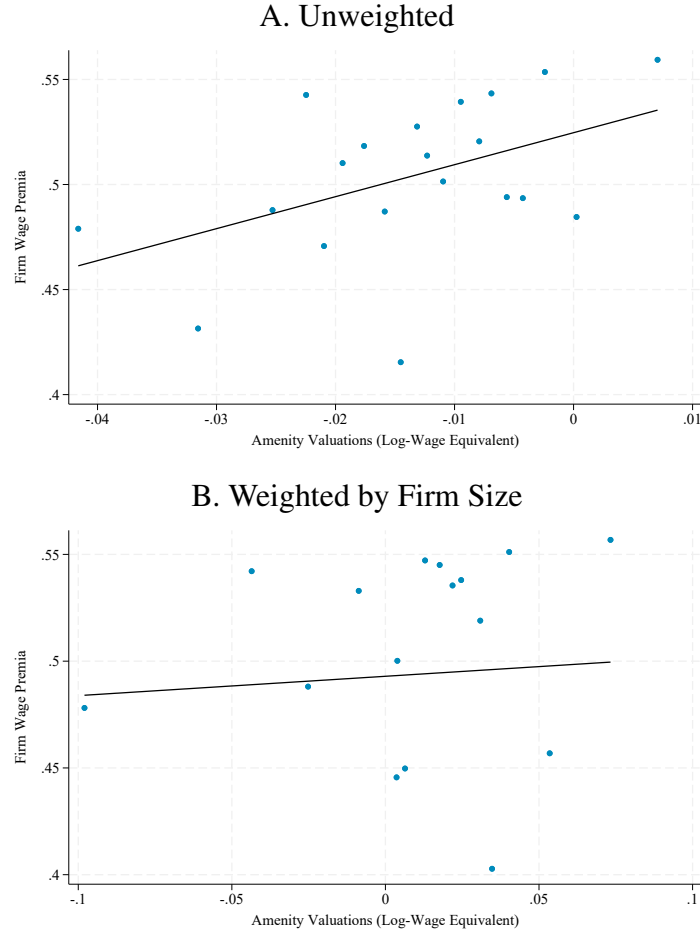
Note: This figure compares the baseline estimates of amenity valuations (in log-wage-equivalent units) to those we obtain when we re-run the model using a discrete choice experiment where we told workers to assume that career growth—future raises and promotions—would be identical across all options. The dots come from a binned scatterplot and the line is the best linear fit. All estimates are relative to the same reference firm. To account for measurement error in the estimation of both types of amenities, we follow a split-sample approach and plot the fitted values of our baseline valuations (predicted values from regressing estimates from one split sample on the other) against the fitted values of our no-growth valuations. Results are weighted by firm size. Versions that use alternative weighting schemes (number of mentions and imputed firm size) are in Appendix Figure D5.

Figure 4: Relationship Between Male and Female Valuations



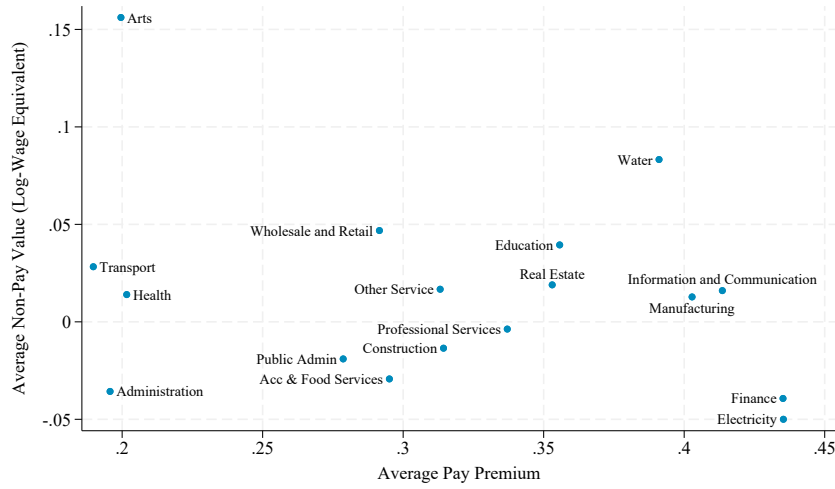
Note: This figure compares the baseline estimates of valuations (in log-wage-equivalent units) for male and female workers. We account for measurement error using a split-sample approach. We first regress the valuations for one half of the workers in the group (male or female workers) on the valuations for the other half. We then plot the predicted values for male workers against those for female workers. Results are weighted by firm size. Versions that use alternative weighting schemes (including imputed firm size and number of mentions) are in Appendix D.4.

Figure 5: Relationship Between Wage Premia and Amenity Valuations



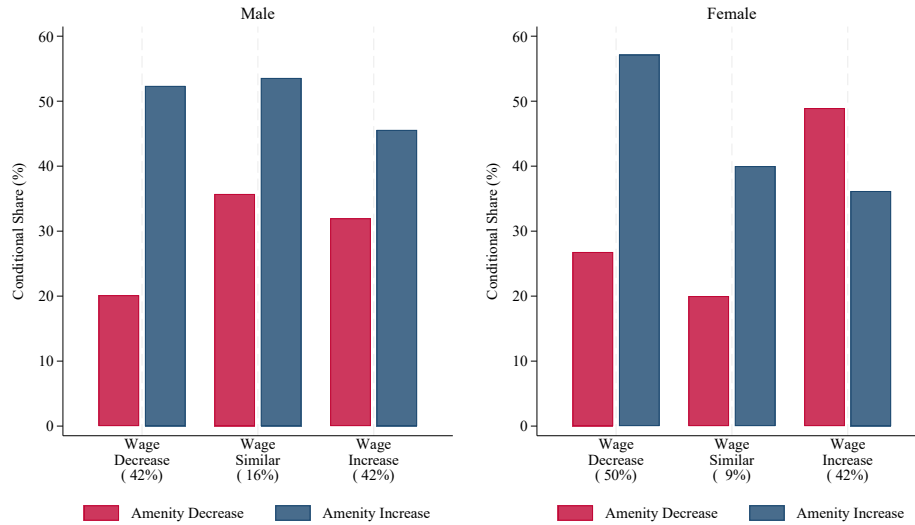
Note: This figure shows binned scatterplots of the relationship between firm wage premia (y-axis) and amenity values in log-wage-equivalent units (x-axis) for firms in our strongly connected set. We first predict amenity values estimated in one random half of the data using the values estimated in the other half and then create a binned scatterplot of the firm wage premia against these predicted values. Amenity values are converted to log-wage equivalents by dividing by the structural logit coefficient $\hat{\beta}$. Panel A is unweighted; the regression in Panel B is weighted by firm size. The estimated slope is 1.522 (s.e. 0.560) in Panel A and 0.091 (s.e. 0.128) in Panel B. Versions that use alternative weighting schemes (including imputed firm size and number of mentions) are in Appendix D.4.

Figure 6: Amenities, Wages, and Values by Sector



Note: This figure shows average amenity values in log-wage-equivalent units (y-axis) and average wage premia (x-axis) by major sector. Amenity values are converted to log-wage equivalents by dividing by the structural logit coefficient $\hat{\beta}$. Within each sector we weight firms by firm size. Versions that use alternative weighting schemes (including imputed firm size and number of mentions) are in Appendix Figure D9.

Figure 7: Classifying Job-to-Job Moves by Sex



Note: Each panel classifies job-to-job transitions by the direction of change in the firm wage premium (horizontal axis) and sex-specific amenity valuations. Bars show the conditional share of movers experiencing an amenity decrease (red) or increase (blue) within each wage-change category. Parenthetical values report the share of movers of that sex in each wage category. The sample includes full-time job movers between 2010 and 2019 whose origin and destination firms both have sex-specific amenity estimates. We drop moves that occur within firms. Appendix Table A5 reports the full transition matrices.

Online Appendix

Firm Pay, Amenities, and Inequality

A Appendix Tables and Figures

Table A1: Reasons Workers Did Not Provide Firm Names

Know companies, but survey interface was too difficult	7%
Not comfortable sharing this information	8%
Could not think of specific companies	38%
No interest in switching	47%
Observations	829

Note: This table tabulates the reasons workers declined to provide the names of firms they would apply to if they wanted to switch firms. The sample includes workers who did not provide firms in the initial survey and did not provide three firms in the follow-up survey. All possible responses are listed.

Table A4: Between-Firm Variance of Total Compensation

	Equal-weighted	Employment-weighted
$\text{Var}(\psi_j)$, observed	0.00993	0.00696
$\text{Var}(a_j/\hat{\beta})$, split-sample signal	0.00166	0.00216
$\text{Cov}(a_j/\hat{\beta}, \psi_j)$	0.00027	-0.00037
$R = \text{Var}(\psi_j + a_j/\hat{\beta})/\text{Var}(\psi_j)$	1.22	1.20
95% confidence interval	[1.03, 1.41]	[0.77, 1.64]
with reliability-corrected $\text{Var}(\psi_j)$	1.26	1.24
$\sqrt{R}$	1.10	1.10

Note: All quantities are in log-wage units. The signal variance of amenity values is the covariance between independent split-sample estimates, divided by $\hat{\beta}^2$. The covariance row uses observed amenity values, which classical measurement error leaves unbiased. The reliability-corrected row multiplies $\text{Var}(\psi_j)$ by 0.85, the split-sample correlation of the wage premia. Confidence intervals bootstrap over firms (1,000 replications).

Table A2: Randomization of Experimental Wage Offers

Variable	<i>p</i> -value
Firm Wage Premium	0.435
Size	0.412
Median Wage	0.362
Mean Wage	0.400
Manufacturing Sector	0.862
Retail Sector	0.141
Professional Sector	0.837
Information Sector	0.146
Transport Sector	0.307
Finance Sector	0.487
HQ in East Germany	0.309
Year Incorporated	0.296
Total Assets/Employee	0.773
Fixed Assets/Employee	0.828
Indicator for Large Employer	0.620
Indicator for Popular Employer	0.856
Brand Recognition	0.319
Page Views (Kununu)	0.191
Pay Score (Kununu)	0.320
Image Score (Kununu)	0.522
Career Score (Kununu)	0.137
Atmosphere Score (Kununu)	0.197
Communication Score (Kununu)	0.495
Cohesion Score (Kununu)	0.133
Work-Life Balance (Kununu)	0.781
Supervisor Behavior (Kununu)	0.577
Interesting Tasks (Kununu)	0.070
Equality (Kununu)	0.547
Treatment of Older Workers (Kununu)	0.350

Note: We assess the randomization of raises to firms by regressing each firm characteristic on the randomized raise offered to that firm. The column presents the *p*-value from a test that the slope on the randomized raise is zero.

Table A3: Effect of Experimental Wage Difference on Pairwise Choice

	All (1)	Male (2)	Female (3)	College (4)	No College (5)
<i>Panel A: Linear Probability Model</i>					
Wage Difference	2.859*** (0.107)	2.870*** (0.129)	2.835*** (0.191)	2.914*** (0.132)	2.757*** (0.181)
R-squared	0.062	0.063	0.062	0.065	0.058
<i>p</i> -value, Male = Female: 0.877					
<i>p</i> -value, College = No College: 0.483					
<i>Panel B: Structural Logit</i>					
Wage Difference	12.08*** (0.50)	12.13*** (0.60)	11.98*** (0.89)	12.33*** (0.62)	11.63*** (0.84)
<i>p</i> -value, Male = Female: 0.886					
<i>p</i> -value, College = No College: 0.501					
Observations	20,740	14,485	6,255	13,550	7,190

Note: The dependent variable is a binary indicator for whether the worker prefers firm j over firm k . The regressor is the difference in randomized raises between the two firms. Panel A reports OLS (linear probability) estimates; Panel B reports the coefficient on the wage difference from a structural logit, which we use as $\hat{\beta}$ in the utility function. Standard errors clustered at the worker level in parentheses. The p -values test whether the wage coefficient is equal across demographic groups using a pooled regression with an interaction term.

Table A5: Classifying Job-to-Job Moves: Sex-Specific Estimates of a_j

	Male				Female			
	Change in Amenities			All Movers	Change in Amenities			All Movers
	Cut	Similar	Raise		Cut	Similar	Raise	
A. Change in Own Wage								
Decrease ($< -2\%$)	26.9%	22.4%	50.7%	19.0%	21.7%	17.4%	60.9%	20.7%
Similar ($\pm 2\%$)	26.7%	17.3%	56.0%	21.2%	27.3%	18.2%	54.5%	19.8%
Increase ($> 2\%$)	28.9%	24.6%	46.4%	59.8%	42.4%	18.2%	39.4%	59.5%
N			353				111	
B. Change in Firm Wage Premium								
Decrease ($< -2\%$)	20.1%	27.5%	52.3%	42.3%	26.8%	16.1%	57.1%	49.6%
Similar ($\pm 2\%$)	35.7%	10.7%	53.6%	15.9%	20.0%	40.0%	40.0%	8.8%
Increase ($> 2\%$)	32.0%	22.4%	45.6%	41.8%	48.9%	14.9%	36.2%	41.6%
N			352				113	
C. Change in Log Median Wage								
Decrease ($< -2\%$)	34.0%	26.3%	39.7%	44.1%	32.8%	19.7%	47.5%	54.0%
Similar ($\pm 2\%$)	19.0%	42.9%	38.1%	5.9%	33.3%	0.0%	66.7%	2.7%
Increase ($> 2\%$)	23.7%	16.9%	59.3%	50.0%	38.8%	16.3%	44.9%	43.4%
N			354				113	
D. Change in Log Mean Wage								
Decrease ($< -2\%$)	33.3%	27.6%	39.1%	44.1%	33.9%	20.3%	45.8%	52.2%
Similar ($\pm 2\%$)	31.8%	27.3%	40.9%	6.2%	0.0%	0.0%	100.0%	3.5%
Increase ($> 2\%$)	22.7%	17.6%	59.7%	49.7%	40.0%	16.0%	44.0%	44.2%
N			354				113	

Note: This table classifies job-to-job transitions by the change in wage and amenity values, separately by sex. Within each panel and sex, the cells in a row are conditional on the wage-change category and sum to 100% across the row; the “All Movers” columns report the unconditional share of movers of that sex in each wage-change category. The sample includes all full-time job movers between 2010 and 2019 who moved between firms for which we observe sex-specific amenity values. We drop moves that occur within firms: establishment switches within a firm leave firm-level premia and amenity values mechanically unchanged. Within each panel, the first four columns report results for male workers and the next four for female workers. Panel A uses changes in workers’ own wages, Panel B changes in the firm wage premium, Panel C changes in log median firm pay, and Panel D changes in log mean firm pay. Pooled (non-sex-specific) results for own wages and firm wage premia are in Table 5.

Table A6: Amenities and the Gender Value Gap: Robustness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Sector FE	IPW Rewtd.	No Growth Valuations	Overall Mean Norm.	Mfg. Norm.	Median Norm.	Overlap Set
A. Firm Wage Premium (ψ_j)							
Female	-0.006 (0.004)	-0.024* (0.013)	-0.005 (0.006)	-0.013** (0.005)	-0.013** (0.005)	-0.013** (0.005)	-0.002 (0.004)
B. Amenity Value ($a_j^g/\hat{\beta}$, log-wage equivalent)							
Female	0.011** (0.005)	0.008 (0.009)	0.041*** (0.010)	0.003 (0.006)	0.005 (0.006)	0.005 (0.006)	0.008 (0.005)
Mincer controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Occupation FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	No	No	No	No	No	No

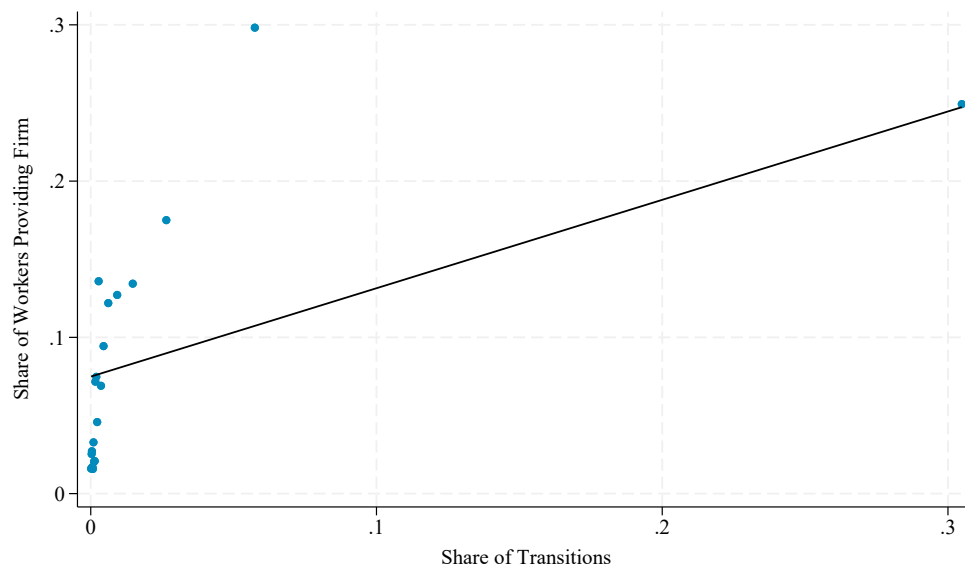
Note: This table examines the robustness of the baseline gender gap specification in Columns 3 and 6 of Table 6. All amenity values are in money-metric (log-wage-equivalent) units. Column 1 replaces occupation fixed effects with sector fixed effects. Column 2 reweights observations using inverse probability weights to match the composition of the full 2% population sample, correcting for selection into the amenity estimation sample. Column 3 uses valuations elicited under the assumption of identical promotion and pay growth trajectories. Columns 4–6 vary how we normalize valuations: overall employment-weighted mean, manufacturing sector, and median firm. Column 7 restricts the sample to the intersection of the male and female connected sets. All columns include Mincer controls (age, experience, education); Columns 2–7 include occupation fixed effects. Standard errors clustered at the firm level in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

Table A7: Female Amenity Advantage by Valuation and Normalization

	Professional Services (baseline anchor)	Overall Mean	Manufacturing Sector	Median Firm
Baseline valuations	0.014** (0.006)	0.003 (0.006)	0.005 (0.006)	0.005 (0.006)
No-growth valuations	0.041*** (0.010)	0.008 (0.010)	0.004 (0.010)	0.002 (0.010)

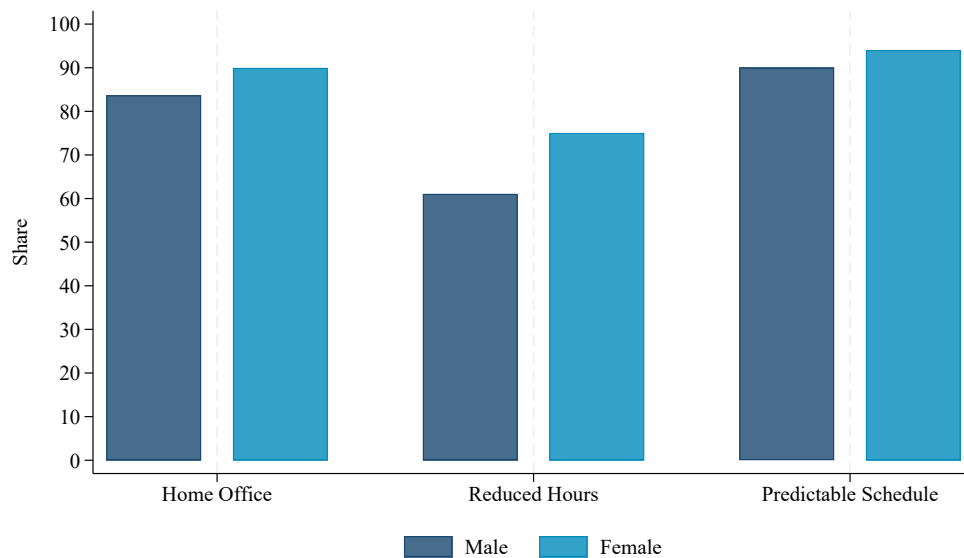
Note: Each cell is the coefficient on the female indicator from the sex-specific amenity-gap regression (equation 2) in money-metric (log-wage-equivalent) units, with Mincer controls and occupation fixed effects, estimated under a different combination of amenity valuation (baseline vs. no-growth) and cross-sex normalization. The professional services normalization is our baseline. The sign is positive throughout, but the magnitude and significance of the level depend on what we normalize to: it is statistically significant only under the professional services normalization. Standard errors clustered at the firm level in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

Figure A1: Relationship Between Stated and Revealed Destinations



Note: For each origin-destination pair we calculate the share of movers leaving the origin firm (2010–2019, including moves through unemployment) who move to a given destination firm. We then calculate the share of surveyed workers at the origin firm who name that destination as a place they would apply if they wanted to switch firms. We present a binned scatterplot of the share of workers naming each destination (y-axis) against the share of movers moving there (x-axis), restricted to pairs with at least one observed transition and weighted by the number of movers from the origin firm. The stated application data are from 2022 (initial survey) and 2024 (follow-up survey).

Figure A2: Gender Differences in Whether Specific Amenities Are Important



Note: This figure shows the percentage of male and female workers in our survey that rate specific amenities (remote work option, predictable hours, and reduced hours option) as “very important” or “extremely important” on a five-point Likert scale ranging from “not important” to “extremely important.”

B Additional Data Sources

This section describes the additional data sources used in the paper.

B.1 IEB Social Security Records

We use the German Social Security records, which are assembled by the Institute for Employment Research (IAB) into the Integrated Employment Biographies (IEB) database, to select the sample of workers to whom we fielded the survey. These data capture all private-sector and public-sector employees with Social Security contributions. They contain information on employee demographics (e.g., gender, age, and education), employer information (e.g., sector and location), and job-spell-based information (e.g., start and end dates, full-time status, daily pay, and occupation). We impute daily pay for individuals whose pay is censored at the Social Security maximum.

We construct three primary samples:

- **Survey Sample:** The survey sample consists of workers in the “CHH Worker Survey” who provided the names of specific firms they would apply to if they wanted to switch firms. We pull employment histories for these workers. We use data from a single cross-section.
- **2% Sample:** We use a 2% sample of the IEB Social Security records. We use the subset of these workers who are in full-time, regular (i.e., not temporary) employment and who are between the ages of 25 and 50. These data cover the period 1975 to 2021; our primary analysis includes data from 2010 to 2019.
- **Movers Sample:** We use the employment histories of all workers in the IEB (100% sample) who switched firms between 2010 and 2019. We restrict to full-time, regular workers between 25 and 50. When we describe “job-to-job” movers, we use the subset of observations where workers moved directly from one firm to another. To identify job-to-job movers, we remove observations where the source is the Employment History (BeH)—a compilation of annual employment summaries required of firms by the government—and where the source of the previous observation is the Benefit Recipient History (LeH)—the IAB’s database on unemployment assistance.

B.2 Linkage of Worker-Provided Firms to Administrative Data

To use data from the worker-provided firm module, we had to clean the firm names that workers provided. In the initial survey, respondents (overall, including those dropped from our main analysis sample) provided 5,828 unique strings. Of these, we could assign 5,016 (86%) to uniquely identifiable firms. We performed this assignment manually through online searches to ensure that the strings matched a unique firm.

Of the 812 strings that could not be uniquely identified, the majority are unspecific firm types (e.g., “City government”, “police”); a small share (10%) represent non-response (e.g., “I do not want to answer”) or unidentifiable content (e.g., “XXX”). Our manual assignment procedure, which corrects for differences in spelling and pools subdivisions of the same firm, assigns the 5,016 identifiable strings to 3,018 unique firms. (These counts cover the initial survey; pooling

both waves, workers named 3,899 unique firms.) Of these 3,018 firms, 2,013 can be matched to establishments in the IAB records. Among the firms that we cannot link to the IAB, 29% are foreign entities and 37% are in the public sector; these firms may not have any workers in Social Security-covered employment in Germany. Because the public sector is not covered by the IEB, we could not have measured firm wage premia for these establishments in any case; prior studies of German wage inequality likewise exclude the public sector. We link 79% of firms named at least twice to the IAB records and obtain the municipality of headquarters for 99% of linked firms. Appendix Table B1 reports how many times each firm was named by workers and how many enter the strongly connected set.

Table B1: Counts by Firm

Times Surveyed	All Firms	Connected Set	Connected Set (Wave 1)
1 Time	2861	251	228
2–4 Times	664	337	302
5–9 Times	210	197	173
10–14 Times	48	48	41
15–19 Times	29	29	23
20–49 Times	51	51	47
50+ Times	36	36	34
Total	3899	949	848

Note: This table tabulates the number of times each firm was provided by a worker and was therefore ranked against other outside firms. The connected-set columns include all 949 firms in the strongly connected set, matching the count reported in the text.

B.3 Imputation of Firm Size

For a subset of firms provided by workers, the number of employees is not directly observable in the administrative data (BHP). Our baseline analyses weight by observed size; imputed sizes enter the robustness checks in Appendix D.4, the descriptive statistics in Table 2, the amenity-correlates figures in Appendix D, and the researcher-provided exhibits. To retain firms with missing size in those checks, we impute firm size using a nested hierarchy of Poisson Pseudo-Maximum Likelihood (PPML) regressions. Poisson models are suited for count-type data and ensure positive predicted sizes.

Methodology. We use a range of firm-level predictors from our administrative and survey data. Let S_j be the observed firm size for firm j and Z_j be a vector of candidate predictors. We estimate specifications of the following form:

$$\mathbb{E}[S_j|Z_j] = \exp(Z_j\gamma)$$

where the exponential form is estimated via PPML to account for the count-based nature of firm size.

For firms with missing size data, we impute the fitted Poisson mean:

$$\text{Firm Size Imputed}_j = \hat{\mu}_j = \exp(Z_j \hat{\gamma}).$$

The fitted mean is strictly positive, so the imputed sizes are always valid analytic weights. Our main results weight by actual firm size from the BHP; imputed sizes appear in appendix exhibits and descriptive tables that include firms not linked to the administrative data.

Predictors and Model Hierarchy. We use the first available prediction from the following hierarchy:

1. Employees in Germany, global employees, survey mention frequency, and total Kununu reviews
2. Employees in Germany, global employees, and survey mention frequency
3. Employees in Germany, survey mention frequency, and total Kununu reviews
4. Survey mention frequency and total Kununu reviews
5. Employees in Germany, global employees, and total Kununu reviews
6. Survey mention frequency only
7. Employees in Germany only
8. Global employees only

The Kununu variables are available only for firms we link to Kununu. The number of global employees comes from Orbis. Following imputation, we replace the imputed value with the true administrative size whenever the latter is non-missing.

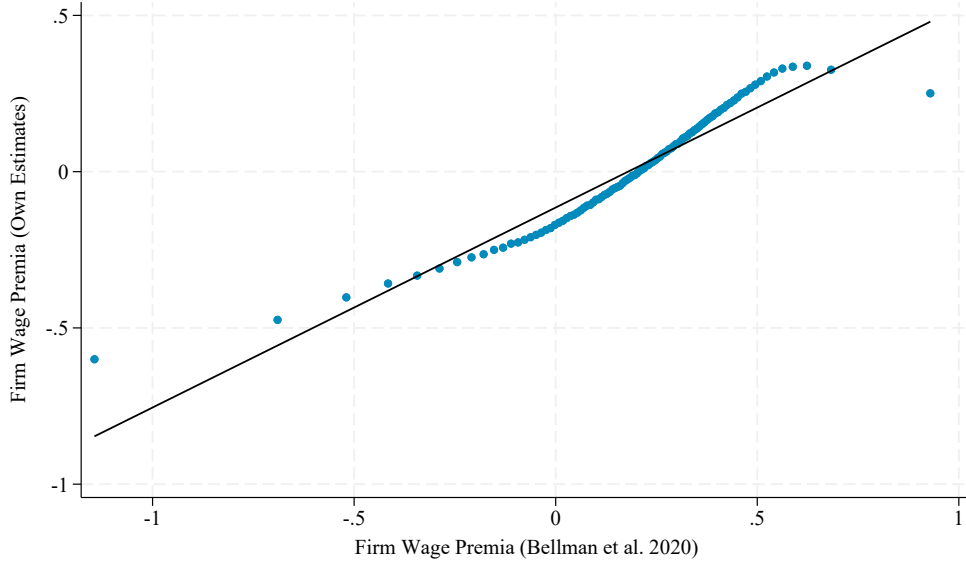
B.4 Estimates of AKM Wage Premia

Most of our analysis uses the AKM wage premia from Bellmann et al. (2020), which are computed from the full sample of workers, have been used by a variety of other researchers, and are provided directly by the IAB. However, in some specifications we use our own estimates so that we can use a split-sample approach to account for noise; this allows us to de-bias estimates of the variance and covariance.

Data Extraction and Initial Processing. From the population of Social Security records we construct a sample that consists of all full-time workers who switched firms between 2010 and 2019. For these workers we have data on wages before and after a transition.

Because wages are top-coded at the Social Security maximum, we impute wages above this threshold. We use this dataset to estimate separate Tobit regressions by year, sex, education group, and age group. The controls are age, the leave-one-out mean log coworker wage, the leave-one-out fraction of coworkers with censored wages, the establishment-level means of years of schooling, the share of university graduates, log wages, and the censoring rate; and an indicator for missing establishment employment. For workers in one-person establishments, the leave-one-out coworker means are set to fixed reference values. There are three education groups: school qualification only, apprenticeship, and university; workers with missing education are assigned to the first group. Age groups are 20–29, 30–39, and 40–60.

Figure B1: Comparison of Firm Wage Premia



Note: This figure is a binned scatterplot that compares our estimates of firm-specific wage premia to those produced by Bellmann et al. (2020). The two sets of estimates use different reference firms for normalization, so the intercept is not zero. The slope of the regression line is 0.99.

Estimation. We then fit the following model:

$$y_{it} = \alpha_i + \psi_{j(i,t)} + x'_{it}\gamma + \varepsilon_{it} \quad (3)$$

where y_{it} is the log daily wage of worker i in period t , α_i is a worker fixed effect, and $j(i, t)$ denotes the employer of worker i in period t . The x_{it} include year–education dummies as well as quadratic and cubic terms in age fully interacted with educational attainment. We link the establishment-level wage premia back to the Social Security data. To obtain firm-level estimates we take the employment-weighted average across all establishments associated with a firm.

Comparison with Other Estimates. Figure B1 compares our firm effect estimates to those of Bellmann et al. (2020) in a binned scatterplot; the two sets of estimates are strongly correlated (a correlation of 0.74). Our own estimates are the noisier of the two: a split-sample exercise—correlating premia estimated on two independent random halves of the data—yields a correlation of 0.85. This is one reason we use the Bellmann et al. (2020) premia in our main analysis; because those are estimated on a larger sample, 0.85 is a conservative lower bound on their reliability when we bias-correct the variance decomposition.

B.5 Orbis

We collect additional information on the provided firms from the Orbis database, which is compiled by the Bureau van Dijk based on firm balance sheet information. The collected firm characteristics include year of incorporation, sector based on the 4-digit NACE industry code, whether the firm

is headquartered in East Germany based on the zip code of its headquarters, and the number of employees.²⁸ To measure firm productivity, we also collect information on firms' fixed assets and total assets. Fixed assets refer to the total amount (after depreciation) of non-current assets (intangible assets, tangible assets, other fixed assets) and thus represent long-term assets that are unlikely to be converted into cash in the near term. Total assets are the sum of fixed assets and current assets (e.g., cash and any assets that will be converted into cash within the year).

For each variable from Orbis, we use the most recent available year. For over 90% of our firms, the most recent information is not older than three years. For fixed and total assets, we CPI-adjust our variables. To find the worker-provided firms in Orbis, we manually match each firm by name and address. We are able to match 79% of worker-provided firms that were named at least twice in the initial survey; we did not attempt this match for firms that were named only once.

B.6 Kununu

To better understand the types of amenities offered by different employers, we use data from Kununu, a leading employer review platform in Germany. Kununu allows current and former employees to rate their workplace across multiple dimensions and to indicate whether their employer offers a specific set of amenities. These data provide a worker-reported source of firm characteristics that complements our survey and administrative data. Because the reviews come from current and former employees, they measure workplace conditions as workers experience them, not as employers describe them.

²⁸Because Orbis draws on firms' balance-sheet information, the number of employees may include employees outside of Germany. We therefore also hand-collect information on the number of employees firms have in Germany from the firms' websites.

Figure B2: Screenshots of Kununu's Review Interface

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Note: This figure provides screenshots of the interface to submit reviews on Kununu, an online platform on which workers rate their current and former employers.

This section describes the firm-level information we collected from Kununu; Figure B2 shows its review interface. We linked firms in the Kununu data to the firms provided by workers based on firm name. Among the worker-provided firms that were named at least twice in the initial survey, we linked 88% to Kununu. As of February 2023 (when we collected the Kununu data), Kununu contained reviews for almost 200,000 German firms. In total, these firms received over 2.5 million ratings and had around 1.5 billion page views.

Kununu provides information on the total number of reviews a company received and the share of reviewers who would recommend the firm to a friend. Each company has an overall rating as well as ratings in different categories, such as pay or company culture. Ratings range from one to five, with higher numbers representing better ratings. When submitting these ratings, reviewers are prompted with the full list of subcategories and can choose between one and five stars for each.

Appendix Table B2 describes the variables we collect from Kununu. To measure how well-known provided firms are, we use the number of page views and number of reviews of each firm. As a measure of firm popularity, we use the share of reviewers who would recommend a firm as an employer to a friend and the overall rating of the firm. Recommendations are not provided for all firms on Kununu. To further characterize provided firms (e.g., high- versus low-paying), we use the numerical ratings of the different subcategories.

B.6.2 Accuracy of Information

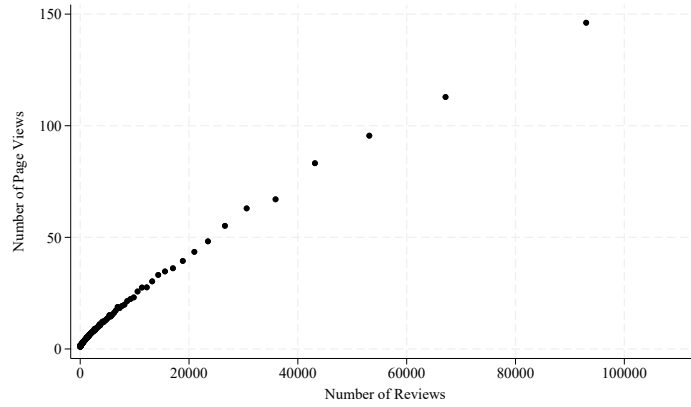
The ratings vary: only 6.4% of firms receive the maximum of five out of five. Firm-specific measures are also internally consistent: a firm's overall rating and the share of workers who would recommend the firm as an employer are highly correlated (Panel B of Appendix Figure B3). The pay-satisfaction rating also tracks pay in the administrative data: firms with higher Kununu pay ratings have higher mean and median wages and higher wage premia, with and without sector fixed effects (Appendix Table B3).

Table B2: Variables Collected from Kununu

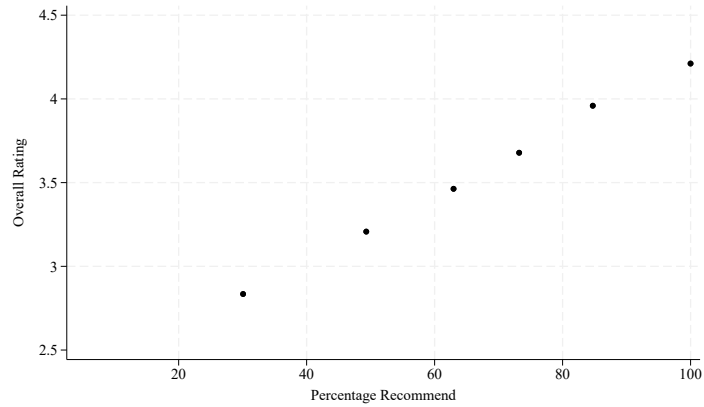
Note: This table presents information on the variables we collected from Kununu on all German firms that are covered by Kununu as of February 2023.

Figure B3: Assessment of Internal Consistency

A. Page Views and Reviews



B. Overall Rating and Percent Recommend



Note: This figure examines the within-Kununu consistency of the measures we collect. Panel A presents a binned scatterplot of a firm's number of reviews against the number of page views it receives. Firms in the top 1% of page views are dropped to make more points visible. Panel B presents a binned scatterplot of the overall Kununu score and the percentage of reviewers who would recommend the firm as an employer. The binned scatterplots are created using the program provided by Cattaneo et al. (2024).

Table B3: Kununu Pay Ratings and Administrative Pay Measures

	Mean Wage		Median Wage		AKM Firm Effect	
	(1)	(2)	(3)	(4)	(5)	(6)
A. Unweighted						
Kununu Pay Rating	26.450*** (2.657)	22.645*** (2.631)	24.317*** (2.430)	20.918*** (2.403)	0.073*** (0.006)	0.066*** (0.006)
R-squared	0.069	0.136	0.070	0.138	0.098	0.151
N	1327	1327	1327	1327	1252	1252
B. Weighted by Firm Size						
Kununu Pay Rating	42.910*** (2.582)	35.435*** (2.610)	37.154*** (2.280)	31.006*** (2.324)	0.056*** (0.005)	0.037*** (0.005)
R-squared	0.172	0.322	0.166	0.306	0.080	0.257
N	1327	1327	1327	1327	1252	1252
Sector FE	No	Yes	No	Yes	No	Yes

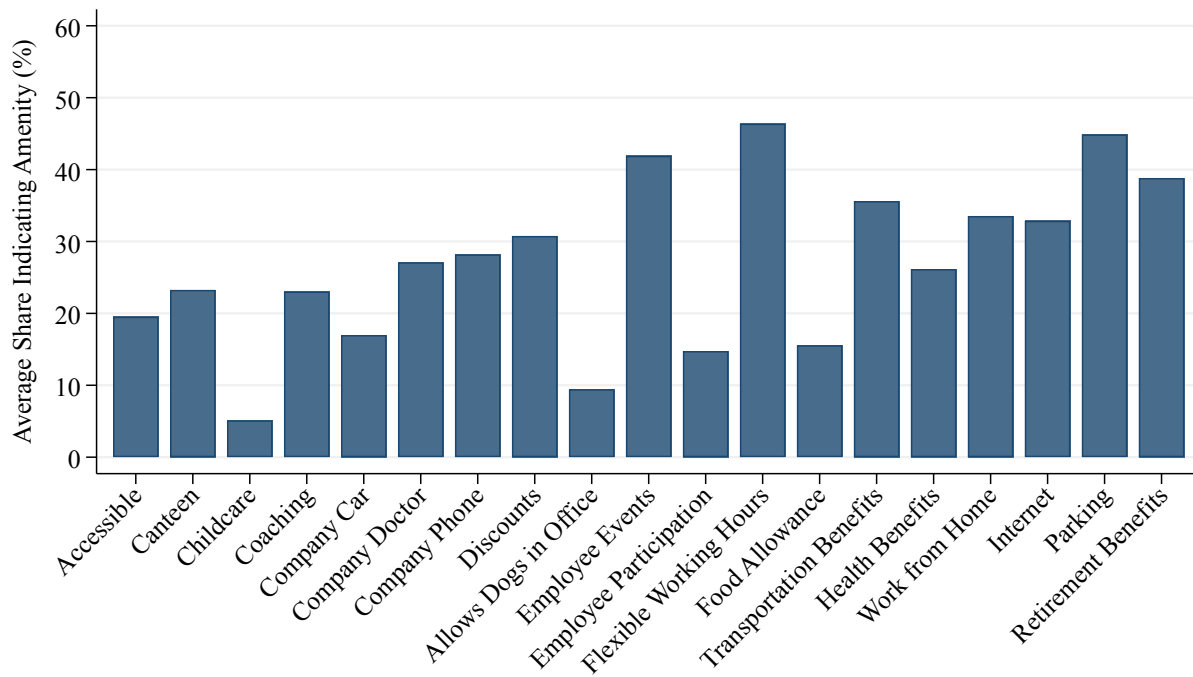
Note: This table reports regressions of administrative pay measures on the firm's average Kununu pay-satisfaction rating, for worker-provided firms matched to Kununu. The outcome is the firm's mean daily wage (columns 1–2) or median daily wage (columns 3–4) from the BHP, or the firm wage premium (columns 5–6) estimated in the IEB. Panel A weights firms equally; Panel B weights by firm size. Even-numbered columns include sector fixed effects. Standard errors in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

B.6.3 Variation in Employer-Provided Amenities

The platform includes indicators for a wide range of specific amenities, such as flexible working hours, remote work options, childcare services, transportation and food benefits, coaching opportunities, and participation in decision-making. Figure B4 reports the average share of reviewers who indicate each amenity, weighted by the number of reviews per firm. The most commonly reported amenities are flexible working hours, parking, employee events, and retirement benefits. Other amenities—such as access to a company doctor, food allowances, and coaching—are less common but still present in a nontrivial share of firms.²⁹

²⁹In cases where no employees indicated any amenities in their reviews, it is unclear whether the firm truly has none of the amenities or the employees skipped the amenities question despite the firm offering amenities. Because most employees indicate at least one amenity, we assume nonresponse to the amenities question means zero amenities if a firm has three or more reviews. Firms with fewer reviews are excluded if no employee responded to the amenities question.

Figure B4: Averages of Kununu Amenity Responses, Weighted by Number of Reviews



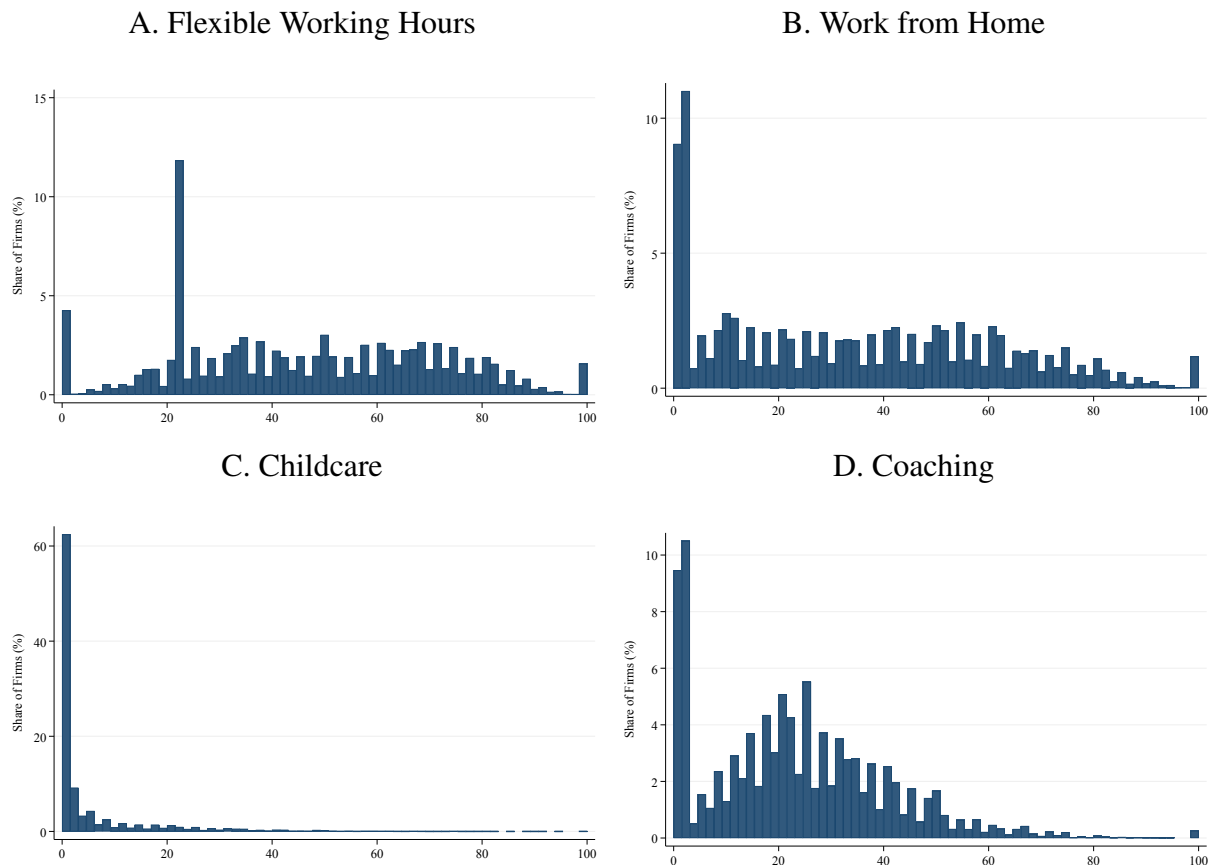
Note: Kununu reports the percentage of reviewers at each firm who indicate the firm offers each amenity. Each bar gives the mean percentage for the indicated amenity, weighted by the number of employee reviews of each firm.

Figure B5 illustrates the share of employees at each firm who report that a given amenity is offered by their employer. We focus on four of the amenities included in the previous graph: flexible working hours, home office, childcare, and coaching. The distributions show mass in intermediate ranges—not concentrated at 0% or 100%. For many firms, employees disagree about whether an amenity is offered, reflecting differences in access or perception.

For example, Panel C shows the distribution of childcare availability. In the review-weighted distribution, most firms have near-zero confirmation rates, consistent with childcare being a relatively rare benefit. This pattern likely reflects limited provision rather than disagreement. Panel D shows a similar pattern for coaching: few firms have unanimous endorsement, but many have partial confirmation across employees.

Even for more common amenities such as flexible working hours and home office options (Panels A and B), we observe a wide spread of confirmation rates across firms. While many firms have a majority of employees reporting availability, relatively few reach full consensus. These patterns are consistent with within-firm variation in the perceived or experienced availability of amenities.

Figure B5: Distribution of Kununu Amenity Responses



Note: Each panel reports the distribution of the amenity indicated. Each bar reports the percentage of firms for which the percentage of reviewers on the x-axis indicated the firm offers the benefit, weighted by the number of employee reviews of each firm.

C Theoretical Appendix

This appendix describes the estimation strategy and shows that it recovers amenity values.

C.1 Setup

The value worker i obtains from working at firm j is:

$$u_{ij} = \beta \log w_{ij} + a_j + \epsilon_{ij} \quad (4)$$

where β is the marginal utility of log wages, w_{ij} is the individual's wage if they work at firm j , a_j is the value of the firm's amenities (in utility units), and ϵ_{ij} is worker i 's idiosyncratic preference for firm j . We assume that the ϵ_{ij} are drawn iid from a Type I extreme value distribution.

Our discrete choice experiments specify proportional changes in the wage, relative to the worker's current log wage ω_i . They specify that the log wage worker i would earn at firm j is:

$$\log w_{ij} = \omega_i + r_{ij},$$

where r_{ij} is the (log) randomized raise associated with firm j and ω_i is the worker's current log wage.

Given the offered wages, worker i prefers firm k to firm j if

$$\beta r_{ij} + a_j + \epsilon_{ij} < \beta r_{ik} + a_k + \epsilon_{ik} \quad (5)$$

$$\epsilon_{ij} - \epsilon_{ik} < \beta(r_{ik} - r_{ij}) + (a_k - a_j). \quad (6)$$

We define the probability a worker prefers firm k to j as:

$$P_{kj} = \Pr[\epsilon_{ij} - \epsilon_{ik} < \beta(r_{ik} - r_{ij}) + (a_k - a_j)] \quad (7)$$

$$= \mathbb{E}_{j,k} \left[\frac{\exp(\beta r_{ik} + a_k)}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)} \right], \quad (8)$$

where the second line comes from the distributional assumptions on ϵ_{ij} , and $\mathbb{E}_{j,k}$ denotes the expectation over the randomized raises (r_{ij}, r_{ik}) offered alongside firms j and k . Following Sorkin (2018), we treat x_k as a latent weight for firm k —the analog of the equilibrium mass of workers at the firm—recovered from the pairwise choice shares. The rank-centrality construction starts from a detailed-balance condition—that the implied flow from firm k to firm j equal the reverse flow—giving

$$x_k P_{jk} = x_j P_{kj} \quad \forall j, k.$$

Summing over j and rearranging—so that only this aggregated condition, not exact pairwise reversibility, is required—we find the solutions $\{x_k\}$ to the linear system

$$x_k = \frac{\sum_j P_{kj} x_j}{\sum_j P_{jk}}. \quad (9)$$

Written in terms of the choice probabilities, each detailed-balance condition reads

$$\mathbb{E}_{j,k} \left[\frac{\exp(\beta r_{ij} + a_j)}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)} \right] x_k = \mathbb{E}_{j,k} \left[\frac{\exp(\beta r_{ik} + a_k)}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)} \right] x_j. \quad (10)$$

The share of workers who prefer j to k (relative to the share of workers who prefer k to j) is larger whenever the gap in utilities is larger. We obtain these $\{x_j\}$ by finding the fixed point of the above linear system.

C.2 Randomization Allows Direct Identification of a_j

The rankings implied by our estimation identify amenity values (not overall values as in Sorkin, 2018).

We observe workers' choices based on comparisons of $\beta r_{ij} + a_j$, not a_j . Let

$$x_k \propto \exp(a_k + \xi_k) \quad (11)$$

where ξ_k is a term that captures potential deviations in what we estimate from a_k . Then the detailed-balance condition becomes

$$\begin{aligned} \mathbb{E}_{j,k} \left[\frac{\exp(\beta r_{ij} + a_j)}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)} \right] \exp(a_k + \xi_k) \\ = \mathbb{E}_{j,k} \left[\frac{\exp(\beta r_{ik} + a_k)}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)} \right] \exp(a_j + \xi_j). \end{aligned} \quad (12)$$

Dividing both sides by $\exp(a_j) \exp(a_k)$ gives

$$\begin{aligned} \mathbb{E}_{j,k} \left[\frac{\exp(\beta r_{ij})}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)} \right] \exp(\xi_k) \\ = \mathbb{E}_{j,k} \left[\frac{\exp(\beta r_{ik})}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)} \right] \exp(\xi_j), \end{aligned} \quad (13)$$

so that

$$\frac{\exp(\xi_k)}{\exp(\xi_j)} = \frac{\mathbb{E}_{j,k} \left[\frac{\exp(\beta r_{ik})}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)} \right]}{\mathbb{E}_{j,k} \left[\frac{\exp(\beta r_{ij})}{\exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)} \right]}. \quad (14)$$

Both expectations share the same random denominator $D \equiv \exp(\beta r_{ik} + a_k) + \exp(\beta r_{ij} + a_j)$. After we apply the approximation $\mathbb{E}[X/Y] \approx \mathbb{E}[X]/\mathbb{E}[Y]$ to the numerator and to the denominator, the common factor $\mathbb{E}[D]$ cancels. This approximation is accurate in our design because the randomized wage component is small enough that its distortion of the nonlinear averaging is limited, as we verify directly in the simulation below. Under this approximation,

$$\frac{\exp(\xi_k)}{\exp(\xi_j)} \approx \frac{\mathbb{E}_k [\exp(\beta r_{ik})]}{\mathbb{E}_j [\exp(\beta r_{ij})]} \approx 1. \quad (15)$$

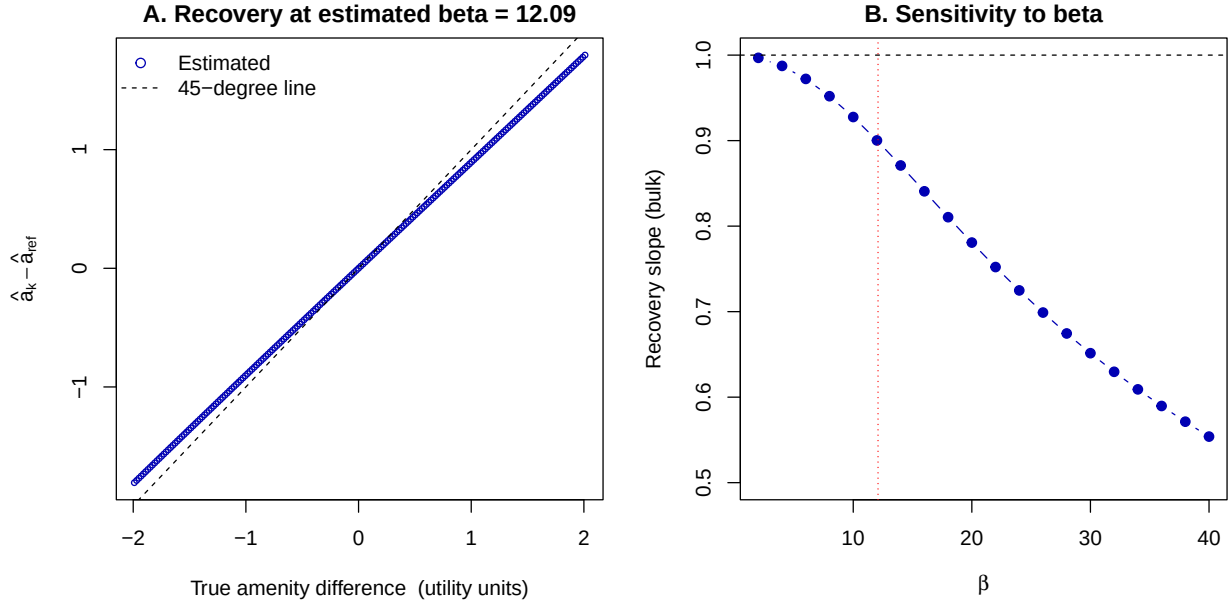
The final step treats r_{ik} and r_{ij} as draws from a common distribution. In the fielded design, we randomized raises conditional on listing position, and firms appear in the order workers named them. A firm’s raise distribution therefore depends on the positions in which workers listed it. Firms named by many workers appear in different positions across lists, which mixes the position-specific distributions. Section 5.1 addresses the residual concern directly: controlling for each firm’s mean randomized raise leaves the estimates essentially unchanged. Therefore

$$x_k \propto \exp(a_k). \quad (16)$$

This is an approximation, not an exact identity. We quantify the distortion from this approximation directly. Panel A of Figure C1 plots estimated against true amenity differences when the wage gap is drawn from its experimental distribution, β is fixed at its estimated value, and amenities span the four-utility-unit range we estimate. This isolates the identification (asymptotic) bias from the wage randomization; it is distinct from finite-sample noise, which we address separately with the split-sample correction. The estimator attenuates amenity differences mildly: the slope is approximately 0.90 over the empirically relevant range, the correlation between estimated and true differences exceeds 0.999, and firm rankings are preserved exactly in this simulation. Because the distortion is close to a linear rescaling, it does not affect the correlation between amenity values and firm wage premia, which is scale-invariant; our central result is unaffected. The attenuation bears only on the money-metric magnitude of amenity dispersion, which it slightly understates, so the true dispersion of amenities is if anything modestly larger than we report.

Panel B reports how this varies with β . The recovery slope is near one for small β and declines smoothly as β grows, because a larger β scales up the experimental wage variation relative to amenity gaps. At our estimated β the slope is 0.90; it would take a value of β near 20—two-thirds above our estimate—to push it below 0.8. The estimator is thus well inside the region where it recovers amenity values accurately.

Figure C1: Accuracy of Estimated Amenity Differences



Note: Panel A plots, for each firm, the amenity difference recovered by the eigenvector estimator (vertical axis) against the true difference (horizontal axis), in utility units relative to a reference firm. The wage gap is drawn from its experimental distribution, β is set to its estimated value (Table A3), and true amenities span the four-utility-unit range we estimate; the dashed line is the 45° line. Panel B plots the recovery slope against β , with the vertical line marking our estimated β . The exercise isolates the asymptotic bias from the wage randomization, separately from finite-sample noise. The estimator attenuates amenity differences mildly at the estimated β and preserves firm rankings exactly.

C.3 Validation of Methodology Using Researcher-Provided Firms

An advantage of the researcher-provided firm module is that we included a relatively small number of firms (30). We can therefore estimate non-wage valuations in two different ways. First, we can follow the standard approach in the DCE literature: we can fit a logit model to workers' preferences and divide by the coefficient on wages. Each pairwise comparison between firms k and j is a binary choice:

$$\Pr(\text{prefer } k \text{ to } j) = \Lambda(\beta(r_{ik} - r_{ij}) + (a_k - a_j)), \quad (17)$$

where Λ is the logistic function.

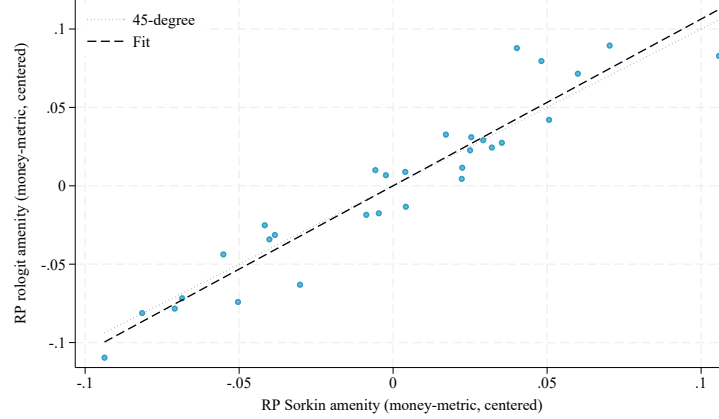
Because we offered workers a menu of firms and asked them to rank them, we fit a rank-ordered logit model to workers' choices. This uses the full ranking each worker provides, rather than a single pairwise choice. The model includes the randomized raise, as well as dummies for each of the firms. We calculate non-wage valuations by dividing each dummy coefficient by the coefficient on the randomized raise.

We can also estimate valuations using the eigenvector approach from Sorkin (2018). Specifically, for this module we construct the 30×30 matrix, in which each entry is a share of workers choosing the row firm over the column firm. We then follow the estimation procedure from the previous section: computing the principal eigenvector, taking logs, and dividing by the wage coef-

ficient.

The two estimators agree. Their correlation is 0.95. A regression of the exact estimates on the eigenvector estimates gives a slope of 1.06(s.e. 0.07), and we cannot reject a slope of one ($p = 0.34$, 30 firms). Figure C2 plots the two against each other.

Figure C2: Exact and Eigenvector Estimates Agree: Researcher-Provided Firms



Note: Each point is one of the 30 researcher-provided firms. The rank-ordered logit and the eigenvector use different base firms—the logit omits one firm, while the eigenvector is normalized to a separate reference firm—so we center each set of valuations on its own mean before plotting. The origin is therefore the mean of each estimator, not a particular firm; both axes are in money-metric (log-wage-equivalent) units. The two estimators agree closely: their correlation is 0.95, and a regression of the exact estimates on the eigenvector estimates gives a slope of 1.06 (s.e. 0.07), which we cannot reject as equal to one ($p = 0.34$).

C.4 Obtaining Money-Metric Values

We convert the eigenvector into money-metric amenity values in two steps: a normalization that fixes the scale of the eigenvector, and a rescaling that places amenities in the same units as wages.

Step 1: Normalization. The eigenvector recovers a term proportional to $\exp(a_k)$ up to the approximation of Appendix C.2,

$$x_k \propto \exp(a_k), \quad (18)$$

so $c \cdot x_k = \exp(a_k)$ for some constant c , and

$$\log(x_k) = a_k - \log(c). \quad (19)$$

The constant $\log(c)$ is not identified: amenity values are pinned down only relative to a reference firm. To see this, note that workers' decision rule,

$$\beta(r_{ij} + \omega_i) + a_j + \epsilon_{ij} < \beta(r_{ik} + \omega_i) + a_k + \epsilon_{ik}, \quad (20)$$

is unchanged if a_j is replaced by $a_j - \log(c)$ at every firm,

$$\beta(r_{ij} + \omega_i) + a_j - \log(c) + \epsilon_{ij} < \beta(r_{ik} + \omega_i) + a_k - \log(c) + \epsilon_{ik},$$

for any c . We fix the scale by normalizing to a single base firm, chosen to have an amenity estimate near the median and held constant across all specifications; cross-sex level comparisons additionally change the normalization, as described below. The log of the normalized eigenvector then gives a_j in the units of the utility function, measured relative to the base firm.

Step 2: Scaling to Money-Metric Units. We place amenities in the same units as wages by dividing by β , the marginal utility of log wages: the money-metric (log-wage-equivalent) value is a_j/β . We estimate β from the structural logit coefficient on the experimental wage difference (Table A3). We apply this scaling wherever amenities are compared with wages—the wage–amenity relationship and the inequality variance ratio (Section 5.1), the classification of job-to-job moves (Section 5.2), and the gender gap (Section 5.3). For cross-group comparisons we divide by a common (pooled) β , so that men’s and women’s amenities are stated in the same wage-equivalent units.

Cross-group levels. Within a group, differences in money-metric amenities are fully identified, because the base-firm constant cancels. Cross-group level comparisons inherit the unidentified normalization: in sex-specific models the estimated gender gap at firm k reflects $(a_k^F - a_k^M) - (a_{\text{base}}^F - a_{\text{base}}^M)$, where the second term is not identified. We address this in Appendix C.5, which writes it as $c^F - c^M$.

C.5 Normalization in Cross-Group Comparisons

The Identification Problem. The discrete choice experiment identifies group-specific amenity valuations $\hat{a}_k^g$ only up to an additive constant within each group $g \in \{M, F\}$:

$$\hat{a}_k^g = a_k^g - c^g, \quad (21)$$

where a_k^g is the true amenity value and c^g is a group-specific constant that pairwise comparisons cannot identify.

1. *Within-group objects are fully identified.* $\hat{a}_j^g - \hat{a}_k^g = a_j^g - a_k^g$ for any two firms j and k ; the constant cancels. Rank orderings, within-group variances, and regressions of $\hat{a}_j^g$ on firm-level covariates are normalization-free.
2. *Cross-group level comparisons require a normalization.* The estimated gender gap at firm k is

$$\hat{a}_k^F - \hat{a}_k^M = (a_k^F - a_k^M) - (c^F - c^M). \quad (22)$$

Any claim about the level of women’s amenities relative to men’s therefore depends on the chosen value of $c^F - c^M$.

Normalization Strategies. A normalization pins down c^g . We consider five alternatives:

1. **Services-sector (baseline).** Set the employment-weighted mean of $\hat{a}_k^g$ among professional services firms to zero within each group; equivalently, this sets c^g to the (unobserved)

employment-weighted mean of the true amenities in services,

$$c^g = \bar{a}_{\text{serv}}^g \equiv \sum_{j \in \text{serv}} s_j a_j^g,$$

where s_j is firm j 's employment share within services. This assumes $\bar{a}_{\text{serv}}^F = \bar{a}_{\text{serv}}^M$: men and women derive, on average, the same amenity value from services firms.

2. **Overall employment-weighted mean.** Set the employment-weighted mean over all firms to zero within each group: $c^g = \bar{a}_{\text{all}}^g$. This assumes men and women derive the same average amenity value across the full economy.
3. **Manufacturing.** Set the employment-weighted mean in manufacturing to zero: $c^g = \bar{a}_{\text{manuf}}^g$.
4. **Median-firm.** Set the amenity value of the median firm (by employment) to zero within each group.
5. **Overlap-set restriction.** Restrict the sample to the intersection of the male and female connected sets, retaining the services-sector normalization from alternative 1. This is a sample restriction rather than a separate normalization; it tests whether the results are driven by firms that appear in only one sex's connected set.

Direction of Bias under the Services Normalization. The services normalization assumes $\bar{a}_{\text{serv}}^F = \bar{a}_{\text{serv}}^M$. If this assumption is violated, substituting $c^g = \bar{a}_{\text{serv}}^g$ into equation (22) gives

$$\hat{a}_k^F - \hat{a}_k^M = (a_k^F - a_k^M) - (\bar{a}_{\text{serv}}^F - \bar{a}_{\text{serv}}^M). \quad (23)$$

Within-sex gradients (e.g., the relationship between a woman's wage premium and her amenity valuation) are normalization-free: the constant c^g cancels within a single group.

Cross-sex level comparisons—including the coefficient on a Female indicator in a regression of $\hat{a}_j^{g(i)}$ on worker demographics—depend on the normalization. Absent other controls, the Female coefficient captures the difference in group means,

$$\theta_1 = (\bar{a}_{\text{women's firms}}^F - \bar{a}_{\text{serv}}^F) - (\bar{a}_{\text{men's firms}}^M - \bar{a}_{\text{serv}}^M),$$

which shifts by exactly $-(\bar{a}_{\text{serv}}^F - \bar{a}_{\text{serv}}^M)$ under the services normalization. With additional controls the group averages become covariate-adjusted means, but the normalization shift is unchanged, since adding a constant to every female value moves the Female coefficient by that same constant. The positive coefficients in Panel B of Table 6 therefore depend on the normalization; we verify their robustness across alternatives in Table A6.

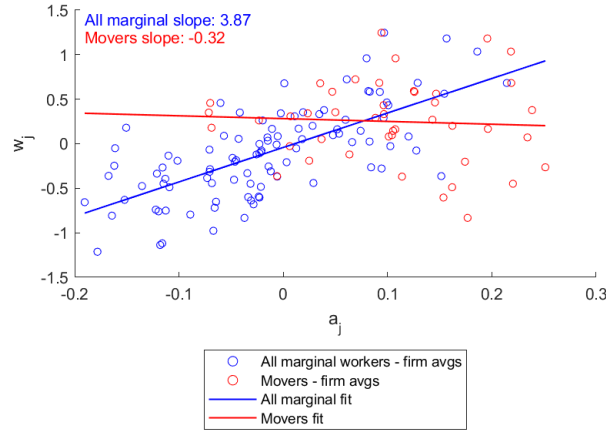
C.6 Estimating Valuations Based on Movers

We fielded the survey to a broad sample of workers rather than to job-to-job movers because the theoretical literature shows that—even when the correlation between wages and amenities is positive—focusing on movers can generate spurious negative correlations (Bonhomme and Jolivet, 2009).

In simulations that assume noiseless observation of workers' firm-specific valuations, this bias

is large.³⁰ In reasonable specifications, focusing only on movers changes the correlation substantially, flipping its sign from the large positive value we assume to a negative one. Figure C3 illustrates the bias for a representative parameterization: across all firms (blue), the slope of the firm wage premium on the firm amenity value is strongly positive; when the same data are restricted to realized job-to-job movers (red), the slope is slightly negative. The results in Section 5.2 confirm that many workers make trade-offs between wages and amenities when switching firms.

Figure C3: Bias from Estimating Wage–Amenity Correlations Using Movers Only



Note: This figure presents results from a simulation in which firms differ in both wage premia ψ_j and amenity values a_j , with a positive population-level correlation between the two. Blue circles plot each firm's true wage premium and amenity value; red circles are firm-level averages of the simulated valuations of realized job movers only. The blue line is the OLS fit across the true firm values; the red line is the fit across the mover-based averages. Conditioning on movers flips the sign of the slope, illustrating the selection mechanism formalized in Bonhomme and Jolivet (2009).

³⁰The assumption that we can observe point estimates of workers' firm-specific valuations without error is strong. Stated valuation approaches are known to yield estimates that are off by a factor of ten (Rodemeier, 2026). Validated approaches, such as discrete choice experiments, do not yield point estimates for each worker (see, e.g., Mas and Pallais, 2017; Wiswall and Zafar, 2018).

D Additional Results and Analysis

D.1 Additional Exercises Validating the Consideration Sets

Table D1 shows that there is a positive relationship between the sector of provided firms and those associated with previous transitions. For instance, among workers in the manufacturing sector, 39% stay in that sector when they make a transition (Panel A); 38% of manufacturing workers in our sample provided a manufacturing firm. There is a positive correlation between the share of workers making a specific sector-to-sector transition and the share of workers providing firms that would imply the same sector-to-sector move (cross-cell correlation 0.57; 0.36 excluding within-sector cells).

Table D1: Stated and Revealed Transitions: Sector

A. Revealed Transitions								
		Destination Firm						
		Manufacturing	Retail	Accommodations and Food Services	Information	Financial	Professional	Other
		(1)	(2)	(3)	(4)	(5)	(6)	(7)
Origin Firm	Manufacturing	39%	12%	2%	2%	1%	7%	37%
	Retail	15%	37%	3%	3%	1%	6%	36%
	Accommodations and Food Services	10%	11%	33%	2%	1%	4%	39%
	Information	9%	9%	1%	39%	2%	14%	25%
	Financial	6%	7%	1%	6%	44%	11%	25%
	Professional	14%	10%	2%	8%	3%	33%	31%
	Other	12%	9%	2%	2%	1%	5%	70%
B. Stated Transitions								
		Stated Firm						
		Manufacturing	Retail	Accommodations and Food Services	Information	Financial	Professional	Other
		(1)	(2)	(3)	(4)	(5)	(6)	(7)
Current Firm	Manufacturing	38%	10%	-	12%	8%	17%	16%
	Retail	19%	24%	-	18%	3%	17%	20%
	Accommodations and Food Services	-	-	-	-	-	-	-
	Information	24%	17%	-	22%	16%	8%	13%
	Financial	25%	12%	-	7%	30%	17%	9%
	Professional	43%	11%	-	13%	8%	13%	12%
	Other	33%	12%	-	8%	4%	12%	31%

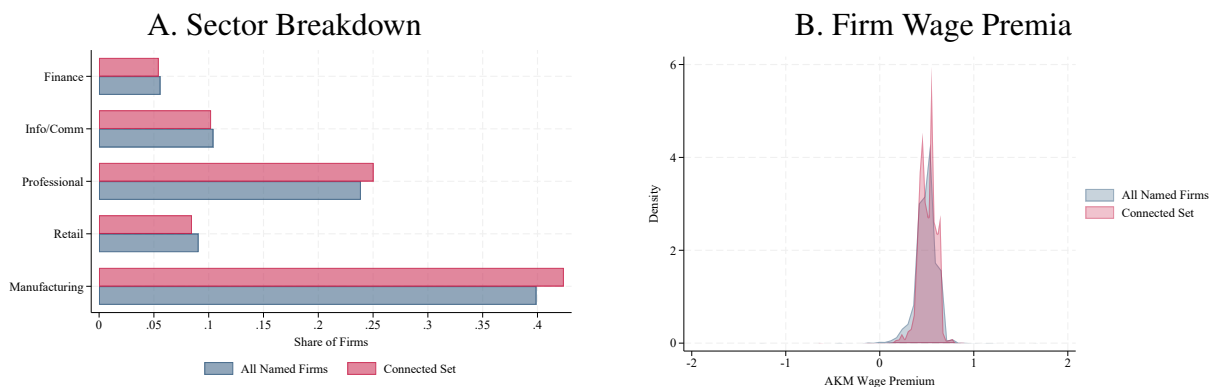
Note: This table examines the characteristics of workers' destination firms, by the characteristics of their origin firms. Panel A uses data on observed mobility and computes statistics at the establishment level; Panel B uses data on workers' provided firms and computes statistics at the firm level. Apart from the worker-provided firms, we do not have a mapping from establishments to firms in the administrative data, so the observed mobility data in Panel A are necessarily reported at the establishment level. Small cells are suppressed for privacy.

D.2 Analysis of the Strongly Connected Set

We can only provide estimates of valuations for firms in the strongly connected set. While the previous section showed that workers provided firms which match, in sectoral composition, the firms that workers from their firm have previously moved to, the strongly connected set is naturally smaller. In particular, it excludes many firms that are named by only a single worker. A natural concern is that selection into the strongly connected set could be correlated with firm characteristics, potentially biasing our estimates of the relationship between wages and amenities.

Appendix Figure D1 shows that the sector distribution and wage premium distribution of firms in the strongly connected set are similar to those among all named firms; in particular, strongly connected set firms are not concentrated among high- or low-wage employers. These comparisons rest on observables; we cannot directly rule out selection into the connected set on unobserved amenities, though the minimum-mentions robustness checks (Appendix D.4) limit its scope.

Figure D1: Provided Firms vs. Strongly Connected Set



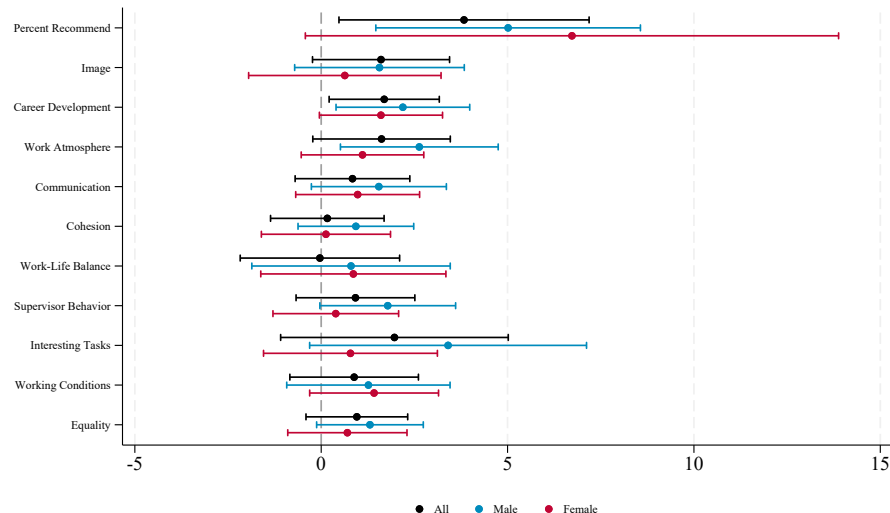
Note: This figure compares the sector distribution (Panel A) and firm wage premia distribution (Panel B) of workers' provided firms to those of the subset in the strongly connected set.

D.3 Amenity Valuations and External Measures of Job Quality

Our primary analysis estimates the overall value of the bundle of non-wage amenities provided by each firm. In this section, we correlate our firm-level amenity valuations with external measures of job quality from Kununu, a leading employer review platform in Germany (described in Appendix B.6).

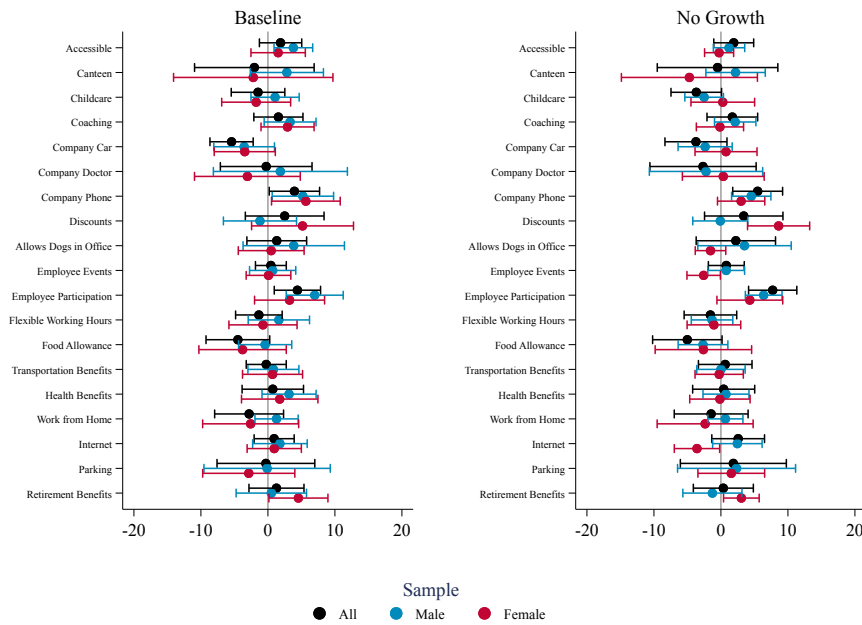
Figure D2 plots coefficients from regressions of standardized Kununu scores on our baseline amenity valuations for all workers, as well as separately for men and women. Firms with higher amenity valuations are significantly more likely to be recommended by their employees, and across nearly all subcategories—including Work Atmosphere, Career Development, and Supervisor Behavior—there is a positive correlation with our valuations. Figure D3 correlates our valuations with amenity provision percentages for specific amenities reported on Kununu. Our index is positively associated with several specific benefits, such as Employee Participation, Coaching, and Company Phones. Because we did not cross-randomize individual amenities (Section 3.4), we read these correlations as validation that our estimates capture real workplace differences rather than as a decomposition of the value of specific attributes. These online reports are also themselves noisy. For the specific-amenity measures in particular, workers at the same firm often disagree about whether it provides a given benefit, which attenuates any correlation with our valuations. Some of this disagreement may not be noise: whether a worker reports that a firm offers an amenity may itself reflect whether she values it.

Figure D2: Correlates of Firm Amenity Values



Note: This figure reports results from regressions of Kununu scores on our baseline amenity values for all workers (black), male workers (blue), and female workers (red). The dots are the coefficient on the amenity valuation (in log-wage-equivalent units); whiskers show 95% confidence intervals. Kununu scores are standardized to have unit standard deviations. Regressions are weighted by imputed firm size (see Appendix B.3).

Figure D3: Correlates of Firm Amenity Values: Kununu Amenities



Note: This figure reports results from regressions of Kununu amenity provision percentages on our amenity values for all workers (black), male workers (blue), and female workers (red). The left panel uses baseline amenity values; the right panel uses alternative valuations that hold wage-growth expectations constant (see Section 4.2). The dots are the coefficient on the amenity valuation (in log-wage-equivalent units); whiskers show 95% confidence intervals. Regressions are weighted by imputed firm size (see Appendix B.3).

D.4 Additional Robustness Checks

This section examines the robustness of our baseline results to alternative weighting schemes and other specification choices. The results in the main text weight by observed firm size. Here we also present specifications that weight by imputed firm size (which allows us to include firms not linked to the BHP establishment database; see Appendix B.3 for details on the imputation procedure) and by the number of times a firm is mentioned in the survey; weighting observations by their sample size is a common approach to dealing with measurement error.

D.4.1 Descriptive Statistics Reported in Section 2

Table D2: Characteristics of Provided Firms: Alternative Weighting Schemes

	All Firms				Connected Set			
	Unweighted	Firm Size	Mentions	Firm Size (Imputed)	Unweighted	Firm Size	Mentions	Firm Size (Imputed)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<u>Linkages</u>								
To Wage Premia	95%	100%	100%	100%	97%	100%	100%	100%
To BHP	97%	100%	100%	100%	98%	100%	100%	100%
Size	29682	149557	76253	149565	34701	149891	76316	149891
(Std. Deviation)	(59653)	(93276)	(78671)	(93273)	(63227)	(93118)	(78674)	(93118)
Size (Imputed)	22733	149565	69887	140018	28385	149891	69987	140730
(Std. Deviation)	(51637)	(93273)	(75381)	(94320)	(56473)	(93118)	(75390)	(94036)
<u>Size (Binned)</u>								
1-10 Workers	3%	0%	0%	0%	2%	0%	0%	0%
11-50 Workers	4%	0%	1%	0%	3%	0%	1%	0%
51-200 Workers	5%	0%	0%	0%	3%	0%	0%	0%
201-1000 Workers	33%	0%	2%	1%	26%	0%	2%	1%
1001-10000 Workers	20%	3%	8%	3%	24%	3%	8%	3%
>10000 Workers	32%	96%	88%	96%	42%	97%	88%	96%
Firm Wage Premium	0.50	0.48	0.51	0.48	0.51	0.48	0.51	0.48
(Std. Deviation)	0.11	0.07	0.08	0.07	0.09	0.06	0.08	0.06
Firm Wage Premium (CHH)	0.16	0.06	0.15	0.06	0.16	0.05	0.15	0.05
(Std. Deviation)	0.22	0.16	0.16	0.16	0.21	0.16	0.16	0.16
<u>Sector</u>								
Manufacturing	40%	44%	58%	43%	42%	44%	58%	43%
Retail	9%	0%	2%	1%	8%	0%	2%	1%
Professional	24%	53%	29%	52%	25%	53%	29%	52%
Information	10%	1%	7%	2%	10%	1%	7%	2%
Finance	6%	0%	1%	0%	5%	0%	1%	0%
Number of Firms	3899	1941	3587	3899	949	603	949	949

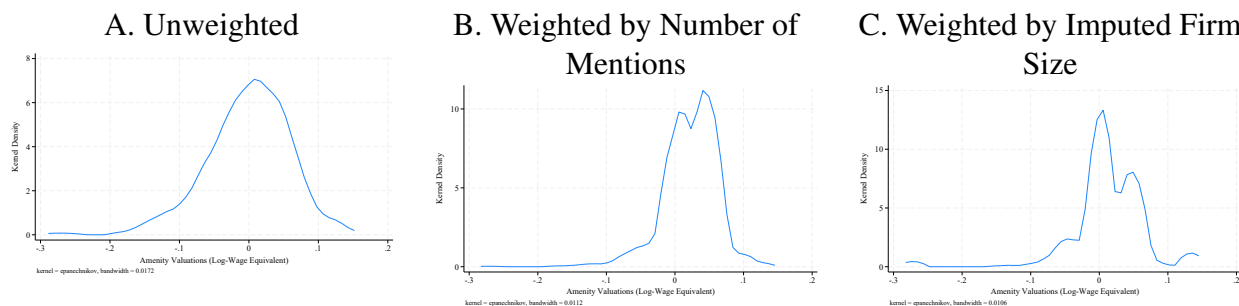
Note: This table describes the firms that workers provided (Columns 1–4) and the subset of these firms in the strongly connected set (remaining columns). Each column indicates how the statistics are weighted. Appendix B.3 describes how we imputed firm size.

D.4.2 Results Reported in Section 4

In Section 4 we describe the distribution of amenity values, as well as how these amenity values agree (or do not agree) across groups. Here we show that these core findings are invariant to weighting scheme. In particular, amenity valuations are highly dispersed; groups disagree about how to value firms; yet the single-index model still has predictive power. We replicate Figure 2 (distribution of amenity values), Table 3 (between-group correlations), and Figure 3 (relationship

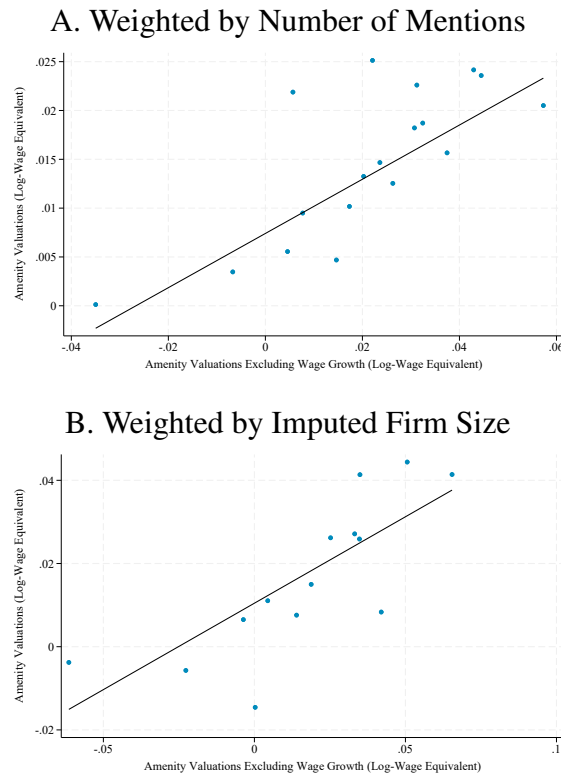
between valuations with and without wage growth). We do not replicate the log firm size regressions in Table 4, as those are naturally unweighted. The Kununu regressions in that table are weighted by firm size and could in principle be replicated; we omit them for brevity. Figure 4 (relationship between male and female valuations) is visually similar under all weighting schemes (Figure D6).

Figure D4: Distribution of Amenity Values: Alternative Weighting Schemes



Note: This figure replicates Figure 2 (amenity values in log-wage-equivalent units) using alternative weighting schemes. Panel A is unweighted; Panel B weights by the number of times the firm is mentioned by workers in the survey; Panel C weights by imputed firm size, which allows firms not linked to the BHP to enter. The baseline figure weights by firm size.

Figure D5: Relationship Between Valuations with and without Wage Growth: Alternative Weighting Schemes



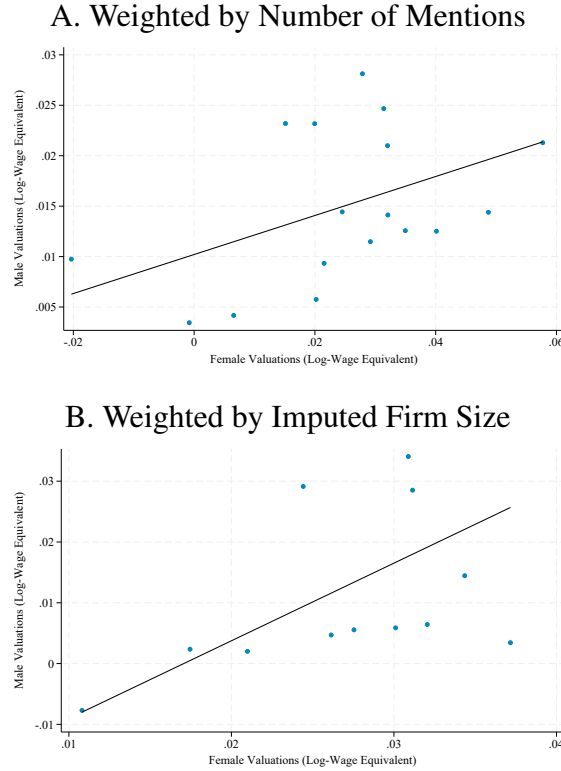
Note: This figure compares the baseline estimates of amenity valuations (in log-wage-equivalent units) to those we obtain when we re-run the model using a discrete choice experiment where we told workers to assume that career growth—future raises and promotions—would be identical across all options. Panel A weights by the number of times firms are mentioned. Panel B weights by imputed firm size (see Appendix B.3). The baseline estimates are in Figure 3.

Table D3: Between-Group Correlations: Weighting and Minimum-Mentions Robustness

	Firm Size (1)	Unweighted (2)	Mentions (3)	≥ 5 Mentions (4)	≥ 10 Mentions (5)
<u>Baseline</u>					
Male vs. Female	0.266	0.104	0.300	0.238	0.282
College vs. No College	0.231	0.090	0.298	0.236	0.314
<u>No Growth</u>					
Male vs. Female	0.294	0.207	0.355	0.288	0.288
College vs. No College	0.411	0.240	0.424	0.407	0.408
Observations	153	209	209	147	101

Note: Pearson correlations between male and female valuations and between college and non-college valuations, for baseline and no-growth amenity values. Columns vary the weighting scheme—weighting by firm size or by the number of mentions, or leaving firms unweighted—or restrict the sample to firms mentioned at least five or at least 10 times. The main text weights by firm size.

Figure D6: Relationship Between Male and Female Valuations: Alternative Weighting Schemes

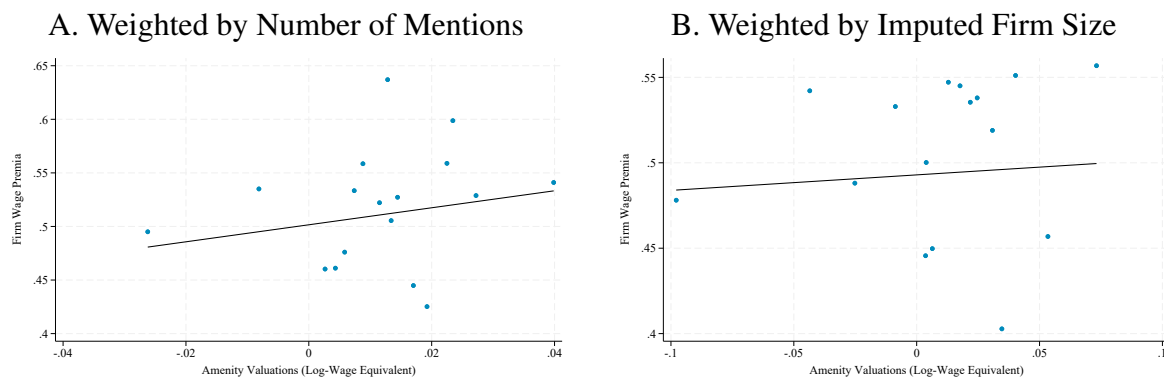


Note: This figure replicates Figure 4 (the relationship between male and female amenity valuations, in log-wage-equivalent units) using alternative weighting schemes. Panel A weights by the number of times firms are mentioned. Panel B weights by imputed firm size (see Appendix B.3). The baseline estimates are in Figure 4.

D.4.3 Results Reported in Section 5

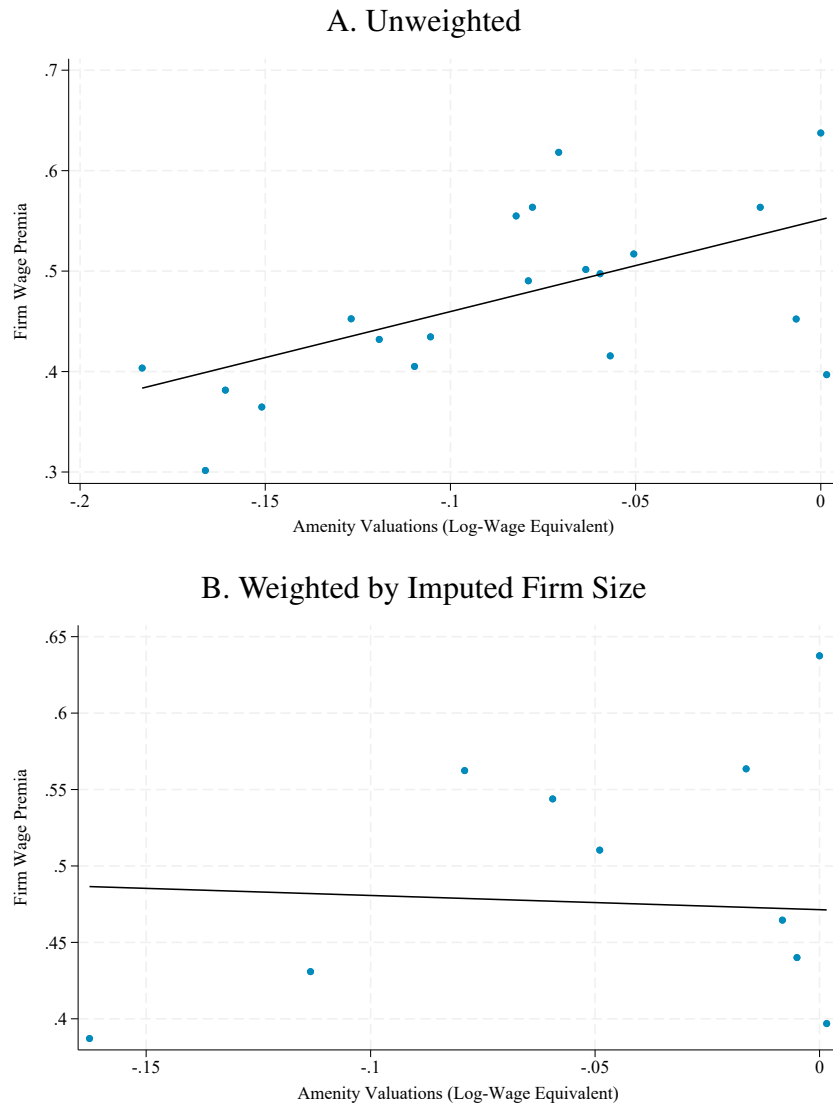
In Section 5 we examine the relationship between amenities, wages, and inequality. Here we replicate Figure 5 (relationship between wage premia and amenity valuations) and Figure 6 (amenities, wages, and values by sector), showing that the core result—valuations do not decline with wage premia, the covariance behind the widening reported in Section 5.1—is robust to alternative weighting schemes. We do not replicate Table 5 (classifying job-to-job moves) or Table 6 (amenities and the gender value gap), as those regressions are conducted at the worker level and are not weighted by firm size.

Figure D7: Relationship Between Wage Premia and Amenity Valuations: Alternative Weighting Schemes



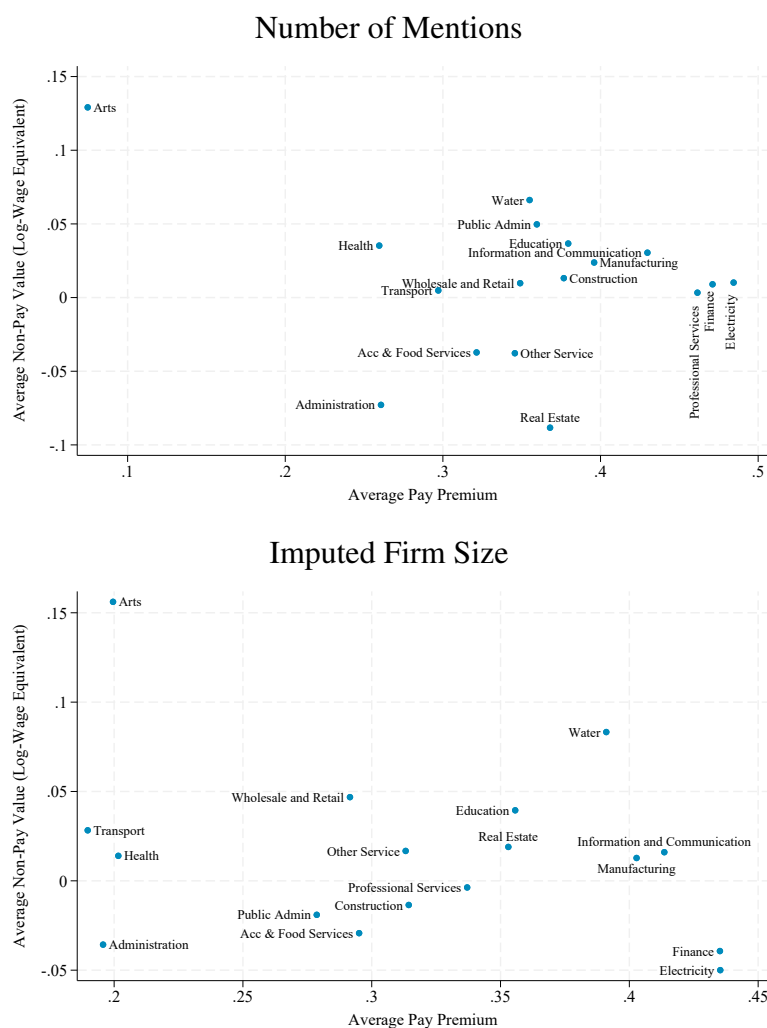
Note: This figure shows binned scatterplots of the relationship between firm wage premia (y-axis) and amenity values in log-wage-equivalent units (x-axis) for firms in our strongly connected set. Panel A weights by the number of times firms are mentioned. Panel B weights by imputed firm size (see Appendix B.3). The baseline estimates are in Figure 5.

Figure D8: Relationship Between Wage Premia and Amenity Valuations: Researcher-Provided Firms



Note: This figure repeats Figure 5 for the researcher-provided firms, using the rank-ordered logit amenity estimates (Appendix C.3) rather than the eigenvector. Each panel is a binned scatterplot of firm wage premia (y-axis) against amenity values in log-wage-equivalent units (x-axis). Panel B weights by imputed firm size. The estimates are positively related when firms are weighted equally and unrelated once weighted by imputed firm size.

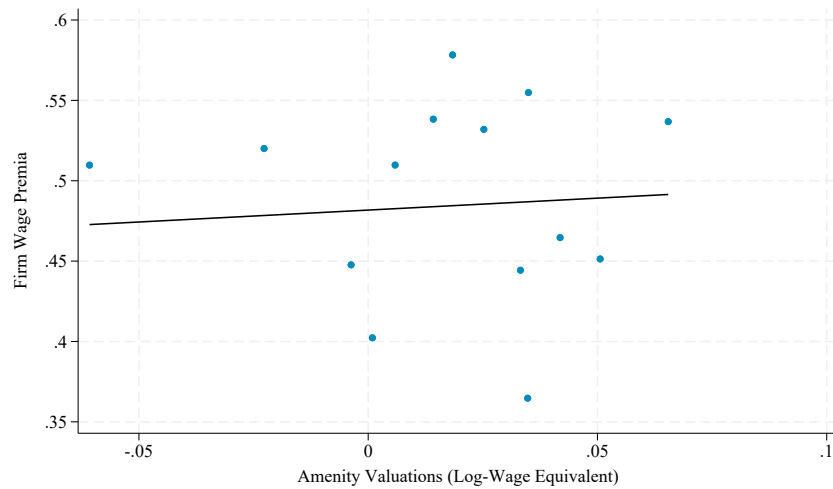
Figure D9: Amenities, Wages, and Values by Sector: Alternative Weighting Schemes



Note: This figure shows the average wage premium (x-axis) and average non-wage amenity value in log-wage-equivalent units (y-axis) by major sector. The top panel weights by the number of times each firm was mentioned. The bottom panel weights by imputed firm size (see Appendix B.3). The baseline estimates are in Figure 6.

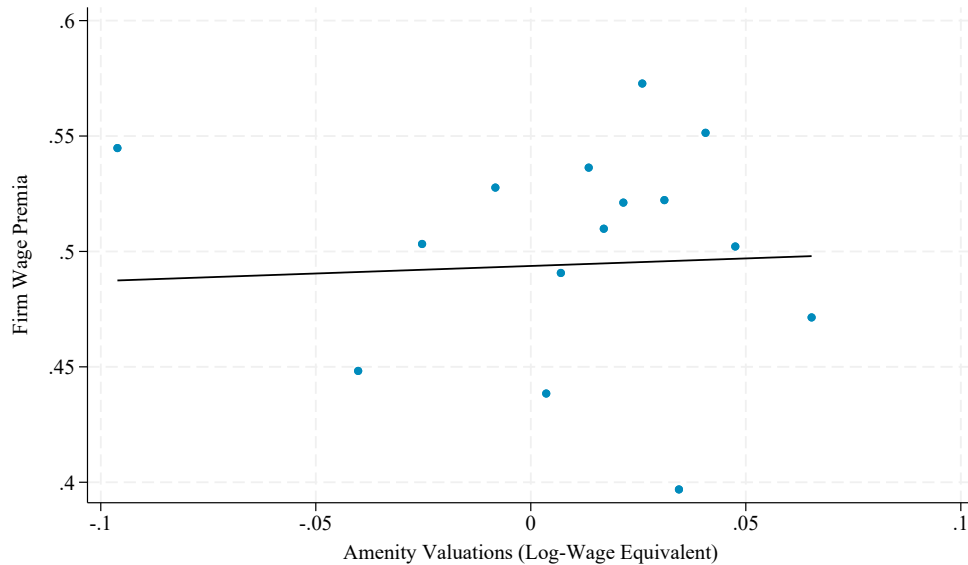
Figure D11 replicates the baseline wage–amenity binscatter (Figure 5) using only amenity estimates from Wave 1 of the survey, in which randomized raises ranged from 5% to 20% with no pay cuts offered. The near-zero slope confirms that the employment-weighted orthogonality does not depend on the introduction of pay cuts in the follow-up wave and that valuations are stable across waves.

Figure D10: Relationship Between Wage Premia and Amenity Valuations: Excluding Wage Growth



Note: This figure replicates the weighted panel of Figure 5 using amenity valuations (in log-wage-equivalent units) estimated under the assumption that career growth is identical across all options. Firms are weighted by employment. Wage premia are from Bellmann et al. (2020), as in Figure 5; results are similar using premia estimated from our mobility sample (Appendix B.4). The relationship between firm wage premia and amenity valuations remains approximately zero, indicating that differences in dynamic expectations do not drive the employment-weighted orthogonality.

Figure D11: Relationship Between Wage Premia and Amenity Valuations: Wave 1 Only



Note: This figure replicates the split-sample binscatter in Figure 5, Panel B, using only amenity estimates from Wave 1 of the survey. In Wave 1, randomized raises ranged from 5% to 20%; no pay cuts were offered. Firms are weighted by employment.

D.4.4 Mobility Results Reported in Section 5.2

Figure D13 presents binscatters of the wage–amenity trade-off among job movers, separately by sex. We plot the change in sex-specific amenity valuations against the change in the firm wage premium. Because the amenity change is the dependent variable, its measurement error inflates the standard errors rather than biasing the slope, and the firm wage premium is externally estimated and precise. On average, workers who move to higher-premium firms experience a decline in amenities. The fitted slope is steeper for women, though the difference is imprecisely estimated.

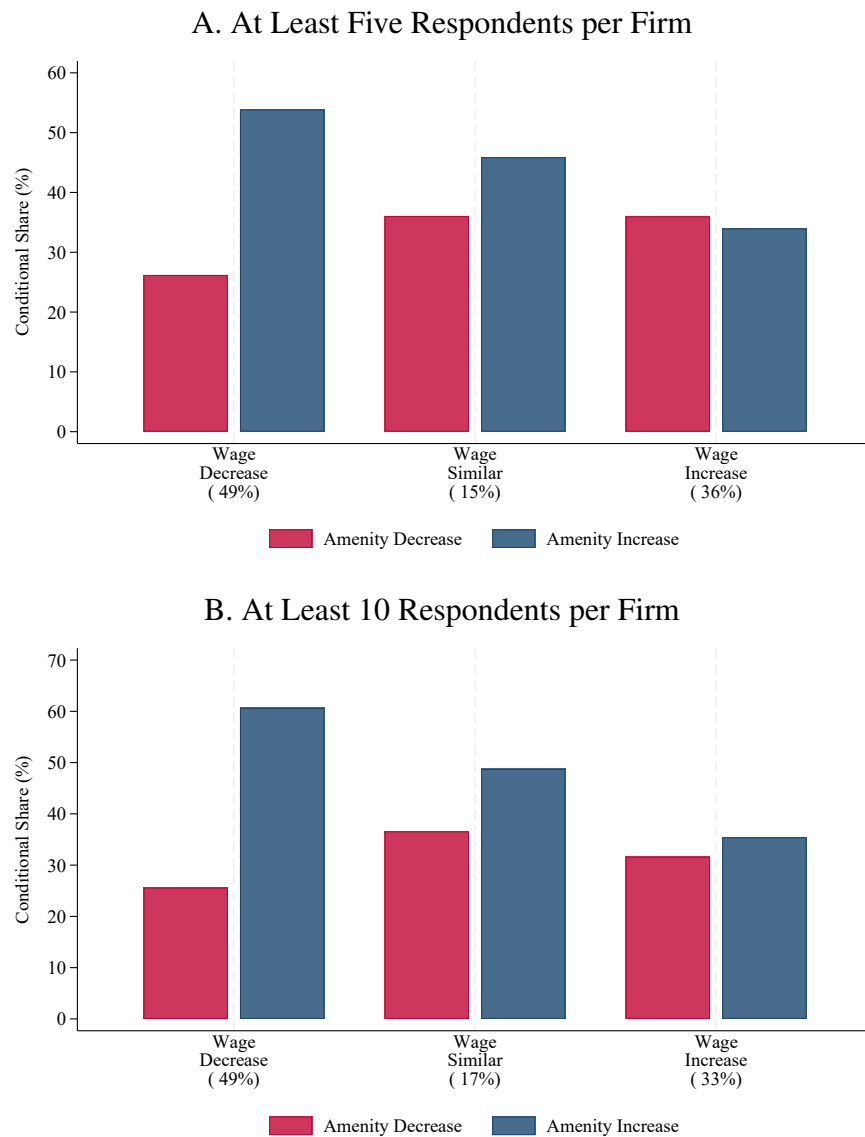
Appendix Tables D4 and D5 present robustness checks on the transition matrix in Table 5. Table D4 restricts to likely voluntary movers (direct job-to-job transitions with no intervening unemployment spell). Table D5 uses a 10% threshold (rather than 2%) to define large changes in wages and amenities. The patterns are similar across both specifications.

Table D4: Classifying Job-to-Job Moves: Voluntary Movers

	Change in Amenities			All Movers
	>2% Cut	<2% Change	>2% Raise	
	(1)	(2)	(3)	
A. Change in Own Wage				
Decrease ($< -2\%$)	28.6%	24.0%	47.4%	17.5%
Similar ($\pm 2\%$)	35.2%	14.4%	50.5%	21.6%
Increase ($> 2\%$)	33.9%	22.0%	44.1%	60.8%
All Movers	33.2%	20.7%	46.0%	99.9%
B. Change in Firm Wage Premium				
Decrease ($< -2\%$)	32.9%	19.9%	47.2%	45.3%
Similar ($\pm 2\%$)	33.3%	17.1%	49.6%	12.9%
Increase ($> 2\%$)	32.3%	24.9%	42.8%	41.8%
All Movers	32.7%	21.6%	45.7%	100.0%
C. Change in Log Median Wage				
Decrease ($< -2\%$)	32.8%	21.3%	45.9%	45.1%
Similar ($\pm 2\%$)	12.2%	63.3%	24.5%	4.9%
Increase ($> 2\%$)	34.8%	17.6%	47.6%	50.0%
All Movers	32.8%	21.5%	45.7%	100.0%
D. Change in Log Mean Wage				
Decrease ($< -2\%$)	32.5%	21.1%	46.4%	44.6%
Similar ($\pm 2\%$)	13.6%	52.5%	33.9%	5.9%
Increase ($> 2\%$)	35.5%	18.1%	46.4%	49.6%
All Movers	32.9%	21.5%	45.7%	100.1%

Note: This table replicates Table 5 restricting to likely voluntary movers—workers who transition directly from one employer to another without an intervening unemployment spell. The four panels vary the measure of the wage change—own wage (Panel A), firm wage premium (Panel B), log median wage (Panel C), and log mean wage (Panel D)—using baseline amenity valuations throughout. Changes are classified as large when they exceed 2%.

Figure D12: Classifying Job-to-Job Moves: Minimum-Mentions Restriction



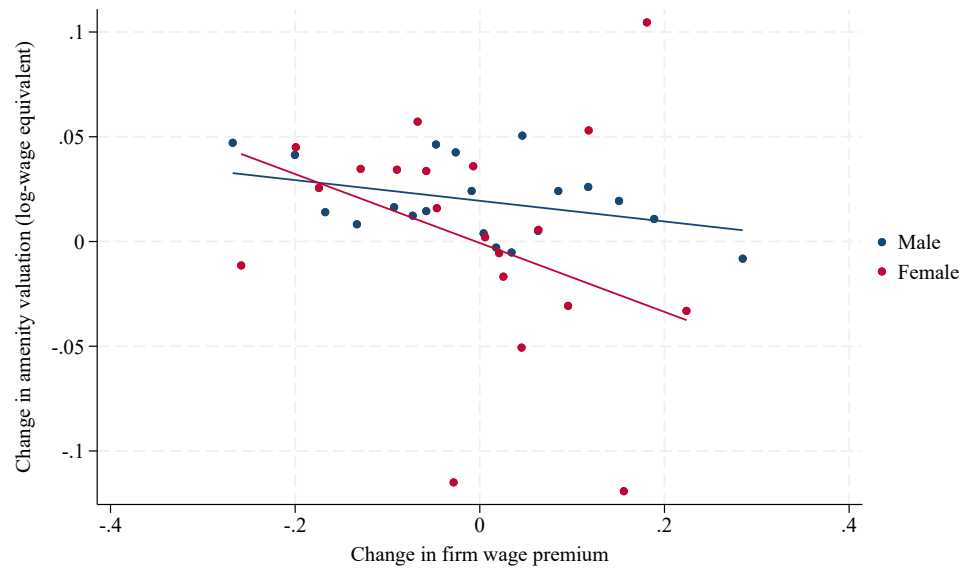
Note: This figure classifies job-to-job moves by the change in the firm wage premium and the change in amenity value, restricting to moves in which both the origin and destination firms are mentioned at least five (Panel A) or at least 10 (Panel B) times. We drop moves that occur within firms.

Table D5: Classifying Job-to-Job Moves: 10% Threshold

	Change in Amenities			All Movers
	>10% Cut	<10% Change	>10% Raise	
	(1)	(2)	(3)	
A. Change in Own Wage				
Decrease ($< -10\%$)	7.5%	84.9%	7.5%	9.3%
Similar ($\pm 10\%$)	6.4%	78.0%	15.6%	43.7%
Increase ($> 10\%$)	8.5%	84.3%	7.2%	47.0%
All Movers	7.5%	81.6%	10.9%	100.0%
B. Change in Firm Wage Premium				
Decrease ($< -10\%$)	4.7%	85.1%	10.2%	21.5%
Similar ($\pm 10\%$)	8.2%	80.4%	11.5%	57.6%
Increase ($> 10\%$)	8.7%	82.2%	9.1%	20.8%
All Movers	7.5%	81.7%	10.7%	99.9%
C. Change in Log Median Wage				
Decrease ($< -10\%$)	5.8%	81.1%	13.1%	36.0%
Similar ($\pm 10\%$)	3.6%	89.2%	7.2%	25.1%
Increase ($> 10\%$)	11.5%	77.2%	11.3%	39.0%
All Movers	7.5%	81.7%	10.9%	100.1%
D. Change in Log Mean Wage				
Decrease ($< -10\%$)	6.2%	80.4%	13.4%	35.7%
Similar ($\pm 10\%$)	2.7%	89.9%	7.4%	25.8%
Increase ($> 10\%$)	11.7%	77.6%	10.7%	38.4%
All Movers	7.4%	81.7%	10.8%	99.9%

Note: This table replicates Table 5 using a 10% threshold (rather than 2%) to define large changes in wages and amenities. The four panels vary the measure of the wage change—own wage (Panel A), firm wage premium (Panel B), log median wage (Panel C), and log mean wage (Panel D)—using baseline amenity valuations throughout.

Figure D13: Wage–Amenity Trade-off Among Movers



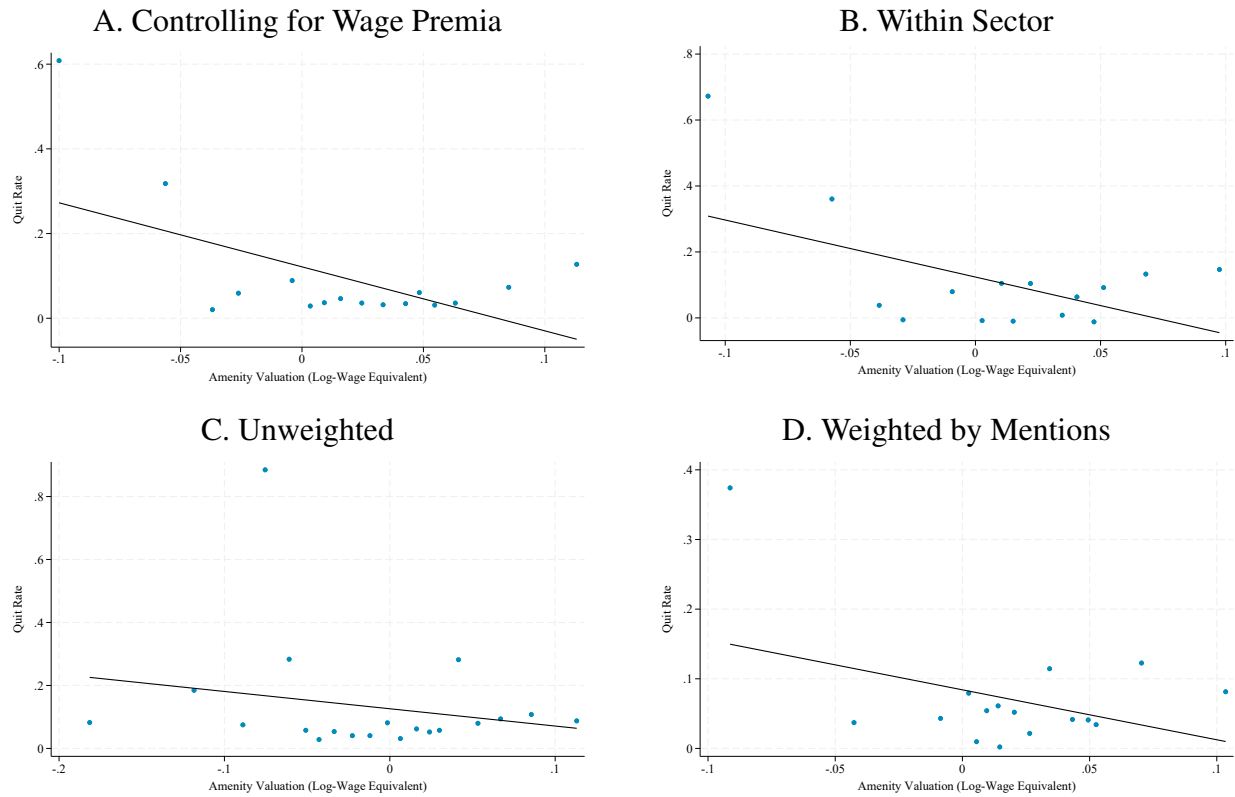
Note: Binned scatterplot of the change in sex-specific amenity valuations (y-axis) against the change in firm wage premium (x-axis) among job movers, separately for men and women. The fitted lines are OLS. Because the amenity change is the dependent variable, its estimation error inflates the standard errors rather than biasing the slope.

D.5 Amenities and Quit Rates

Compensating-differentials models predict that firms offering better amenities should, all else equal, experience lower separation rates. We provide descriptive evidence on this prediction using administrative data on job-to-job transitions from the IEB. We construct a firm-level measure of “quit rates” by counting the number of workers who leave each firm and move directly to another employer (i.e., without an intervening unemployment spell), divided by firm size. This is an imperfect proxy for voluntary separations: not all direct job-to-job moves are voluntary, and some separations involving short unemployment spells may be. Nonetheless, the measure captures the intensive margin of worker retention that models of amenity provision emphasize.

Figure D14 presents binned scatterplots of quit rates against amenity valuations, residualized on firm wage premia. Firms with higher amenity valuations have lower quit rates, conditional on wages: the slope is negative in every baseline specification—across weighting schemes—but never statistically significant. The pattern appears overall (Panel A), within sectors (Panel B), and under alternative weighting schemes (Panels C and D). Because the exercise is cross-sectional, the quit-rate proxy is noisy, and the estimates are imprecise, we present these patterns as descriptive.

Figure D14: Amenity Valuations and Quit Rates



Note: Binned scatterplots of firm-level quit rates against amenity valuations (in log-wage-equivalent units), controlling for firm wage premia. Quit rates are the firm's annual direct job-to-job separations divided by employment, averaged over 2010–2019 (IEB). Panels A and B weight firms by firm size; Panel B additionally controls for sector fixed effects. Panels C and D replicate Panel A unweighted and weighted by the number of times the firm is mentioned in the survey. The fitted slope is -1.51 (robust s.e. 1.19) in Panel A and -1.73 (s.e. 1.44) in Panel B; slopes are negative but statistically insignificant in every baseline specification.

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D.6 Results from the Researcher-Provided Firm Module

The results in the main text use data from a module in which we asked workers to provide the names of specific firms, and then to rank hypothetical offers from these firms under researcher-randomized raises. For the analysis in Caldwell, Haegele and Heining (2025), we designed a parallel module focusing on researcher-selected firms. We can use these data to probe the robustness of our core findings. We discuss the relationship between these researcher-provided valuations and firm wage premia—both unweighted and employment-weighted—in the main text (Section 5.1); the supporting figures appear in this appendix (Figure D8).³¹

We estimate firm-specific amenity valuations via rank-ordered logit, separately by demographic group. All valuations are normalized relative to the same base firm (a manufacturing firm), which is the omitted category in the rank-ordered logit. Because the same base firm is used for all subsamples, differences across groups in valuations relative to that base firm are identified; absolute cross-group levels are not.

Table A6 reports correlations and formal tests of equality in valuations from the researcher-provided module. Because many workers evaluated the same researcher-selected firms, these estimates are more precise than those from worker-provided firms. Panel A uses all ranked firms; Panel B restricts to firms workers indicated they would consider applying to (“marginal workers”). Columns (1)–(2) report pairwise correlations between group-specific valuations. Columns (3)–(5) report p -values from paired t -tests, Kolmogorov–Smirnov tests, and a joint χ^2 test accounting for estimation error (delta method).

³¹The relationship with firm wage premia uses 29 of the 30 researcher-provided firms, as one firm cannot be linked to the administrative wage-premium data.

Table A6: Between-Group Heterogeneity in Valuations: Researcher-Provided Module

<i>Panel A: All Workers</i>					
	Correlations		<i>p</i> -values		
	Pearson	Spearman	Paired <i>t</i>	K-S	Joint χ^2
	(1)	(2)	(3)	(4)	(5)
Male vs. Female	0.424	0.430	0.000***	0.007***	0.000***
College vs. No College	0.770	0.792	0.001***	0.236	0.501
Firms	30	30			
<i>Panel B: Marginal Workers</i>					
	Correlations		<i>p</i> -values		
	Pearson	Spearman	Paired <i>t</i>	K-S	Joint χ^2
	(1)	(2)	(3)	(4)	(5)
Male vs. Female	-0.069	-0.087	0.741	0.388	0.758
College vs. No College	0.036	0.206	0.610	0.035**	0.717
Firms	30	30			

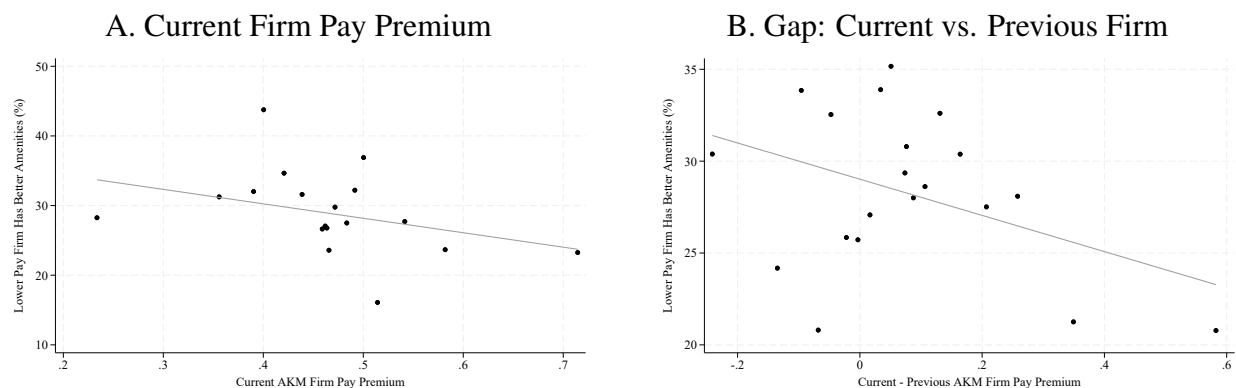
The wage–amenity relationship in this module is positive when firms are weighted equally and near zero once weighted by imputed employment—far from the negative slope that offsetting would require (Figure D8). The rank-ordered logit used here agrees with the baseline eigenvector estimator (Appendix C.3).

D.7 Beliefs About the Correlation Between Wages and Amenities

Caldwell, Haegele and Heining (2025) document that, in the follow-up survey, we asked workers whether they thought a firm paying 30% above market offered amenities that were better, the same, or worse than those offered by a firm paying 10% above market. Only 29.2% of workers said the higher-paid firm offered worse non-wage amenities; 42.8% said the amenities were the same, and 28.0% said they were better ($N = 1708$). Here we describe the predictors of this belief. Because we did not ask whether the gap in non-wage amenities was enough to offset the gap in wages, we interpret this as a descriptive investigation of the share of workers who believe there is a negative correlation between wages and amenities.

Panel A of Appendix Figure A15 shows that, within sector, there is a negative correlation between these beliefs and the pay premium of the respondent's current firm. Panel B shows a similar negative correlation with the gap between the pay premia of a worker's current and previous firm, consistent with the idea that workers who move to lower-pay firms are more likely to believe that high wages come with lower amenities.

Figure A15: Heterogeneity in Beliefs About $\text{Cov}(a_j, \psi_j)$ by Firm



Note: These figures present binned scatterplots plotting the share of respondents who say the lower-pay firm has better amenities against the pay premium of the respondent's current firm (Panel A) or the gap between the pay premia of their current and previous firm (Panel B). Both control for sector fixed effects.

E Survey Logistics

This appendix provides additional details on how we selected and invited survey participants. We also describe how we randomly assigned incentives to participate and discuss survey response patterns. This section replicates some of the results and discussion presented in the appendix to Caldwell, Haegele and Heining (2026), which analyzes the implementation and response rates to the initial survey. This appendix also provides the questionnaires.

E.1 Implementation of the Initial Survey

E.1.1 Sample Construction

We used German Social Security records to identify participants for the survey. Our eligible pool consisted of workers who were—as of December 30, 2020—between the ages of 25 and 50, employed full-time, and at their current establishment for eight or fewer years. To manage the large number of letters, we mailed the survey in batches. For the first batch, we drew 82,500 workers (75% of the batch) from the set of eligible workers at firms that participated in the firm survey linked to IAB records in Caldwell, Haegele and Heining (2026). We oversampled from surveyed firms to achieve adequate power for the analysis in Caldwell, Haegele and Heining (2026). We selected the remaining 25% (N=27,500) at random from (a random 5% sample of) workers at non-surveyed firms. The entire second batch (N=25,000) was selected from the random 5% sample of eligible workers at non-surveyed firms. We mailed reminders to a random 25% subset of individuals in the first batch who had not responded to the initial invitation when we prepared the second mailing. In spring 2024, we invited all respondents from the initial survey who provided panel consent to participate in a follow-up survey.

E.1.2 Invitations

After we identified workers for inclusion in the survey, a specialized department at the IAB pulled their addresses. The IAB fielded the survey and the director of the IAB signed the invitation to participate. We mailed invitations to the initial survey between June 2022 and December 2022. We described the survey to potential respondents as a scientific study on pay progression in Germany. To manage the large number of letters, we staggered the mailings.

We invited respondents via mail instead of e-mail or phone because the Federal Employment Agency in Germany (*Bundesagentur für Arbeit*) only has e-mail and phone numbers of individuals who have recently been unemployed or have participated in re-employment measures. Postal addresses are available for all workers. In the invitation, we informed respondents that the survey would take approximately 10 minutes to complete. Appendix Figure E1 shows (a translation of) the wording of the invitation; the example shown is for a worker randomized into the high-gift-card treatment.

In one batch of the invitations, a printing error mixed up endorsement and cover letters in some cases: two different addresses (endorsement person a, cover person b) ended up in one envelope. We sent letters to the addresses provided on the endorsement letter, but the cover letter information included the name and password of a different individual. If an individual who received an in-

Figure E1: Original Invitation (English Translation)



Institut für Arbeitsmarkt- und Berufsforschung
Regensburger Str. 104 · Re100 407 · 90478 Nürnberg

Bei Rückfragen wenden Sie sich bitte an:

Dr. Jörg Heining
Regensburger Str. 104
90478 Nürnberg
E-Mail: lf-befragung@iab.de

«First name» «Last name»
«Street address»
«Zip code» «City»

Anschreiben-ID: «anschreiben_id»
Nürnberg, «date»

Scientific Study on Wage Flexibility in Germany

Hello «First name» «Last name»,

The Institute for Employment Research (IAB) is conducting a scientific study to understand how the salary progression of employees in Germany is changing. We are therefore interested in your experience in the labor market and would like to invite you to a survey. By participating, you support the IAB in advising political decision-makers and thus help to improve economic and social policy in Germany.

In a nutshell - we will interview you via the Internet

The survey will not take more than **10 minutes** of your time. Your participation is of course voluntary and anonymous. To get to the survey, you can use the display **QR code** or click on the following link:

<https://umfragen.iab.de/goto/LF-befragung>



Your personal password for participation is: «survey_password»

Participating pays off

As a thank you for your participation, we are raffling off 1,500 vouchers, each with a value of 10 euros.

Your information is confidential

The safety of your personal data is important to us. We assure you that your information will be treated with strict confidentiality in accordance with the statutory data protection regulations and will only be used for scientific purposes. Your answers cannot be linked to your person. You will find additional data protection explanations attached.

Thank you for your cooperation and for your trust!

Kind regards

Prof. Bernd Fitzenberger Ph.D.
Direktor des Instituts für Arbeitsmarkt- und Berufsforschung (IAB)

correct mailing participated in the survey, they would have been linked to the wrong respondent's record. Based on inspection of the frequency of this error in letters that were returned due to incorrect addresses (which is an independent issue, but allows us to analyze the frequency of the error), we calculate that at most 431 people among the set of 17,772 recipients in this batch were likely affected (2.4%), assuming the share among returns is the same as in the overall sample. We cannot identify the affected individuals and therefore exclude no responses. Recipients may have realized the mix-up and declined to participate. Because the error affected only recipients of endorsement letters (which we randomly assigned), it may have reduced the response rate for this group.

E.1.3 Exogenous Variation in Incentives to Participate

To characterize patterns of non-response, we introduced random variation in individuals' incentives to participate. We did this through (1) randomized financial incentives, (2) randomized endorsement letters, and (3) randomized follow-up.

Gift Card Lottery. We included information on the gift-card lottery with the initial invitation for random subsets of the first 110,000 workers selected for batch 1. We selected 10,000 workers to be in a raffle for a 5-euro gift card and 20,000 workers to be in a raffle for a 10-euro gift card. Workers were informed about the lottery in the cover letter, which provided the gift card value, as well as the number of gift cards that we would raffle off. After the survey closed in January 2023, the IAB conducted the gift card lottery by randomizing among participants who started the survey. Winners were informed either via e-mail or mail, according to the preferences indicated in the survey.

Endorsement Letters. The endorsement letter was assigned to a subset of 82,500 of the first 110,000 workers in batch 1 and was included with the initial invitation to participate. The letter was signed by one of the 2021 Nobel Prize winners in economics, identified both as one of the 2021 Laureates and as a previous collaborator of the IAB. The letter highlighted the importance of scientific labor market research and urged recipients to complete the survey.

Reminder Mailings. There were 99,698 initial non-responders. We randomly selected 25,000 to receive a follow-up letter. The reminder letters had nearly identical wording to the initial invitations, but reminded individuals that they had previously been invited to participate in the survey. The letters included the same information on data protection as before. Individuals who had been included in the gift card raffle in the initial invitation received reminders of their offer to participate. Individuals who had been randomized to receive endorsement letters did not receive a second endorsement letter.

E.1.4 Balance Check

Appendix Table E1 describes the workers we invited for the survey. As Columns 2 and 3 indicate, conditional on the strata used for selection (whether an individual is at a surveyed firm), there is no difference in the characteristics of eligible workers and those invited to participate in our survey. Columns 4 and 5 compare those selected for the treatment and control groups for the endorsement letters and gift-card treatments. Conditional on the strata used to assign these treatments,

the groups are balanced on observed characteristics: the joint test does not reject, though manufacturing employment differs at the 1% level in the gift-card comparison.³² Column 6 shows that among those eligible to receive a reminder (initial non-responders in batch 1), there is no difference between those selected to receive a reminder and those not selected.

E.1.5 Response Rate and Consent

The survey was open until January 15, 2023. We received 13,680 total responses. Subtracting the number of letters that could not be delivered, the survey had an effective response rate of 11.4%. This response rate is much higher than those of other surveys at the IAB that invite respondents for the first time (Haas et al., 2021). Among the 13,680 individuals who started the survey, 11,868 completed it; this represents a completion rate of 87%.³³ The median response time among individuals who completed the survey was 9 minutes.

We asked participants for their consent to link their answers to the employer-employee data at the IAB. We have 10,134 complete responses with linkage consent, which we link to the IAB records. While this direct consent is necessary under German privacy laws to link the survey data to other data sources, we analyze the raw, unlinked data for both consenters and non-consenters. We also asked participants for their consent to participate in follow-up surveys. 8,416 respondents who provided consent for this linkage also provided consent to be contacted for future survey waves. Among the 10,134 linkage consenters, this represents a panel consent rate of 83%.

E.1.6 Impact of Randomized Incentives on Response Rates

Appendix Table E2 shows that neither the gift cards nor the endorsement letter had a statistically significant (or economically meaningful) impact on response rates in the initial survey. By contrast, the reminder message increased response rates by 4 percentage points among workers who did not initially respond to the survey. Both incentives were visible only to individuals who opened the mailer; the null effects suggest that much of the initial non-response reflects individuals who never opened the invitation.

E.1.7 Selection into Non-Response

We follow Dutz et al. (2021) in analyzing the characteristics of compliers. We restrict attention to individuals in batch 1 of the initial survey, because this is the batch in which we implemented the randomization of reminders and incentives. The complier categories below are defined by the reminder randomization. We do not characterize endorsement letter or gift card compliers because these incentives did not have a significant impact on response rates. Appendix Table E3 describes three populations of workers. Column 1 describes “early always takers”: those who responded to the survey before we mailed the reminder. Column 2 describes the “late always takers”: those whom we randomized into not receiving a reminder, but who nonetheless responded after we

³²We divided individuals into two groups based on their federal state of residence; gift cards were randomized within the second of these groups. We randomly assigned endorsement letters without regard to state.

³³We define a response as complete if a respondent clicked through to the (second to last) question eliciting consent for participating in a follow-up survey. We do not require respondents to have answered every question to be counted as complete responses. The survey did not require individuals to respond to particular questions.

Table E1: Randomization of Survey Invitations and Incentives

	Eligible Mean	Selection		Wave 1 Randomization		
		Wave 1 - Eligible	Wave 2 - Eligible	Lottery - No Lottery	Letter - No Letter	Reminder - No Reminder
	(1)	(2)	(3)	(4)	(5)	(6)
<u>Demographics</u>						
Female	0.32	0.63	0.33	0.48	0.41	0.32
Age	33.02	0.29	0.94	0.75	0.73	0.82
German Citizen	0.77	0.22	0.65	0.75	0.67	0.22
College Education	0.34	0.59	0.47	0.06	0.35	0.39
Apprenticeship	0.45	0.90	0.67	0.64	0.51	0.69
Daily Earnings	129.12	0.22	0.39	0.23	0.47	0.09
<u>Occupation Group</u>						
Manager	0.04	0.96	0.45	0.95	0.71	0.57
Recent Entrant	0.34	0.86	0.84	0.95	0.06	0.01
<u>Sector</u>						
Manufacturing	0.32	0.86	0.72	0.01	0.74	0.44
Retail	0.10	0.47	0.40	0.53	0.47	0.41
Professional	0.10	0.95	0.42	0.08	0.44	0.70
Establishments		24928	21248	7253	19600	7204
Workers		110000	25000	30000	82500	25000
F-statistic		0.693	0.444	1.558	0.982	1.039
p-value		0.747	0.937	0.104	0.460	0.408

Note: This table documents that the random assignment of survey invitations and incentives was successful. Column 1 describes workers eligible for inclusion in our worker survey. Each entry in Columns 2 to 6 presents the p -value for a two-sided test of the coefficient from a regression of the characteristic in each row on the treatment in each column, controlling for the strata used for random assignment. P -values are calculated using robust standard errors. At the bottom of the table we present the F -statistic and associated p -value from a test of whether all of the coefficients are jointly zero. Columns 2 and 3 show that, conditional on the strata used for selection (whether an individual worked at a surveyed firm in 2020), selected individuals are not statistically distinguishable from non-selected individuals. Columns 4 to 6 show that, conditional on the strata used for random assignment, individuals selected to receive each of the three types of incentives are largely balanced: the joint tests do not reject, though manufacturing employment differs at the 1% level in the gift-card comparison. We only randomized these incentives to workers in batch 1.

Table E2: Impact of Randomized Incentives on Response Rates in the Initial Survey

	Endorsement	Gift Card		Reminder
	Letter	Level	Binary	
	(1)	(2)	(3)	(4)
Treatment	0.000 (0.002)	-0.000 (0.000)	-0.002 (0.002)	0.040*** (0.001)
Observations	109995	109995	109995	99698

Note: This table analyzes the effect of the randomized incentives on the likelihood that invited individuals completed the survey and provided linkage consent. Each coefficient stems from a separate regression of survey completion on an indicator for the respective incentive, conditional on the strata used for random assignment. Column 1 focuses on endorsement letters. Columns 2 and 3 focus on gift cards. Column 4 focuses on the survey reminder. Robust standard errors are presented in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

mailed the reminders. Column 3 describes the reminder compliers. As Column 2 indicates, virtually all of the always takers responded before we mailed the reminder. Column 3 indicates that the reminder compliers are, relative to the early always takers, somewhat more likely to be male or covered by a collective bargaining agreement.

We also examine selection into providing consent to link survey responses to administrative data and to be invited to participate in future surveys. Appendix Table E4 compares the characteristics of invited individuals (Column 1) to those who completed the survey and provided linkage consent (Column 2) and to those who additionally provided consent to participate in future surveys (Column 4). Columns 3 and 5 compare the samples in Columns 2 and 4 to the samples in Columns 1 and 2, respectively. We find modest differences in respondent characteristics with respect to gender, age, occupation type, and sector. For instance, while the female share is 30% among invited individuals, it is 32% among those who responded and provided consent. German citizens were much more likely to respond than non-citizens, likely because they are more comfortable with the survey language and may feel a greater obligation to contribute to research on the German labor market. We also find that more educated, higher-earning workers were more likely to respond, perhaps because they are more familiar with the IAB. Columns 6 and 7 show that workers who responded to the follow-up survey were again positively selected on citizenship, education, and earnings.

While there is some positive selection into the survey, our final sample includes workers at a broad set of firms, including low-pay firms (Figure E2).

Table E3: Characteristics of Reminder Compliers and Always Takers in the Initial Survey

	Always Takers		Reminder Compliers
	Early	Late	
	(1)	(2)	(3)
Shares	0.05	0.000	0.03
Female	0.27	0.50	0.24
Age	33.85	34.36	34.02
German	0.94	0.93	0.91
College Degree	0.63	0.64	0.60
Apprenticeship	0.31	0.21	0.34
Daily Pay	186.04	202.40	184.97
Censored Pay	0.26	0.29	0.29
Hours Worked	40.63	41.39	40.37
CBA	0.61	0.71	0.65
<u>Sector</u>			
Manufacturing	0.56	0.64	0.56
Retail	0.07	0.00	0.05
Professional	0.17	0.14	0.20
<u>AKM Fixed Effects</u>			
Person Effect	4.47	4.33	4.44
Firm Effect	0.52	0.58	0.54
Risk Preferences	6.22	5.57	6.19
Outside Options	1.41	1.14	1.41
Did Not Provide Names	0.45	0.50	0.50

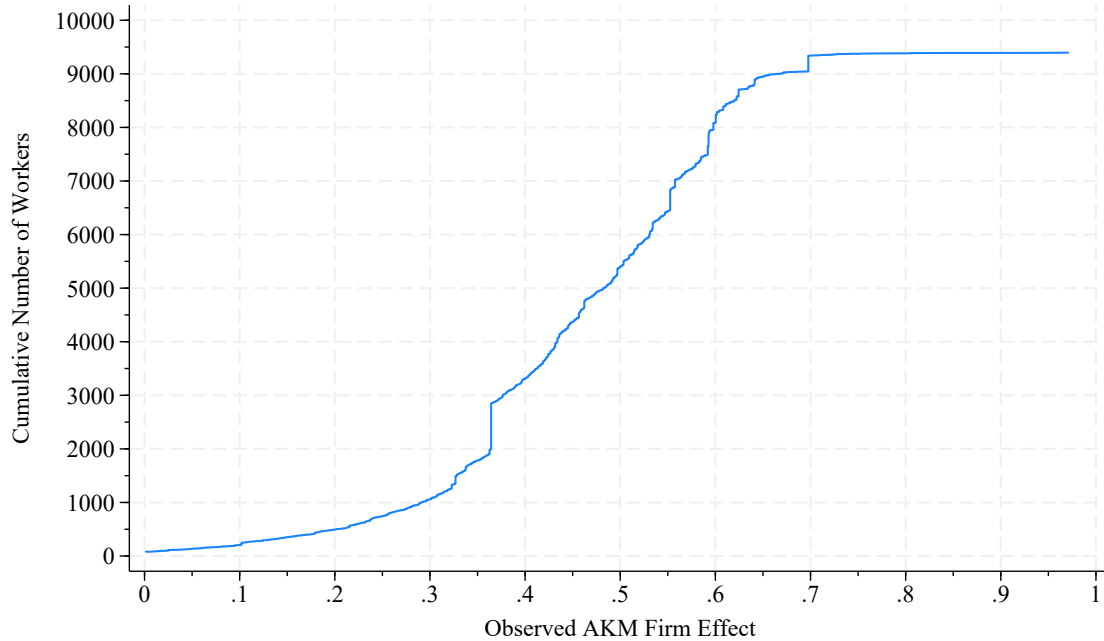
Note: This table compares the characteristics of early and late always takers to reminder compliers. Early always takers are workers who responded before we mailed the (randomized) reminders. Late always takers are workers in the control group who responded after we mailed the reminders. Reminder compliers are workers who responded only because they received a reminder; following Dutz et al. (2021), we estimate the reminder compliers' average characteristics via an instrumental variables regression with $Y_i(1-R_{i1})R_{i2}$ as the outcome variable, $(1-R_{i1})R_{i2}$ as the endogenous variable, and Z_i as the instrument. Y_i is the characteristic of interest, R_{i1} is an indicator for responding before we mailed the reminders, R_{i2} is an indicator for responding after we mailed the reminders, and Z_i is an indicator for having received a reminder. Shares reported in Row 1 are out of all workers invited to respond to the initial survey (Column 1), those who neither responded earlier nor received a reminder (Column 2), and those who did not respond earlier but did receive a reminder (Column 3).

Table E4: Non-Response and Consent

	Linkage Consent			Panel and Linkage		Responded to Follow-Up	
	Invited Mean	Difference		Mean	Rel. Linked	Difference Rel.	
	(1)	Mean	Rel. Invited	(4)	(5)	Mean	Invited
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Demographics</u>							
Female	0.30 (0.46)	0.32 (0.46)	0.02 (0.00)	0.32 (0.47)	0.01 (0.01)	0.31 (0.46)	-0.01 (0.01)
Age	33.63 (6.59)	33.33 (6.23)	-0.32 (0.06)	33.33 (6.14)	-0.02 (0.17)	33.41 (6.16)	0.14 (0.14)
German Citizen	0.81 (0.39)	0.92 (0.27)	0.12 (0.00)	0.92 (0.26)	0.03 (0.01)	0.94 (0.24)	0.02 (0.01)
College Education	0.39 (0.49)	0.59 (0.49)	0.22 (0.01)	0.60 (0.49)	0.07 (0.01)	0.65 (0.48)	0.07 (0.01)
Apprenticeship	0.45 (0.50)	0.33 (0.47)	-0.12 (0.00)	0.32 (0.47)	-0.05 (0.01)	0.29 (0.45)	-0.06 (0.01)
Daily Earnings	146.03 (60.77)	169.79 (56.71)	25.69 (0.59)	170.92 (56.67)	6.61 (1.50)	175.02 (55.27)	7.27 (1.24)
<u>Occupation Group</u>							
Manager	0.05 (0.21)	0.06 (0.24)	0.02 (0.00)	0.06 (0.24)	0.01 (0.01)	0.07 (0.25)	0.01 (0.01)
Recent Entrant	0.26 (0.44)	0.20 (0.40)	-0.06 (0.00)	0.20 (0.40)	-0.02 (0.01)	0.19 (0.39)	-0.02 (0.01)
<u>Sector</u>							
Manufacturing	0.44 (0.50)	0.46 (0.50)	0.03 (0.01)	0.46 (0.50)	0.00 (0.01)	0.47 (0.50)	0.01 (0.01)
Retail	0.10 (0.30)	0.09 (0.28)	-0.01 (0.00)	0.09 (0.28)	0.01 (0.01)	0.09 (0.28)	0.00 (0.01)
Professional	0.10 (0.29)	0.13 (0.34)	0.04 (0.00)	0.13 (0.34)	0.01 (0.01)	0.13 (0.34)	0.00 (0.01)
Establishments	42705	3556		2983			1457
Observations	134995	10134		8416			3664

Note: This table compares the characteristics of workers we invited to complete the survey (Column 1) to workers who completed the survey and provided consent to link their responses to the administrative data (Column 2), to the subset of these workers who additionally provided consent to participate in follow-up surveys (Column 4), and those who participated in the follow-up survey (Column 6). Columns 3, 5, and 7 present the strata-adjusted differences between the samples indicated in the header and sub-header. For instance, Column 3 reports the difference between workers who completed the survey and provided linkage consent to the set of invited workers. Robust standard errors are presented in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

Figure E2: Sample Size by AKM Firm Effect



Note: This figure shows the cumulative number of workers in our sample (y-axis) who, at the time of sampling, worked at an establishment whose AKM firm effect was at or below the value indicated on the x-axis. Workers whose firms do not have associated estimates are not included in this graph.

E.2 Implementation of the Follow-Up Survey

In spring 2024, we invited all respondents from the initial survey who provided panel consent to participate in a follow-up survey. Participants who did not provide an e-mail address in the initial survey were contacted via mail by the IAB. The invitation letter closely followed the initial survey letter but reminded respondents of their prior participation and introduced the follow-up. Among the participants who did provide an e-mail address in the initial survey, we randomized the initial mode of contact: 25% received an invitation sent by the IAB via e-mail and 75% first received an invitation letter via mail (identical to those participants who did not provide an e-mail) and then received another invitation via e-mail. Within each group of respondents, we randomly selected 75% to receive a further reminder (either by mail or e-mail).

The survey was open for 14 weeks and received a 50% response rate. The completion rate was 92%. The median response time among individuals who completed the survey was 9 minutes. Among those who completed the follow-up survey, 90% provided linkage consent.

E.2.1 Impact of Randomized Incentives on Response Rates

We asked individuals in the initial wave of the survey whether we could contact them for a follow-up survey. We also asked these individuals whether we could contact them via e-mail. Most workers indicated they would prefer to be contacted by e-mail in the follow-up survey. In the

Table E5: Impact of Randomized Incentives on Response Rates

	Provided an E-Mail			
	No	Yes		
	(1)	(2)	(3)	(4)
Reminder Letter	0.079*** (0.019)			
Initial Letter		0.230*** (0.015)		0.232*** (0.015)
Reminder E-mail			0.070*** (0.016)	0.077*** (0.016)
Constant	0.360*** (0.017)	0.270*** (0.013)	0.393*** (0.014)	0.210*** (0.018)
Observations	3405	5011	5011	5011

Note: This table analyzes the effect of the randomized incentives on the likelihood that invited individuals completed the follow-up survey and provided linkage consent. In the initial survey, we asked individuals whether we could contact them for a follow-up survey and whether we could contact them via e-mail (which they then provided). Column 1 focuses on workers who did not provide an e-mail address with which to contact them for the follow-up survey. Columns 2 to 4 focus on workers who did provide an e-mail address for this purpose. Each column presents the coefficients from a regression among workers in the indicated sample; the dependent variable is an indicator for whether the individual responded to the follow-up survey and provided consent to link this new response to the administrative records; the independent variables are an indicator for the survey incentive(s) indicated in the rows and a constant. Robust standard errors are presented in parentheses. Levels of significance: * 10%, ** 5%, and *** 1%.

follow-up survey, we grouped individuals by whether they provided an e-mail address and targeted incentives accordingly.

Overall, about 50% of workers we invited to complete the follow-up survey did so. Appendix Table E5 shows that each of the reminders in this follow-up had a significant impact on response rates. Column 1 examines the response rates of individuals in the letter (non-e-mail) sample. For these workers, we randomized a single incentive: the provision of a reminder letter (as in the initial survey wave). As Column 1 indicates, those who did not receive the reminder (25% of the sample) had a response rate of just under 40%; the reminder boosted response rates by 8 percentage points.

Among workers who provided an e-mail address, we cross-randomized two incentives. First, we mailed a random subset (75%) of these individuals a letter before we e-mailed them regarding the survey. Second, we sent an e-mail reminder to a random subset (75%, cross-randomized) of these individuals. Both incentives were highly effective. The initial letter boosted response rates by 23 percentage points (Column 2). The e-mail reminder boosted response rates by 7 percentage points (Column 3).

E.2.2 Selection into Non-Response

Analogous to the initial survey, we analyze the characteristics of compliers for the follow-up survey following Dutz et al. (2021). Appendix Table E6 describes three groups of individuals: early always takers (Column 1), late always takers (Column 2), and reminder compliers (Column 3).

As in the initial survey, almost all of the always takers responded before the reminder was sent. Column 3 indicates that the reminder compliers are very similar to the early always takers.

Table E6: Characteristics of Reminder Compliers and Always Takers in the Follow-Up Survey

	Always Takers		Reminder
	Early	Late	Compliers
	(1)	(2)	(3)
Shares	0.35	0.01	0.13
Female	0.32	0.50	0.31
Age	33.37	32.55	33.36
German	0.94	0.91	0.93
College Degree	0.65	0.77	0.61
Apprenticeship	0.29	0.14	0.33
Daily Pay	175.12	157.38	171.34
Censored Pay	0.22	0.18	0.19
Hours Worked	40.45	40.48	40.52
CBA	0.56	0.33	0.61
<u>Sector</u>			
Manufacturing	0.47	0.36	0.48
Retail	0.09	0.09	0.09
Professional	0.13	0.18	0.13
<u>AKM Fixed Effects</u>			
Person Effect	4.45	4.41	4.42
Firm Effect	0.46	0.44	0.45
Risk Preferences	6.09	5.68	6.22
Outside Options	1.40	1.45	1.41
Did Not Provide Firm Names	0.44	0.41	0.43

Note: This table compares the characteristics of early and late always takers to reminder compliers for the follow-up survey. Early always takers are workers who responded before we mailed the (randomized) reminders. Late always takers are workers in the control group who responded after we mailed the reminders. Reminder compliers are workers who responded only because they received a reminder; following Dutz et al. (2021), we estimate the reminder compliers' average characteristics via an instrumental variables regression with $Y_i(1-R_{i1})R_{i2}$ as the outcome variable, $(1-R_{i1})R_{i2}$ as the endogenous variable, and Z_i as the instrument. Y_i is the characteristic of interest, R_{i1} is an indicator for responding before we mailed the reminders, R_{i2} is an indicator for responding after we mailed the reminders, and Z_i is an indicator for having received a reminder. The shares reported in Row 1 are unconditional shares of all workers invited to respond to the follow-up survey.

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F Questionnaire

Although we included questions for workers who indicated they were self-employed or non-employed at the time of the survey, the survey targeted employed workers. We exclude self-employed and non-employed individuals from our analysis (in all papers) as the number of workers in these categories was low.

F.1 English Translation of Questionnaire

This section provides the questionnaires of both the initial and follow-up survey. From both surveys, we omit questions about negotiation that were designed for and described in detail in Caldwell, Haegeler and Heining (2026). The appendix of that paper provides more information on those questions. We first provide the English translation of both questionnaires, followed by the original wording in German.

F.1.1 Main Survey

1. In order to keep the survey short, we would like to include data in the evaluation that are available at the Federal Employment Agency in Nuremberg. This is, for example, information about your previous employment.

We strictly comply with all data protection regulations or these are strictly controlled by the IAB. All information and answers are evaluated and presented anonymously, i.e. without name, address and contact information. Nobody can therefore recognize which answers you have given. There is also no transfer of data that would identify you. Further information on data protection can be found <https://www.iab.de/de/befragungen/lohnflexibilitaet-in-deutschland.aspx>.

The evaluation is carried out for purely scientific purposes, i.e., not for commercial purposes such as advertising or marketing. Your consent is of course voluntary.

Do you agree to your data being linked?

- a. Yes, I agree to the link
 - b. No, I do not agree to the link
2. What best describes your current main employment?
 - (a) Employed with social security contributions
 - (b) Self-employed
 - (c) Not employed
 - (d) Other

Module: Employment and Bargaining

Submodule: Employed or missing

3. When did you first join your current company?

- (a) In the past 6 months
 - (b) 6-12 months ago
 - (c) 1-2 years ago
 - (d) 2-3 years ago
 - (e) >3 years ago
4. Is your current position covered by a CBA agreement (i.e., are you paid according to a CBA)?
 - (a) Yes
 - (b) No
 - (c) I don't know
 5. How many hours do you work in a typical week? {fill in}
 6. At the time that you first applied to your current company, did you know anyone who worked at the company?
 - (a) Yes
 - (b) No, but I knew someone who used to work at the company
 - (c) No, I did not know anyone
 7. At the time that you applied, did you know what salary you would earn?
 - (a) I had no or very little idea
 - (b) I only had a rough idea what is paid in my region or sector
 - (c) I had at least a rough idea what this company pays for the position
 - (d) I knew exactly what this company pays for the position
 8. During the past six months, have you done any of the following? Please select all that apply.
 - (a) Looked at job postings
 - (b) Updated public resume or employment information (e.g., Xing, LinkedIn)
 - (c) Reached out to people in my network for information about potential job opportunities
 - (d) Applied to a job at another company
 9. During the past six months, did anyone reach out to you to provide information about potential job opportunities (e.g., sent you a job opening or offered to provide a referral)?
 - (a) Yes
 - (b) No
 10. During the past six months, did you receive any job offers from other companies?
 - (a) Yes
 - (b) No
 11. During the past six months, did your company offer you a salary increase without you asking?
 - (a) Yes
 - (b) No
 12. During the past six months, did you actively ask for an increase in salary?
 - (a) Yes
 - (b) No

Submodule: Self-Employed or Non-Employed

13. How many hours do you work in a typical week? {fill in}
14. During the past six months, have you done any of the following? Please select all that apply.

- o Looked at job postings
 - o Updated public resume or employment information (e.g., Xing, LinkedIn)
 - o Reached out to people in my network for information about potential job opportunities
 - o Applied for jobs
15. During the past six months, did anyone reach out to you to provide information about potential job opportunities (e.g. sent you a job opening or offered to provide a referral)?
- (a) Yes
 - (b) No
16. During the past six months, did you receive any job offers?
- (a) Yes
 - (b) No

Module: Worker-Provided Firms

17. Suppose you planned to move to a new company in the next {one/three/six} months. What are companies that you would consider applying to? Please list three companies that you would consider applying to and that hire employees in positions like yours (e.g. “Place-Holder Inc”). These can be companies without current job vacancies.
- (a) {fill in Company 1}
 - (b) {fill in Company 2}
 - (c) {fill in Company 3}
 - (d) I do not want to answer this question [exclusive option]
18. **[Employed]** What do you think your gross annual pay would be if you worked at these companies in a position similar to your current one?
- (a) {fill in Company 1}
 - (b) {fill in Company 2}
 - (c) {fill in Company 3}
19. What best describes how certain you feel about your pay estimates?

Company Name	Very Uncertain	Uncertain	Somewhat Certain	Certain	Very Certain
{fill in Company 1}					
{fill in Company 2}					
{fill in Company 3}					

20. Some of these companies may have subsidiaries in multiple locations. Which location did you have in mind?

Company Name	Location
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

21. Do you know any employees at these companies (e.g. former coworkers, friends and family)?

	Company Name	Yes, I know current employees	Yes, I know former employees	No, I do not know anyone who ever worked at this company
22.	{ fill in Company 1 }			
	{ fill in Company 2 }			
	{ fill in Company 3 }			

23. Suppose one of your colleagues got offers from the following companies for a position like yours. Which company do you think offers the highest gross annual pay? Please rank the offers from 1 (highest pay) to 3 (lowest pay).

Company Name	Ranking
{ fill in Company 1 }	
{ fill in Company 2 }	
{ fill in Company 3 }	

24. **[Employed]** Suppose you can remain at your current company or switch to any of the companies you provided and immediately receive the raise specified below. Please rank the following job offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{ fill in Company 1 } with a { randomized raise }	
{ fill in Company 2 } with a { randomized raise }	
{ fill in Company 3 } with a { randomized raise }	
Remain at Current Firm at Current Pay	

25. **[Self-Employed]** Suppose you can remain self-employed or switch to any of the companies you provided and immediately receive the raise specified below. Please rank the following job offers from 1 (most likely to take) to 4 (least likely to take).

Option	Ranking
{ fill in Company 1 } with a { randomized raise }	
{ fill in Company 2 } with a { randomized raise }	
{ fill in Company 3 } with a { randomized raise }	
Remain Self-Employed at Current Pay	

26. **[Non-Employed]** Suppose you can remain non-employed or join any of the companies you provided and immediately receive the increase in income specified below. Please rank the following job offers from 1 (most likely to take) to 4 (least likely to take).

Option	Ranking
{ fill in Company 1 } with a { randomized income increase }	
{ fill in Company 2 } with a { randomized income increase }	
{ fill in Company 3 } with a { randomized income increase }	
Remain Non-Employed	

27. **[Employed]** Now we would like to ask you to re-rank the firms under 2 scenarios. First, suppose that you have the same three offers, but would be able to have the same **commute/route** to work. Please rank the offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{fill in Company 1} with a {randomized raise}	
{fill in Company 2} with a {randomized raise}	
{fill in Company 3} with a {randomized raise}	
Remain at Current Firm at Current Pay	

28. **[Employed]** Now suppose that you have the same set of offers, but instead of the commute staying the same your **career growth** (future raises/promotions) are identical at each option. Please rank the offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{fill in Company 1} with a {randomized raise}	
{fill in Company 2} with a {randomized raise}	
{fill in Company 3} with a {randomized raise}	
Remain at Current Firm at Current Pay	

Module: Researcher-Provided Firms

29. Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.
- (a) Company 1
 - (b) Company 2
 - (c) Company 3
 - (d) Company 4
 - (e) Company 5
 - (f) Company 6
 - (g) Company 7
30. **[Employed/Self-Employed]** We are now interested in your perception of job opportunities available at different companies. What do you think your gross annual pay would be if you worked at these companies in a position similar to your current one?

Company Name	Pay in Euros
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

31. **[Non-Employed]** We are now interested in your perception of job opportunities available at different companies. What do you think your gross annual pay would be if you worked at these companies in a position similar to your most recent one?

Company Name	Pay in Euros
{fill in Company 1}	
{fill in Company 2}	
{fill in Company 3}	

32. **[Employed]** Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Job Offer	Ranking
{ fill in Company 1 } with a { randomized raise }	
{ fill in Company 2 } with a { randomized raise }	
{ fill in Company 3 } with a { randomized raise }	
Remain at Current Firm at Current Pay	

33. **[Self-employed]** Now suppose you can remain self-employed or switch to any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Option	Ranking
{ fill in Company 1 } with a { randomized raise }	
{ fill in Company 2 } with a { randomized raise }	
{ fill in Company 3 } with a { randomized raise }	
Remain Self-employed at Current Pay	

34. **[Non-employed]** Now suppose you can remain non-employed or join any of the companies listed below and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Option	Ranking
{ fill in Company 1 } with a { randomized income increase }	
{ fill in Company 2 } with a { randomized income increase }	
{ fill in Company 3 } with a { randomized income increase }	
Remain Non-employed	

35. In your opinion, how easy would it be for you to obtain a job offer from a different company that you would prefer to your current position?
- (a) Very Easy
 - (b) Easy
 - (c) Difficult
 - (d) Very difficult
36. If you had to move to a new company within the next {one/three/six} months, what best describes where you would search for a job? This only applies to positions **without** a home office option.
- (a) I would search for jobs which are not further away from my place of residence than {fill-in} Kilometers
37. Finally, we would like to ask you to assess yourself. Are you generally a person who is willing to take risks or do you try to avoid taking risks? Please choose a value on the scale below, where the value 0 means “not at all willing to take risks” and the value 10 means “very willing to take risks”.
- (a) 0 (Not at all willing to take risks) 1 2 3 4 5 6 7 8 9 10 (Very willing to take risks)

Module: Consent

38. To thank participants for their time, we are raffling off {1500}/{1000} gift cards with a value of {10}/{5} Euros. If you would like to participate in the raffle, please provide your email

- address below. _____ {open-field that accepts e-mail addresses only}
39. Your information supports IAB's research on employees' salary progression in Germany. Since current labor market dynamics are rapidly changing, we would like to survey you again in a few months. Would you like to continue to providing well-founded information on employees' circumstances in Germany? Then we ask you to provide us with your email address so that we can question you again. Of course, we will only use the e-mail address to write to you for our survey. You can of course unsubscribe from the survey at any time. To which email address can we send our invitation?
 - (a) {open field that accepts e-mail addresses only}
 - (b) I do not want to be contacted by email.
 40. May we then invited you again by mail?
 - (a) Yes
 - (b) No

F.1.2 Follow-up Survey

1. In order to keep the survey short, we would like to include data in the evaluation that are available at the Federal Employment Agency in Nuremberg. This is, for example, information about your previous employment. We strictly comply with all data protection regulations or these are strictly controlled by the IAB. All information and answers are evaluated and presented anonymously, i.e., without name, address and contact information. Nobody can therefore recognize which answers you have given. There is also no transfer of data that would identify you. Further information on data protection can be found <https://www.iab.de/de/befragungen/lohnflexibilitaet-in-deutschland.aspx>. The evaluation is carried out for purely scientific purposes, i.e. not for commercial purposes such as advertising or marketing. Your consent is of course voluntary. Do you agree to your data being linked?
 - (a) Yes, I agree to the linkage
 - (b) No, I do not agree the linkage
2. Have you started a new job at a different company since {month_survey1 2022/2023}?
 - (a) Yes
 - (b) No
3. [Q2=Yes] What best describes your current main employment?
 - (a) Employed with social-security contributions
 - (b) Self-employed
 - (c) Not employed
 - (d) Other
4. During the past six months, have you done any of the following? Please select all that apply.
 - (a) Looked at job postings
 - (b) Updated public resume or employment information (e.g., Xing, LinkedIn)
 - (c) Reached out to people in my network for information about potential job opportunities
 - (d) Applied to a job at another company

5. During the past six months, did anyone reach out to you to provide information about potential job opportunities (e.g., sent you a job opening or offered to provide a referral)?
 - (a) Yes
 - (b) No
6. During the past six months, how many job offers from other companies did you receive?
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3 or more

Module: Worker-Provided Firms

7. Next, we are interested in how you perceive employment opportunities in Germany. Even if you don't know something exactly, we would like to ask you to share your best guess with us. Suppose you planned to move to a new company in the next {one/three/six} months. What are companies that you would consider applying to? Please list three companies that you would consider applying to and that hire employees in positions like yours (e.g., "Place-Holder Inc"). These can be companies without current job vacancies.
 - (a) {fill in Company 1}
 - (b) {fill in Company 2}
 - (c) {fill in Company 3}
 - (d) I do not want to answer this question [exclusive option]
8. **[Q7=d]** What best describes why you did not provide any company names in the previous question?
 - (a) I am not comfortable sharing this information
 - (b) There are companies I would consider, but the survey interface was too difficult
 - (c) I could not think of specific companies
 - (d) I have no interest in switching companies and have not thought about it at all
9. **[Q7 has only 1 or 2 firms provided]** What best describes why you did not provide three company names in the previous question?
 - (a) I am not comfortable sharing this information
 - (b) There are more companies I would consider, but the survey interface was too difficult
 - (c) I could not think of any other specific companies
 - (d) I have no interest in switching companies and have not thought much about it
10. **[Q7 is not missing]** What do you think your **gross annual pay** would be if you worked at these companies in a position similar to your current position?

Company	Gross Annual Pay in Euros
{Company 1}	
{Company 2}	
{Company 3}	

11. **[Q7 is not missing]** Suppose you can remain at your current company or switch to any of the companies you provided and immediately receive the raise specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Job Offer	Ranking
{ Company 1 } with a { randomized raise }	
{ Company 2 } with a { randomized raise }	
{ Company 3 } with a { randomized raise }	
Remain at Current Firm at Current Pay	

[Note: Raises were randomized within groups.]

Module: Researcher-Provided Firms

12. Suppose you planned to move to a new company in the next {one/three/six} months. Would you consider applying to any of these? Please select all that apply.

- (a) Company 1
- (b) Company 2
- (c) Company 3
- (d) Company 4
- (e) Company 5
- (f) Company 6
- (g) Company 7

13. What do you think your **gross annual pay** would be if you worked at these companies in a position similar to your current position?

Company	Gross Annual Pay in Euros
{ Company 1 }	
{ Company 2 }	
{ Company 3 }	

14. **[Q13 is missing]** Even if you are unsure, we are interested in your guess of how these companies compare with respect to what they pay. Please rank the companies from 1 (highest pay) to 3 (lowest pay).

Company	Ranking
{ Company 1 }	
{ Company 2 }	
{ Company 3 }	

15. Suppose you can remain at your current company or switch to any of the companies listed below and immediately receive the pay specified below. Please rank the following job offers from 1 to 4 where 1 is the offer you are most likely to take and 4 is the offer you are least likely to take.

Job Offer	Ranking
{ Company 1 } with a { randomized raise }	
{ Company 2 } with a { randomized raise }	
{ Company 3 } with a { randomized raise }	
Remain at Current Firm at Current Pay	

[Note: Raises were randomized within groups.]

16. Now we would like to ask you to rerank the companies under two scenarios. First suppose that you have the same three offers but would be able to have the same **commute** to work as

before. Please rank the offers from 1 (most likely to take) to 4 (least likely to take).

Job Offer	Ranking
{ Company 1 } with a { randomized raise }	
{ Company 2 } with a { randomized raise }	
{ Company 3 } with a { randomized raise }	
Remain at Current Firm at Current Pay	

[Note: Raises were randomized within groups.]

17. Now suppose that you have the same set of offers, but instead of the commute staying the same you knew that the **training and learning opportunities** at each of the three firms was the same as at your current company. Please rank the offers from 1 (most likely to take) to 4 (least likely to take)

Job Offer	Ranking
{ Company 1 } with a { randomized raise }	
{ Company 2 } with a { randomized raise }	
{ Company 3 } with a { randomized raise }	
Remain at Current Firm at Current Pay	

[Note: Raises were randomized within groups.]

Module: Applications

18. Suppose you planned to move to a new company in the next {one/three/six} months, how many different companies do you think would you apply to?
- (a) The number of companies I would apply to is: _____
19. [Randomize who sees Q19 (50%) and who sees Q20/Q21 (50%)] In your opinion, what are the main reasons that make employees hesitant to switch jobs? Please choose the two most important reasons.
- (a) Personal ties to coworkers
 - (b) Location
 - (c) Pay
 - (d) Benefits/ company culture
 - (e) Reluctance to undergo changes
 - (f) Lack of opportunities at other companies
20. Suppose you are comparing job opportunities at two different companies: Company 1 pays 10% above the market average and Company 2 pays 30% above the market average. Which company do you think attracts more qualified applicants per opening?
- (a) Company 1
 - (b) Company 2
 - (c) Both attract the same number of applicants
21. Which company do you think provides better non-wage amenities (e.g., home office, child-care subsidy)?
- (a) Company 1
 - (b) Company 2
 - (c) Both provide the same non-wage amenities

22. Imagine you were to discover that other companies in your area pay a {randomized percentage} more than your current employer. How likely is it that you would start applying for jobs at other companies?
 - (a) Please type in a number between 0% (I would definitely not apply) and 100% (I would start applying immediately): []%
23. How sure would you have to be that you would get a job at one of the higher-paying companies for you to start applying for a new job?
 - (a) At least 90% sure
 - (b) At least 75% sure
 - (c) At least 50% sure
 - (d) At least 25% sure
 - (e) I would start applying in any case
24. When considering a potential job opportunity, how important are each of the following features to you? [Dropdown scale with 1 - not important at all, 2 - slightly important, 3 - moderately important, 4 - very important, 5 - extremely important]
 - (a) Opportunity for reduced hours
 - (b) Home office option
 - (c) Predictable schedule (including overtime)
25. To what extent do you agree with the following statements? [Dropdown scale 1 (totally disagree) 2 3 4 5 6 7 (totally agree)]
 - (a) "I have confidence in my capabilities."
 - (b) "If I were comparing companies to apply to, I would factor in the amount of competition from other applicants."
 - (c) "I would be reluctant to apply for a job if the probability I would get an offer is low."

Module: Person-Specific Information

26. At the end of the survey, we are interested in your personal background. Are you married?
 - (a) Yes
 - (b) No
27. Do you have children?
 - (a) Yes
 - (b) No

F.2 Original German Questionnaire

F.2.1 Initial Survey

Module: Consent

1. Um die Befragung kurz zu halten, würden wir gerne bei der Auswertung Daten einbeziehen, die bei der Bundesagentur für Arbeit in Nürnberg vorliegen. Dabei handelt es sich zum Beispiel um Informationen zu Ihrer bisherigen Berufstätigkeit. Wir halten alle datenschutzrechtlichen Bestimmungen streng ein bzw. diese werden vom IAB streng kontrolliert. Alle Angaben und Antworten werden in anonymisierter Form, also

ohne Namen, Anschrift und Kontaktinformationen, ausgewertet und dargestellt. Niemand kann daher erkennen, welche Antworten Sie gegeben haben. Es erfolgt auch keine Weitergabe von Daten, die Ihre Person erkennen lassen. Weitere Informationen zum Datenschutz finden Sie unter <https://www.iab.de/de/befragungen/lohnflexibilitaet-in-deutschland.aspx>). Die Auswertung erfolgt zu rein wissenschaftlichen Zwecken, das heißt nicht für kommerzielle Zwecke wie Werbung oder Marketing. Ihre Zustimmung ist selbstverständlich freiwillig. Sind Sie mit der Zuspiegung Ihrer Daten einverstanden?

- (a) Ja, ich bin mit einer Zuspiegung einverstanden.
 - (b) Nein, ich bin mit einer Zuspiegung nicht einverstanden.
2. Was trifft am besten auf Ihre aktuelle Haupterwerbstätigkeit zu?
- (a) Sozialversicherungspflichtig beschäftigt
 - (b) Selbstständig
 - (c) Nicht erwerbstätig
 - (d) Sonstiges

Module: Employment and Bargaining

Submodule: Employed or missing

3. Bitte beantworten Sie folgende Fragen in Bezug auf Ihren Hauptarbeitgeber. Seit wann sind Sie in Ihrem jetzigen Unternehmen beschäftigt?
- (a) Seit weniger als 6 Monaten
 - (b) Seit 6-12 Monaten
 - (c) Seit 1-2 Jahren
 - (d) Seit 2-3 Jahren
 - (e) Seit mehr als 3 Jahren
4. Ist Ihre Stelle tarifgebunden (d.h. werden Sie nach Tarifvertrag bezahlt)?
- (a) Ja
 - (b) Nein
 - (c) Ich weiß nicht
5. Wie viele Stunden arbeiten Sie in einer typischen Woche?{fill in}
6. Als Sie sich zum ersten Mal bei Ihrem jetzigen Unternehmen beworben haben, kannten Sie jemanden, der bei diesem Unternehmen beschäftigt war?
- (a) Ja.
 - (b) Nein, aber ich kannte jemanden, der in der Vergangenheit bei diesem Unternehmen beschäftigt war.
 - (c) Nein, ich kannte niemanden.
7. Wussten Sie zum Zeitpunkt Ihrer Bewerbung welches Gehalt Sie verdienen würden?
- (a) Ich hatte keine oder nur sehr wenig Ahnung.
 - (b) Ich hatte nur eine ungefähre Vorstellung davon, was in meiner Region oder Branche bezahlt wird.
 - (c) Ich hatte zumindest eine ungefähre Vorstellung, was dieses Unternehmen für die Stelle bezahlt.
 - (d) Ich wusste genau, was dieses Unternehmen für die Stelle bezahlt.

8. In den vergangenen sechs Monaten haben Sie Folgendes getan? Bitte wählen Sie alle zutreffenden Antworten aus.
- (a) Stellenausschreibungen angesehen
 - (b) Aktualisierten Lebenslauf oder Beschäftigungsinformationen online gestellt (z.B. über Xing, LinkedIn)
 - (c) Personen in meinem Netzwerk kontaktiert, um Informationen zu potentiellen Jobangeboten zu erhalten
 - (d) Sich auf eine Stelle in einem anderen Unternehmen beworben
9. In den vergangenen sechs Monaten hat Sie jemand mit Informationen zu potentiellen Jobangeboten kontaktiert (z.B. Stellenausschreibungen zugeschickt oder Ihnen angeboten, eine Empfehlung für Sie auszusprechen)?
- (a) Ja
 - (b) Nein
10. In den vergangenen sechs Monaten haben Sie Stellenangebote von anderen Unternehmen erhalten?
- (a) Ja
 - (b) Nein
11. In den vergangenen sechs Monaten..... hat Ihr Unternehmen Ihnen eine Gehaltserhöhung angeboten, ohne dass Sie danach gefragt haben?
- (a) Ja
 - (b) Nein
12. In den vergangenen sechs Monaten..... haben Sie proaktiv nach einer Gehaltserhöhung gefragt?
- (a) Ja
 - (b) Nein
13. Wie viele Stunden arbeiten Sie in einer typischen Woche{ fill in }
14. In den vergangenen sechs Monaten haben Sie Folgendes getan? Bitte wählen Sie alle zutreffenden Antworten aus.
- (a) Stellenausschreibungen angesehen
 - (b) Aktualisierten Lebenslauf oder Beschäftigungsinformationen online gestellt (z.B. über Xing, LinkedIn)
 - (c) Personen in meinem Netzwerk kontaktiert, um Informationen zu potentiellen Jobangeboten zu erhalten
 - (d) Auf eine Stelle beworben
15. In den vergangenen sechs Monaten hat Sie jemand mit Informationen zu potentiellen Jobangeboten kontaktiert (z.B. Stellenausschreibungen zugeschickt oder Ihnen angeboten, Empfehlung für Sie auszusprechen)?
- (a) Ja
 - (b) Nein
16. In den vergangenen sechs Monaten haben Sie Stellenangebote erhalten?

- (a) Ja
- (b) Nein

Module: Worker-Provided Firms

17. Als Nächstes interessiert uns, wie Sie Beschäftigungsmöglichkeiten in Deutschland wahrnehmen. Auch wenn Sie etwas nicht genau wissen, möchten wir Sie bitten, Ihre Einschätzung mit uns zu teilen. Angenommen, Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln. Bei welchen Unternehmen würden Sie erwägen, sich zu bewerben? Bitte geben Sie drei Unternehmen an, bei denen Sie sich bewerben würden und die Beschäftigte in Stellen wie Ihrer einstellen (z.B. „Musterunternehmen GmbH“). Das können auch Unternehmen ohne aktuelle Stellenangebote sein.

- (a) {Unternehmen 1 eintragen}
- (b) {Unternehmen 2 eintragen}
- (c) {Unternehmen 3 eintragen}
- (d) Ich möchte diese Frage nicht beantworten [exklusive Option]

18. **[Employed]** Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

19. Was beschreibt am besten, wie sicher Sie sich bei Ihren Gehaltsschätzungen sind?

Unternehmen	Sehr unsicher	Unsicher	Etwas sicher	Sicher
{Unternehmen 1}				
{Unternehmen 2}				
{Unternehmen 3}				

20. Einige dieser Unternehmen haben möglicherweise Niederlassungen in mehreren Orten. An welchen Ort hatten Sie jeweils gedacht?

Unternehmen	Ort
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

21. Kennen Sie Mitarbeiter bei diesen Unternehmen (z.B. ehemalige Kollegen, Freunde oder Familienmitglieder)?

Unternehmen	Ja, ich kenne derzeitige Mitarbeiter	Ja, ich kenne ehemalige Mitarbeiter	Nein, ich kenne niemanden, der jemals bei diesem Unternehmen beschäftigt war
{Unternehmen 1}			
{Unternehmen 2}			
{Unternehmen 3}			

22. Nehmen Sie nun an, einer Ihrer Kollegen bekommt Jobangebote von folgenden Unternehmen für eine Stelle wie Ihre. Was glauben Sie, welches Unternehmen bietet das höchste Bruttojahresgehalt? Bitte ordnen Sie die Angebote von 1 (höchstes Gehalt) bis 3

Unternehmen	Rangfolge
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

(niedrigstes Gehalt).

23. **[Employed]** Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {randomisierter Gehaltserhöhung}	
{Fill-in 2} mit {randomisierter Gehaltserhöhung}	
{Fill-in 3} mit {randomisierter Gehaltserhöhung}	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

24. **[Self-Employed]** Angenommen, Sie könnten selbstständig bleiben oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {randomisierter Gehaltserhöhung}	
{Fill-in 2} mit {randomisierter Gehaltserhöhung}	
{Fill-in 3} mit {randomisierter Gehaltserhöhung}	
Mit aktuellem Gehalt selbstständig bleiben	

25. **[Non-employed]** Angenommen, Sie könnten weiterhin nicht erwerbstätig sein oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort die unten angegebene Einkommenserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {randomisierter Einkommenserhöhung}	
{Fill-in 2} mit {randomisierter Einkommenserhöhung}	
{Fill-in 3} mit {randomisierter Einkommenserhöhung}	
Weiterhin keine Erwerbstätigkeit	

26. **[Employed]** Nun möchten wir Sie bitten, die Rangfolge der Unternehmen unter zwei Szenarien neu einzustufen. Nehmen Sie zuerst an, Sie hätten wieder die drei Jobangebote zur Auswahl, aber Sie hätten den gleichen Arbeitsweg wie bisher. Bitte ordnen Sie die Angebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {randomisierter Gehaltserhöhung}	
{Fill-in 2} mit {randomisierter Gehaltserhöhung}	
{Fill-in 3} mit {randomisierter Gehaltserhöhung}	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

27. **[Employed]** Statt dem Arbeitsweg, nehmen Sie nun an, dass Ihre Karriereentwicklung (zukünftige Gehaltserhöhungen/Beförderungen) für alle Optionen identisch ist. Bitte ordnen Sie die Angebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Fill-in 1} mit {randomisierter Gehaltserhöhung}	
{Fill-in 2} mit {randomisierter Gehaltserhöhung}	
{Fill-in 3} mit {randomisierter Gehaltserhöhung}	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

Module: Researcher-Provided Firms

28. Falls Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln, würden Sie erwägen, sich bei folgenden Unternehmen zu bewerben? Bitte kreuzen Sie alle zutreffenden Antworten an.

- (a) Unternehmen 1
- (b) Unternehmen 2
- (c) Unternehmen 3
- (d) Unternehmen 4
- (e) Unternehmen 5
- (f) Unternehmen 6
- (g) Unternehmen 7

29. **[Employed or Self-Employed]** Uns interessiert nun, wie Sie Stellenangebote in verschiedenen Unternehmen wahrnehmen. Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

30. **[Non-Employed]** Uns interessiert nun, wie Sie Stellenangebote in verschiedenen Unternehmen wahrnehmen. Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie Ihrer letzten Stelle arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Unternehmen 1}	
{Unternehmen 2}	
{Unternehmen 3}	

31. **[Employed]** Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{ Unternehmen 1 } mit { randomisierter Gehaltserhöhung }	
{ Unternehmen 2 } mit { randomisierter Gehaltserhöhung }	
{ Unternehmen 3 } mit { randomisierter Gehaltserhöhung }	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

32. **[Self-Employed]** Angenommen, Sie könnten selbstständig bleiben oder zu einem der genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenig-

sten annehmen).	Jobangebot	Rangfolge
	{ Unternehmen 1 } mit { randomisierter Gehaltserhöhung }	
	{ Unternehmen 2 } mit { randomisierter Gehaltserhöhung }	
	{ Unternehmen 3 } mit { randomisierter Gehaltserhöhung }	
	Mit aktuellem Gehalt selbstständig bleiben	

33. **[Non-Employed]** Angenommen, Sie könnten weiterhin nicht erwerbstätig sein oder zu einem der genannten Unternehmen wechseln und erhalten sofort die unten angegebene Einkommenserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

34.	Jobangebot	Rangfolge
	{ Unternehmen 1 } mit { randomisierter Einkommenserhöhung }	
	{ Unternehmen 2 } mit { randomisierter Einkommenserhöhung }	
	{ Unternehmen 3 } mit { randomisierter Einkommenserhöhung }	
	Weiterhin keine Erwerbstätigkeit.	

35. Was glauben Sie, wie einfach wäre es für Sie, ein Stellenangebot von einem anderen Unternehmen zu erhalten, das Sie Ihrer jetzigen Stelle vorziehen würden?

- (a) Sehr einfach
- (b) Einfach
- (c) Schwierig
- (d) Sehr schwierig

36. Wenn Sie {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} das Unternehmen wechseln müssten, was beschreibt am besten, wo Sie nach einer Stelle suchen würden? Es geht dabei ausschließlich um Stellen ohne Home-Office Option.

- (a) Ich würde nach Stellen suchen, die nicht weiter entfernt von meinem Wohnort sind als {eintragen} Kilometer

37. Abschließend interessiert uns Ihre Selbsteinschätzung. Sind Sie generell ein risikobereiter

Mensch oder versuchen Sie Risiken zu vermeiden? Verwenden Sie dazu bitte eine Skala von 0 bis 10. Der Wert 0 bedeutet „gar nicht risikobereit“ und der Wert 10 „sehr risikobereit“. Mit den Werten dazwischen können Sie Ihre Einschätzung abstufen.

(a) 0 (gar nicht risikobereit) 1 2 3 4 5 6 7 8 9 10 (sehr risikobereit)

Module: Consent

38. Als Dankeschön für Ihre Teilnahme verlosen wir {1500}/{1000} Geschenkkarten im Wert von {10}/{5} Euro. Wenn Sie an der Verlosung teilnehmen möchten, geben Sie bitte Ihre E-Mail-Adresse an. {open field that accepts e-mail addresses only}
39. Ihre Angaben unterstützen das IAB bei seiner Forschung zu Gehaltsentwicklung von Beschäftigten in Deutschland. Da sich die aktuellen Arbeitsmarktdynamiken stetig verändern, möchten wir Sie gerne in einigen Monaten erneut befragen. Wollen Sie auch weiterhin dazu beitragen, fundierte Informationen zur Gehaltsentwicklung in Deutschland bereitzustellen? Dann bitten wir Sie, uns Ihre E-Mail-Adresse zur Verfügung zu stellen, damit wir Sie erneut zu einer Befragung einladen können. Die Teilnahme daran ist freiwillig. Natürlich werden wir die E-Mail-Adresse ausschließlich nutzen, um Sie für unsere Befragung anzuschreiben. Sie können sich selbstverständlich jederzeit wieder aus der Befragung abmelden. An welche E-Mail-Adresse dürfen wir unsere Einladung schicken?
 - (a) {open field that accepts e-mail addresses only}
 - (b) Ich möchte nicht per E-Mail kontaktiert werden.
40. Dürfen wir Sie dann postalisch erneut einladen?
 - (a) Ja
 - (b) Nein

F.2.2 Follow-up Survey

1. Um die Befragung kurz zu halten, würden wir gerne bei der Auswertung Daten einbeziehen, die bei der Bundesagentur für Arbeit in Nürnberg vorliegen. Dabei handelt es sich zum Beispiel um Informationen zu Ihrer bisherigen Berufstätigkeit. Wir halten alle datenschutzrechtlichen Bestimmungen streng ein bzw. diese werden vom IAB streng kontrolliert. Alle Angaben und Antworten werden in anonymisierter Form, also ohne Namen, Anschrift und Kontaktinformationen, ausgewertet und dargestellt. Niemand kann daher erkennen, welche Antworten Sie gegeben haben. Es erfolgt auch keine Weitergabe von Daten, die Ihre Person erkennen lassen. Weitere Informationen zum Datenschutz finden Sie unter <https://www.iab.de/de/befragungen/lohnflexibilitaet-in-deutschland.aspx>). Die Auswertung erfolgt zu rein wissenschaftlichen Zwecken, das heißt nicht für kommerzielle Zwecke wie Werbung oder Marketing. Ihre Zustimmung ist selbstverständlich freiwillig. Sind Sie mit der Zuspiegelung Ihrer Daten einverstanden?
 - (a) Ja, ich bin mit einer Zuspiegelung einverstanden.
 - (b) Nein, ich bin mit einer Zuspiegelung nicht einverstanden.
2. Haben Sie seit {month_survey1 2022/2023} eine neue Stelle bei einem anderen Unternehmen angetreten?

- (a) Ja
 - (b) Nein
3. [Q2=1] Was trifft am besten auf Ihre aktuelle Haupterwerbstätigkeit zu?
- (a) Sozialversicherungspflichtig beschäftigt
 - (b) Selbstständig
 - (c) Nicht erwerbstätig
 - (d) Sonstiges
4. In den vergangenen sechs Monaten... ..haben Sie Folgendes getan? Bitte wählen Sie alle zutreffenden Antworten aus.
- (a) Stellenausschreibungen angesehen
 - (b) Aktualisierten Lebenslauf oder Beschäftigungsinformationen online gestellt (z.B. über Xing, LinkedIn)
 - (c) Personen in meinem Netzwerk kontaktiert, um Informationen zu potentiellen Jobangeboten zu erhalten
 - (d) Sich auf eine Stelle in einem anderen Unternehmen beworben
5. In den vergangenen sechs Monaten.....hat Sie jemand mit Informationen zu potentiellen Jobangeboten kontaktiert (z.B. Stellenausschreibungen zugeschickt oder Ihnen angeboten, eine Empfehlung für Sie auszusprechen)?
- (a) Ja
 - (b) Nein
6. In den vergangenen sechs Monaten.....wie viele Stellenangebote von anderen Unternehmen haben Sie erhalten?
- (a) 0
 - (b) 1
 - (c) 2
 - (d) 3 oder mehr

Module: Worker-Provided Firms

7. Als Nächstes interessiert uns, wie Sie Beschäftigungsmöglichkeiten in Deutschland wahrnehmen. Auch wenn Sie etwas nicht genau wissen, möchten wir Sie bitten, Ihre Einschätzung mit uns zu teilen. Angenommen, Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln. Bei welchen Unternehmen würden Sie erwägen, sich zu bewerben? Bitte geben Sie drei Unternehmen an, bei denen Sie sich bewerben würden und die Beschäftigte in Stellen wie Ihrer einstellen (z.B. „Musterunternehmen GmbH“). Das können auch Unternehmen ohne aktuelle Stellenangebote sein.
- (a) {Unternehmen 1 eintragen}
 - (b) {Unternehmen 2 eintragen}
 - (c) {Unternehmen 3 eintragen}
 - (d) Ich möchte diese Frage nicht beantworten [exklusive option]
8. [Q7=d] Was beschreibt am besten, warum Sie in der vorherigen Frage keine Unternehmen angegeben haben?
- (a) Ich möchte diese Informationen nicht teilen

- (b) Es gibt Unternehmen, die ich in Betracht ziehen würde, aber die Eingabe in der Umfrage war zu kompliziert
- (c) Mir fallen keine konkreten Unternehmen ein
- (d) Ich habe kein Interesse daran, das Unternehmen zu wechseln und habe darüber bis jetzt nicht nachgedacht
9. **[Q7 has only 1 or 2 provided firms]** Was beschreibt am besten, warum Sie in der vorherigen Frage nicht drei Unternehmen angegeben haben?
- (a) Ich möchte diese Informationen nicht teilen
- (b) Es gibt Unternehmen, die ich in Betracht ziehen würde, aber die Eingabe in der Umfrage war zu kompliziert
- (c) Mir fallen keine weiteren, konkreten Unternehmen ein
- (d) Ich habe kein Interesse daran, das Unternehmen zu wechseln und habe darüber bis jetzt nicht nachgedacht
10. **[Q7 is not missing]** Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{Fill-in 1}	
{Fill-in 2}	
{Fill-in 3}	

11. **Q7 is not missing** Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der von Ihnen genannten Unternehmen wechseln und erhalten sofort das unten angegebene Gehalt. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{Unternehmen 1} mit {randomisierter Gehaltserhöhung}	
{Unternehmen 2} mit {randomisierter Gehaltserhöhung}	
{Unternehmen 3} mit {randomisierter Gehaltserhöhung}	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

[Note: Raises were randomized within groups.]

Module: Researcher-Provided Firms

12. Falls Sie planen {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln, würden Sie erwägen, sich bei folgenden Unternehmen zu bewerben? Bitte kreuzen Sie alle zutreffenden Antworten an.
- (a) Firm 1
- (b) Firm 2
- (c) Firm 3
- (d) Firm 4
- (e) Firm 5
- (f) Firm 6
- (g) Firm 7
13. Was denken Sie, wie hoch wäre Ihr Jahresbruttogehalt, wenn Sie bei diesen Unternehmen in einer ähnlichen Stelle wie bisher arbeiten würden?

Unternehmen	Jahresbruttogehalt in Euro
{ Unternehmen 1 }	
{ Unternehmen 2 }	
{ Unternehmen 3 }	

14. **[Q13 is missing]** Auch wenn Sie sich nicht sicher sind, interessieren wir uns sehr für Ihre Einschätzung, wie diese Unternehmen im Hinblick auf das Gehalt abschneiden. Bitte ordnen Sie die Unternehmen von 1 (höchstes Gehalt) bis 3 (niedrigstes Gehalt) ein.

Unternehmen	Rangfolge
{ Unternehmen 1 }	
{ Unternehmen 2 }	
{ Unternehmen 3 }	

15. Angenommen, Sie könnten in Ihrem aktuellen Unternehmen bleiben oder zu einem der genannten Unternehmen wechseln und erhalten sofort die unten angegebene Gehaltserhöhung. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{ Unternehmen 1 } mit { randomisierter Gehaltserhöhung }	
{ Unternehmen 2 } mit { randomisierter Gehaltserhöhung }	
{ Unternehmen 3 } mit { randomisierter Gehaltserhöhung }	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

[Note: Raises were randomized within groups.]

16. Nun möchten wir Sie bitten, die Rangfolge der Unternehmen unter zwei Szenarien neu einzustufen. Nehmen Sie zuerst an, Sie hätten wieder die drei Jobangebote zur Auswahl, aber Sie hätten den gleichen Arbeitsweg wie bisher. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{ Unternehmen 1 } mit { randomisierter Gehaltserhöhung }	
{ Unternehmen 2 } mit { randomisierter Gehaltserhöhung }	
{ Unternehmen 3 } mit { randomisierter Gehaltserhöhung }	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

[Note: Raises were randomized within groups.]

17. Statt dem Arbeitsweg, nehmen Sie nun an, dass die Training- und Weiterbildungsmöglichkeiten der drei Unternehmen die gleichen sind wie in Ihrem aktuellen Unternehmen. Bitte ordnen Sie die folgenden Jobangebote von 1 (am ehesten annehmen) bis 4 (am wenigsten annehmen).

Jobangebot	Rangfolge
{ Unternehmen 1 } mit { randomisierter Gehaltserhöhung }	
{ Unternehmen 2 } mit { randomisierter Gehaltserhöhung }	
{ Unternehmen 3 } mit { randomisierter Gehaltserhöhung }	
Im jetzigen Unternehmen bleiben mit aktuellem Gehalt	

[Note: Raises were randomized within groups.]

Module: Applications

18. Falls Sie planen würden {im nächsten Monat/ in den nächsten drei Monaten/ in den nächsten sechs Monaten} in ein neues Unternehmen zu wechseln, bei wie vielen verschiedenen Unternehmen würden Sie sich bewerben?
 - (a) Die Anzahl der Unternehmen, bei denen ich mich bewerben würde, ist: _____
19. **[Randomize who sees Q19 (50%) and who sees Q20/Q21 (50%)]** Was sind Ihrer Meinung nach die Hauptgründe, die Mitarbeitende zögern lassen, den Arbeitsplatz zu wechseln? Bitte wählen Sie die zwei wichtigsten Gründe aus.
 - (a) Persönliche Bindung zu den Arbeitskollegen
 - (b) Standort
 - (c) Gehalt
 - (d) Nebenleistungen/ Unternehmenskultur
 - (e) Abneigung gegen Veränderungen
 - (f) Fehlende Möglichkeiten bei anderen Unternehmen
20. Angenommen, Sie vergleichen die Stellenangebote bei zwei verschiedenen Unternehmen: Unternehmen 1 zahlt 10% über dem Durchschnitt im Arbeitsmarkt und Unternehmen 2 zahlt 30% über dem Durchschnitt im Arbeitsmarkt. Welches Unternehmen zieht Ihrer Meinung nach mehr qualifizierte Bewerbende pro Stelle an?
 - (a) Unternehmen 1
 - (b) Unternehmen 2
 - (c) Beide ziehen die gleiche Anzahl von Bewerbenden an
21. Welches Unternehmen bietet Ihrer Meinung nach die besseren Nebenleistungen (z. B. Home Office, Kita-Zuschuss)?
 - (a) Unternehmen 1
 - (b) Unternehmen 2
 - (c) Beide bieten die gleichen Nebenleistungen
22. Nehmen sie an, Sie stellen fest, dass andere Unternehmen in Ihrer Gegend {randomisierter Prozentsatz} mehr bezahlen als Ihr aktueller Arbeitgeber. Wie wahrscheinlich ist es, dass Sie sich auf eine neue Stelle bei einem anderen Unternehmen bewerben würden?
 - (a) Bitte geben Sie eine Zahl zwischen 0% (ich würde mich auf keinen Fall bewerben) und 100% (ich würde mich sofort bewerben) an: ____%.
23. Wie sicher müssten Sie sein, dass Sie eine Stelle bei einem der besser bezahlenden Unternehmen bekommen würden, um sich für einen neuen Job zu bewerben?

- (a) Mindestens 90% sicher
 - (b) Mindestens 75% sicher
 - (c) Mindestens 50% sicher
 - (d) Mindestens 25% sicher
 - (e) Ich würde mich in jedem Fall bewerben
24. Falls Sie einen neuen Job in Betracht ziehen, wie wichtig ist jede der folgenden Eigenschaften für Sie? [Dropdown scale with 1 - überhaupt nicht wichtig, 2 - Etwas wichtig, 3 - Mäßig wichtig, 4 - Sehr wichtig, 5 - äußerst wichtig]
- (a) Möglichkeit für reduzierte Arbeitszeiten
 - (b) Homeoffice-Option
 - (c) Planbare Arbeitszeiten (einschließlich Überstunden)

Module: Person-Specific Information

25. Am Ende der Umfrage interessieren wir uns für Ihren persönlichen Hintergrund. Sind Sie verheiratet?
- (a) Ja
 - (b) Nein
26. Haben Sie Kinder?
- (a) Ja
 - (b) Nein